

Martial Arts Hand Wrap Problem

EG Terminators

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1. Module 1

1.1 Introduction

1.1.1 Background

Hands wraps have been used in various formats for thousands of years as a form of protection for the wearer or in some circumstances as a weapon. Modern day hand wrapping is generally done with cotton or similar fabric strips to protect the hands while participating in combat sports that involve striking such as boxing or Muay Thai.

1.1.2 Team Problem Definitions

Reusable wrapping materials, including athletic wraps, elastic bandages, compression wraps, and other similar fabric supports, are widely used in athletic, medical, and related applied industries. After each use, these wraps must be re-rolled for storage and later use. The re-rolling process is typically manual and highly dependent on the individual's technique, which results in variable tightness, alignment, and compactness.

Inconsistent re-rolling can cause wraps that are uneven, loosely wrapped, and difficult to apply quickly, and are prone to tangling. In situations and environments where they have to be applied quickly and consistently, these issues can reduce readiness and increase preparation time.

The problem definition is: How might we improve efficiency, consistency, and usability of re-rolling reusable and flexible wrap materials to enhance quality of storage, and readiness for a variety of situations?

1.1.3 Context of the Problem

1.1.3.1 *Sources and Root Causes*

Manually hand-rolling wraps are a very common process for athletes. This process is normally done after a session, or after washing the wraps. By doing so, the wraps are prepared for reapplication later on. The time-consuming nature of the activity itself is enough to draw questions about efficiency and effectiveness. By asking questions regarding how the team may reduce the time it takes to roll a wrap, increase the efficiency of the wrap-rolling process, as well as the effectiveness, the team was able to approach the problem definition. This problem definition grew over the course of the project, and as a result became more specific over time.

1.1.3.2 *Key stakeholders*

The scope of this project regards sports athletes who utilize hand-wraps for training. In addition to this, more stakeholders may be the coaches of the athletes as well as the owners of the dojos hosting the training. The relevance of each stakeholder relative to the team problem is variable. The athletes who are manually rolling their wraps after every session are the most affected by the problem. Menial tasks like this reduce the time spent training and can frustrate the athletes. This stakeholder is affected more by the problem than any other stakeholder. The coaches are the next most impacted stakeholders. Though they are not the individuals who are rolling the wraps themselves, they rely on the athlete's ability to roll them. In other words, coaches depend on a degree of consistency, such that the athletes can come in, train, leave, roll their wraps, and come back in for the next session. Lastly, the dojo owners are the next most impacted stakeholder. These individuals may utilize a solution to the team problem to

bring in more customers. Though the problem is less impactful on this group, a solution has the potential to bring positive changes for them.

1.1.4 Existing Solutions

The problem we have defined does not have a perfect solution at this time. The research in this report will show that there are many existing solutions that all have advantages and disadvantages. Aside from hand rolling the wraps there are primarily two types of solutions, mechanical and electrical.

Hand rolling wraps is by far the most common method of preparing them for reuse. This provides the user with the most control of the quality of roll up but as discussed throughout this report the primary downsides are time and fatigue. There are mechanical solutions that involve a crank and wheel structure that rolls the wrap onto a bar which it is removed from once completed such as the Meister portable hand wrap roller [32]. These can be handheld or attached to a wall. The user still needs to carefully feed the wrap into the mechanism as it is cranked to prevent folding and a messy roll.

Multiple versions of the power drill style electric roller were discovered during research such as the Title Boxing wrap wrench [45]. These provide a quicker and more user-friendly process. However, the electric rollers also require the user to carefully feed the wrap as it is rolled to prevent a sloppy roll.

Overall, the existing solutions provide some options to wrap users that desire more efficiency. However, none of the options appear to have a good solution for the issue of the material folding up on itself as mentioned previously they require manual feeding. This is an opportunity for a better solution that can provide quick and neat rolling with minimal effort.

1.1.5 Ancillary Issues

Different ancillary issues have been identified regarding hand wrapping, some that work to our benefit. One of the issues that comes up is dealing with some of the cultural or societal beliefs about the product. The community that takes part in martial arts may see hand wrapping as something they believe should be done by hand and will not be open to any additional devices to do it for them. They may see it as a sign of weakness or a lack of effort and dedication in the skill. This could easily be a difficult market to convince that whatever we create is something that they need.

One of our other ancillary issues that we identified is the sustainability of the product that we are trying to create. If it is something that is powered off battery whether rechargeable or replaceable, it could be seen as wasteful. This wastefulness could be seen by the consumer who might decide that they instead want to go with a product that uses no power. In addition, the convenience is slightly lacking if the item needs to be recharged at any point. However, with environmentalists, this product might lose popularity.

1.2 Stakeholder Outreach/Problem Validation

Table 1-1: Daniel's Stakeholder Interview with Jorge Portillo

	Daniel Benson
Who did you interview?	Jorge Portillo
What did you ask?	1. What do you do with your hand wraps after training?

	- How often do you wash your hand wraps? - Do you ever replace your hand wraps and why? - How long does it take for your hand wraps to dry? - How do you store your hand wraps?
What did you learn?	I learned the basics of caring for hand wraps to keep them clean and knowing when to replace them. I learned after training you need to put the wraps completely unraveled in a mesh bag, so they dry, wash them once a week after three training sessions, and replace them around every six months. Overall, everything about caring for hand wraps from a stakeholder's side.

Table 1-2: Ryan's Stakeholder Interview with a Fighter at his Gym

	Ryan Cherry
Who did you interview?	A fighter at my gym, asked to not share his name
What did you ask?	- How many sets of hand wraps do you own? - How many times do you use each before washing them? - Have you used any tools to re-roll them? - About how long does it take you to re-roll them?
What did you learn?	- He has 5 or 6 sets of hand wraps and wears each once or twice before washing them. - If he trains multiple times in 1 day, he doesn't wear the same wraps for the sessions - He has used a rolling tool, but doesn't own one so he re-rolls them by hand - Using a tool takes him about 30 seconds, so a little over a minute per set - By hand It takes him around 1 minute per wrap, so around 2.5 minutes to re-roll 1 set

	○ He usually does this while watching TV •
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Table 1-3: Max's Stakeholder Interview with Lyam Hassell

	Max Golterman
Who did you interview?	Lyam Hassell
What did you ask?	1. How long have you been using hand wraps in your training? 2. What problems or challenges do you notice when using hand wraps? 3. After training, what do you usually do with your hand wraps? 4. How easy or difficult is it to roll your hand wraps back up after a session? Can you briefly describe the process and how long it takes? 5. Does this problem arise for you outside of martial arts? 6. Is there anything else about hand wraps, either the wearing or rolling process that you think is important?
What did you learn?	• He began kickboxing in 2021 and continued it for about a year before moving to Brazilian Ju Jitsu. • Putting on hand wraps is a finicky process and easy to make uncomfortable, also they are itchy for him. • After removal he wads them up to wash in a mesh bag. • Rolling his 16 foot wrap back up is a slow painstaking process and the wrap often folds on itself. • Some people use the time rolling as social time with other gym members or roll in front of the tv.

Table 1-4: Revell's Stakeholder Interview with Mye Choy and Gabe Cumbess

	Revell Jenkins
Who did you interview?	An EMS professional and Professor, Mye Choy, and a home-use patient, Gabe Cumbess, who used an elastic compression bandage for a sprained ankle over a four-week period
What did you ask?	• How are bandages stored and prepared before deployment? • Do bandages ever need to be re-rolled or adjusted before use? • If re-rolling happens, how much time does it add? • Have you observed inconsistencies in rolling or storage? • Are there standards dictating how bandages must be rolled, packaged, or stored? • How is bandage rolling taught in training? • Does organization impact response efficiency? • If you could improve one aspect of bandage storage or preparation, what would it be? For the patient: • What injury did you use the bandage for and for how long? • Did you re-roll it yourself, and how often? • How easy was it to re-roll neatly? • Did it stay compact or loosen over time? • Was it easy to deploy again? • How was it stored? • Did packaging present difficulty?

	What would you improve?
What did you learn?	The EMS professional reported that bandages are stored in sterile packaging and aren't re-rolled; instead, contaminated bandages are discarded. However, packaging accessibility and fast use under stress are critical, especially when hands are shaking or covered in blood. The home use patient reported that re-rolling the bandage multiple times per day and observed that it gradually lost its compact shape, which required adjustments before reuse. Packaging wasn't problematic for at-home use, but long-term durability and shape retention were both identified as concerns.

Table 1-5: Tjay's Stakeholder Interview with Aiden Martin

	Terrence Micciche-Hall
Who did you interview?	Aidan Martin
What did you ask?	1. Have you ever used hand wraps? 2. What kind of hand wraps did you use? 3. Have you ever fought? 4. Have you ever injured yourself in a fight? 5. Do you ever experience fatigue after fighting? 6. Do hand-wraps make your punches feel more supported?

	7. What do you know about hand-wraps?
What did you learn?	Within this interview, the interviewee revealed a traumatic event caused by not using hand-wraps. In self-defense, the interviewee accidentally punched a solid object. This caused multiple bones in their hand to break. This supports a possible link between hand health and wearing hand wraps.

1.3 Individual Team Member Sections

1.3.1 Ryan Cherry

1.3.1.1 *Research*

The research in this section relates to the time-consuming nature of rolling hand wraps by hand, and how various people at a Muay Thai gym approach rolling their own hand wraps.

This was done in various forms, including an interview with a Muay Thai fighter with several years of experience. A site visit was also conducted to participate in a Muay Thai class and observe the state of the student's hand wraps before application, their procedure for putting them away, and to ask them questions related to the rolling of hand wraps to get a general idea of:

1. What percent of people re-roll their hand wraps before using?
2. What percent of people re-roll them by hand?
3. What percent of people use a tool to re-roll them?
4. What are their thoughts on re-rolling in general?

These observations were made and questions asked to get a better idea of how a larger population of participants handle re-rolling hand wraps. This group all had their hand wraps rolled and ready for application at the start of class and stored them unrolled after use. Of this group, none had used a tool and so they all re rolled them by hand. While one individual didn't mind the process, the others found it tiresome and were interested in something that could save them time and effort.

In my interview with the fighter, he mentions that he has multiple sets of wraps and must roll each multiple times a week due to the frequency of his training. He mentions that he has become much faster and more efficient at rolling over the years but that he still does not enjoy having to do it. He estimates that it takes him a little over 20 minutes to roll 6 sets and that he does so while watching TV to pass the time. While he has used a tool to roll them and prefers it, he still hasn't purchased one.

I also made a video [20] demonstrating the process of re-rolling hand wraps to showcase the amount of time required to roll a single wrap by hand and give more context as to why a solution is necessary, considering many people own multiple pairs.

Another source of information I found useful was various Reddit posts about people's thoughts and experiences with rolling hand wraps. In one post, a user asks others' thoughts on the process, saying "...does anyone also find that the whole rolling, re-rolling, and wrapping of hand wraps to be quite an ordeal? Used to just throw my washed wraps into my bag but it makes wrapping hands during training to be insanely slow as compared to when it's rolled up..." [21], while another user responds with his DIY solution stating "I took a wire clothes hanger...and cut off a straight section 8 inches long, folded it in half so that both ends are touching, and then put it into a chuck of a wireless electric drill...What used to take me about 30 minutes now takes less than 5" [21]. These demonstrate that not only do people find the process tiresome, but enough so that they've sought to create a solution themselves. In another post [22], multiple users recommend various products to re-roll them.

To get a better idea of how existing solutions handle the problem, I found videos of different tools dedicated to re-rolling hand wraps [23]. While one of these tools appears to save time when compared to rolling by hand, the other product seems to require a similar amount of time and effort to doing so by hand.

1.3.1.2 *Remaining Unknowns*

Although the general consensus of those interviewed and the online sources indicate most people do not enjoy the process of hand rolling hand wraps, no definitive source was found to prove that this is a large-scale issue to be solved. A further step to be confident that a solution is required would be to engage more stakeholders outside of the gym visited.

1.3.1.3 *Summary*

The research in this section indicates that re-rolling hand wraps is commonly viewed as a time-consuming and undesirable task. Observations from a Muay Thai class showed that participants typically store wraps unrolled after use and re-roll them by hand before training, with no one reporting the use of a dedicated tool. Most individuals observed found the process tiresome and expressed interest in a way to reduce the time and effort required. This aligns with the interview findings, in which an experienced fighter reported rolling multiple sets several times per week and estimated that rolling six sets takes a little over 20 minutes, even after becoming more efficient over time. Online discussion posts further reinforce these findings, with users describing the process as an "ordeal," creating DIY solutions, and recommending products designed to speed up re-rolling. Overall, the evidence suggests there is consistent frustration with hand-rolling wraps and interest in tools that improve efficiency, although additional outreach beyond the gym visited would be needed to confirm the scale of the issue.

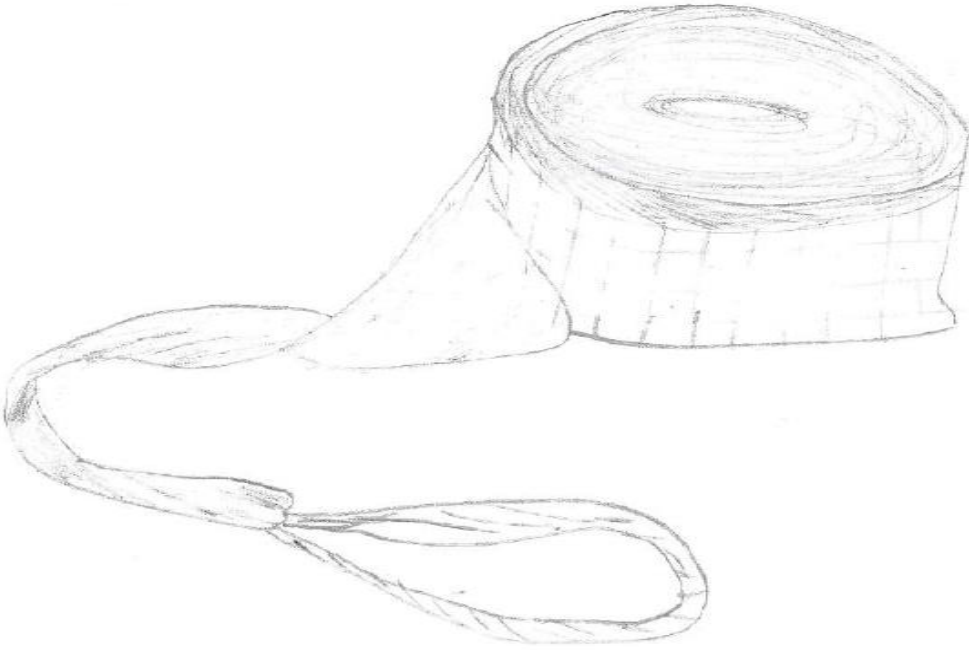


Figure 1- 1: Partially Rolled Hand Wrap



Figure 1- 2: Un-Wrapping a Hand Wrap

1.3.2 Terrence Micciche Hall

1.3.2.1 Research

The research of this section regards the need for a solution to the defined problem. More specifically, the unformatted questions that will be considered throughout this section are:

1. How do athletes feel after exercising?
2. Why are wrapping hands and wrists important when fighting?

These questions are not meant to be answered directly but rather used as a platform for understanding how to generally consider the process of finding a solution to the defined problem. While considering these questions, Red Rocks Community College (RRCC) librarian Tanner King and RRCC student Aiden Martin were contacted and interviewed. Tanner King provided a plethora of useful resources, two of which were academic articles related to hand and wrist injuries in MMA, and Aidan Martin's interview helped to solidify a justification for a solution to the defined problem. Finally, a visit to Peak Kickboxing was conducted to visually examine how many combat sport athletes are using hand wraps.

The first academic article revealed that there does not seem to be a higher predilection to hand or wrist injuries in sports such as MMA as compared to other combat athletes [1]. Though it is conceded by the author that there is a lack of data regarding injuries in MMA, more specifically for the hands and wrist. Later in the article, it is described that when comparing MMA gel gloves to other types of gloves

used in combat sports, there does not seem to be a pattern of injury based on the size or type of the glove [1]. This validates further research into the necessity of hand-wraps.

From the second academic article, a theme developed in the first article is perpetuated. Most protective equipment, such as gel gloves or hand wraps, have not been sufficiently tested to support any claims of increased support or injury prevention [2]. Though, hand and wrist injuries are very common in combat athletes [2]. In response to this, the author had conducted their own experimental study and found an immensely promising conclusion for determining the efficacy of using hand-wraps and gloves. This conclusion was that hand wraps provide measurable impact absorption, even when compared to the more effective gel gloves [2]. The need for combat athletes to wear hand-wraps for protected is thus validated.

Next is the interview with Aiden Martin, which was a short, yet insightful one. This student has never used hand wraps, and he finds them to be rather impractical. This is because the fights that he has participated in are primarily described “street-fights,” where an individual normally does not wear any sort of protective gear. He describes that every time after he fights, his hands always swell and his joints feel very sensitive. However, this alone does not make him a key stakeholder in the search for the need for a solution to the defined problem. In one of his most memorable fights, one which he describes as in the interest of in-the-moment self-preservation, he had broken multiple bones in his hand. Aidan describes the injury as such, “The reason my hand broke was because I hit the floor, and it did not absorb the shock,” [3]. Figure 1- 3 below depicts the interviewee after breaking their hand.

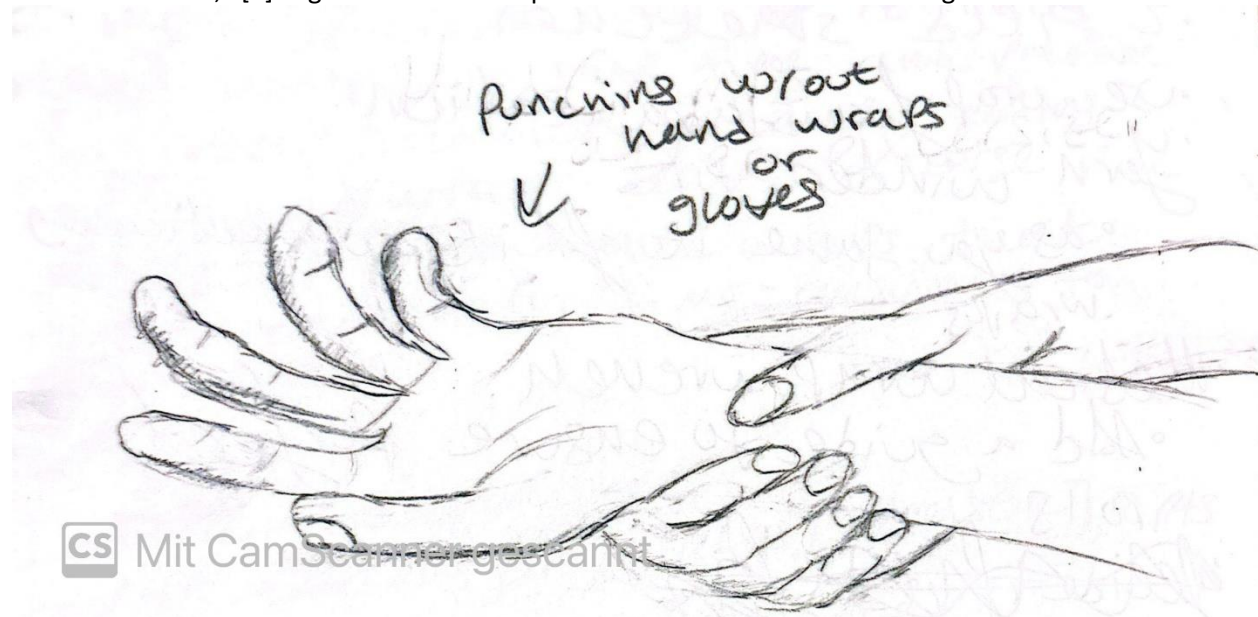


Figure 1- 3: Hands Without Proper Wraps

While in the case of self-preservation, such that an individual should not have the time to apply protective gear before fighting, there is no argument that the fighters should have spent more time considering the health of their hands and wrist. However, anecdotal evidence from Aidan, a streetfighter who has never worn hand-wraps in their life, suggests that an individual is more likely to harm their hands and wrists if not wearing anything to absorb the shock-impact from a punch.

Finally, there was a trip to the Peak Kickboxing dojo. While there was no interview or meaningful communication conducted here, something very useful stemmed from it: every individual utilized hand-

wraps. This information continues to justify the need for a solution to the defined problem, as it is more apparent than ever that athletes in combat sports, such as Kickboxing, utilize hand-wraps.

1.3.2.2 Remaining Unknowns

After the research phase of the need for a solution, there still exists much that has yet to be answered. Most particularly, there still is no concrete justification for why hand-wraps are necessary to wear in combat sports. While the lack of studies is fairly concerning, analysis of key stakeholders' interviews, in addition to experimental studies found in academic articles, still suggest that there is room for argument regarding the importance of hand wraps. Moving forward, a very important question should be asked: are hand wraps paired with gel gloves more likely to prevent hand and wrist injury in combat sports than only gel gloves alone? Answering this question should take a separate experimental study.

1.3.2.3 Summary

In summary, while there is a lack of formal studies regarding hand and wrist injuries in combat sports, such as MMA, it is apparent from select academic articles as well as interviews that hand-wraps have the potential to be useful for protecting hand-health[2]. In addition to this, an on-site visit to a dojo shows that hand-wraps are used in martial arts. This justifies the need for a solution to the defined team problem. Figure 1- 4 below illustrates a rolled-up hand-wrap.



CS Mit CamScanner gescannt

Figure 1- 4: A Rolled Hand Wrap

1.3.3 Revell Jenkins

1.3.3.1 *Research*

Scholarly literature shows that the mechanical performance of elastic compression bandages depends on material structure and tensile behavior. Research has shown that bandages sold within the same commercial category can exhibit significantly different tensile properties due to variations in construction parameters [4]. Their findings indicate that different levels of stretch must be applied to achieve required therapeutic pressure, meaning compression output is directly influenced by handling and elongation.

Similarly, Kwon et al. characterized compression bandage performance based on interface pressure and mechanical rigidity [5]. Their research demonstrated that pressure profiles are governed by tension–elongation relationships, reinforcing that applied stretch controls compression effectiveness.

Clinical literature further emphasizes the importance of sufficient and controlled compression during emergency treatment. Naimer and Chemla reported that elastic adhesive dressings must be wrapped until adequate pressure is achieved to terminate hemorrhage, and the application technique directly influences effectiveness [6]. In time-sensitive scenarios, compression must be applied quickly and reliably.

Together, these sources establish that elastic compression bandages are performance-sensitive devices whose effectiveness depends on predictable handling, controlled application, and proper rewrapping practices, especially in home use.

1.3.3.2 Stakeholder Engagement

An EMS professional was interviewed to understand bandage storage and deployment practices in emergency contexts [7]. In these settings, bandages are stored in sterile, sealed packaging, and are not re-rolled. If a bandage becomes unsterile, it is discarded rather than reused. This finding eliminates manual re-rolling as a workflow issue in professional emergency environments. As shown in Figure 1- 5, improperly rolled wraps can reduce organization, increase deployment difficulty, and negatively affect usability in emergency situations.

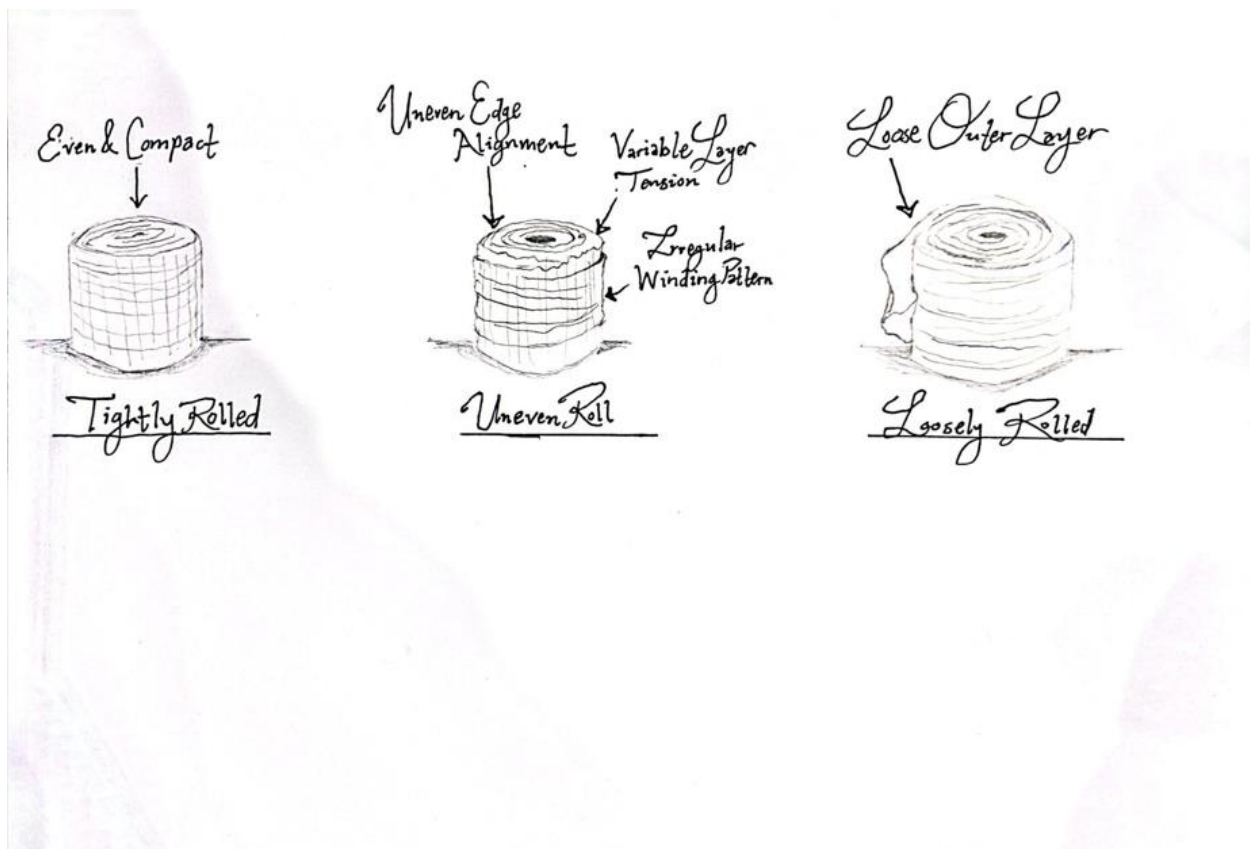


Figure 1- 5: Properly and Improperly Rolled Wraps

However, significant usability challenges were identified in packaging and rapid deployment. Packaging may be difficult to open under stress when hands are shaking or covered in blood. Small pull tabs can fail, increasing the risk of accidental drops and contamination. Storage organization and compactness were also identified as critical, as intervention in severe bleeding scenarios may be required within one to two minutes [7]. These findings validate packaging accessibility and deployment readiness as meaningful constraints in emergency contexts.

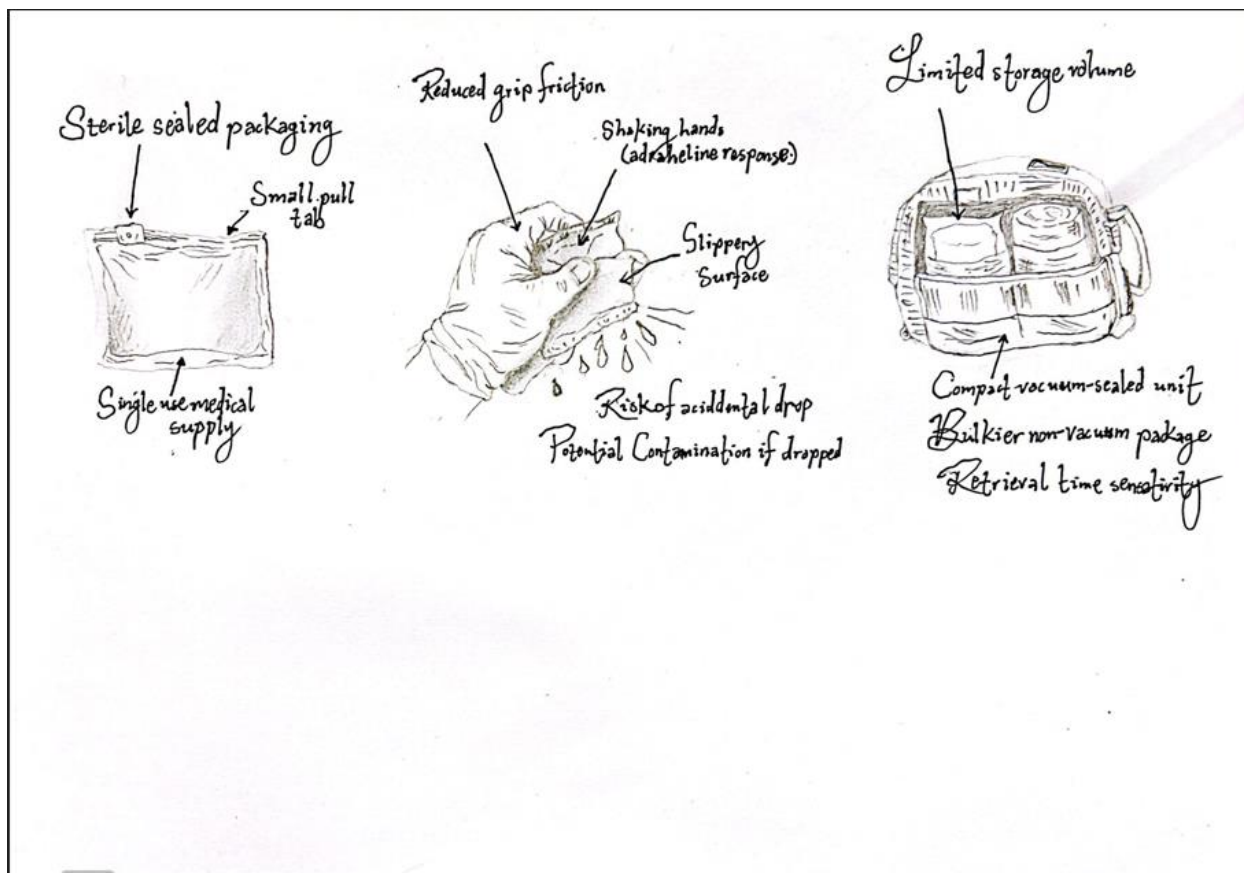


Figure 1- 6: Proper Medical Bandage Packaging

A home-use patient was also interviewed regarding repeated use of an elastic compression bandage for a sprained ankle over a four-week period [8]. The patient reported re-rolling the bandage approximately three times per day. While re-rolling itself was not difficult, the bandage began loosening after the first week and no longer retained its original compact shape. Prior to reuse, the roll required adjustment and tightening. The bandage was stored in a drawer between uses, and packaging did not present difficulty during initial opening. The patient suggested that material durability and shape retention were areas for improvement [8].

These findings indicate that usability challenges differ across stakeholder groups. In emergency contexts, packaging accessibility and rapid deployment are primary concerns. In home-use contexts, repeated re-rolling and long-term shape retention affect redeployment convenience and user satisfaction. Together, these interviews support the need to evaluate packaging design, storage configuration, and material durability as components of the broader usability problem.

1.3.3.3 Remaining Unknowns

Although scholarly research confirms that compression bandage performance depends on controlled stretch and proper application, several practical uncertainties remain regarding packaging, storage organization, and deployment usability in both emergency and home-use environments.

The following unknowns require further validation through stakeholder engagement:

- The extent to which packaging design affects deployment speed under stress
- The frequency of packaging-related errors, such as dropping or contaminating bandages during opening
- Whether storage configuration meaningfully impacts retrieval time in emergency scenarios
- How home users store reusable elastic compression bandages between uses
- The frequency with which patients manually re-roll bandages and whether this affects redeployment convenience
- Whether packaging bulk or compactness limits organization, portability, or carry capacity

While emergency medical personnel prioritize sterility and often discard contaminated bandages rather than re-roll them, it remains unclear how packaging design and storage structure influence operational efficiency. Additionally, the usability impact of manual re-rolling in home settings has not yet been fully quantified.

Further interviews with both emergency professionals and patient stakeholders are necessary to determine whether these factors represent minor inconveniences or meaningful workflow inefficiencies that justify design intervention.

1.3.3.4 Summary

Existing literature demonstrates that the performance of elastic compression bandages depends on controlled stretch, material behavior, and consistent application. Variations in tensile properties and tension–elongation relationships influence the pressure delivered during use, confirming that proper handling is necessary for predictable performance outcomes.

Stakeholder engagement indicates that in emergency settings, bandages are stored in sterile packaging and are not re-rolled. However, usability challenges were identified in packaging design, storage organization, and rapid deployment under stress. Difficult-to-open packaging and bulk storage methods may affect accessibility and response efficiency in time-sensitive situations. In home-use contexts, additional uncertainty remains regarding how reusable bandages are stored, redeployed, and perceived by users.

At this stage, the problem is supported by mechanical and clinical research, as well as preliminary stakeholder findings related to deployment readiness. Further engagement with both emergency professionals and patient users will determine whether packaging and storage usability represent meaningful inefficiencies that justify design development.

1.3.4 Max Golterman

1.3.4.1 Research

The research in this section relates to the existence of the previously defined problem outside of the world of martial arts. To keep the scope narrow, this report will only encompass other athletic disciplines and issues ancillary to those activities. This includes, but is not limited to weightlifting, team and individual sports, and equestrian sports.

First, we met with RRCC Librarian Tanner King to receive his assistance with finding research materials online. These come in several forms including scholarly articles on athletic injuries, videos on different wraps, and some current solutions associated with the overarching problem of returning those

wraps to their proper rolled form. Secondly, to get an idea of the difficulty and time-consuming nature of rolling up a martial arts hand wrap we borrowed one from a martial artist and rolled it up ourselves. The process took approximately two minutes, and it was difficult to roll it neatly without having the wrap folded on itself lengthwise.

At home, to continue getting a feel for the difficulty of rolling up similar objects, I performed the same experiment with a roll of dog waste bags which had become unrolled. In this experiment I found the same issue of having a difficult time keeping the roll neat and preventing the material from folding on itself. To prevent that a good amount of patience was required which made for slow, painstaking rolling. This hands-on research demonstrates that one of the primary issues associated with our problem is the neatness of the roll. In our section on existing solutions, we will show that many of the devices currently available do not account for this difficulty. This is especially true with the existing solutions that utilize a powered rolling device. Some of the manual devices have a secondary piece that feeds the material into the rolling apparatus which we will attempt to use as inspiration in our solution.

The subject of my stakeholder interview was Lyam Hassell a martial artist. In recent years Lyam has focused on training Brazilian Ju Jitsu but trained kick boxing for nearly a year where he used hand wraps. Lyam found the process of hand wrapping difficult to do correctly as they were often too tight, or too loose and had the tendency to fold up on themselves. His experience was that after removal he would ball the loose wrap up to differentiate dirty from clean. When washing he would place the loose wrap in a mesh bag otherwise it would become a twisted clump. To prepare them for the next use, he would roll them by hand carefully which was a tedious process due to the 16-foot length of the wrap.

The problem exists in the sport of weightlifting for both professional and everyday participants. To support joints and protect themselves from injury, weightlifters will often utilize knee and elbow wraps. During a site visit to the Highlands Ranch Recreation Center, there were gym users with different knee devices such as wraps and braces being worn. We learned that weightlifters performing back squat exercises with knee wraps on did see a 10% increase in peak power on one repetition [9]. However, the study concluded that while the wraps increase mechanical output, they also changed the range of motion which could possibly impact knee integrity. As a result, they recommended looking at the source of perceived joint weakness in training.

A video from Youtube.com user Clay Butler shows a current solution related to our problem of rolling the wrap back up for the next use [10]. This solution appears to be a specialized attachment for an unknown brand of power drill. The attachment goes into the output end of the drill just like any drill bit would and locks in via the drill's locking mechanism. This could translate to most brands of power drills. The specialized bit has an elongated U with two metal bars extending away from the point of attachment. The knee wrap is fed into the U-shaped metal prong and when the drill trigger is pulled, it winds the wrap up into a roll. With this method the user is required to guide the wrap as it is rolled up slowly or else the wrap could fold on itself, get askew or get jammed in the mechanism. Figure 1- 7 shows a sketch of the knee wrap rolling device from the above referenced video.

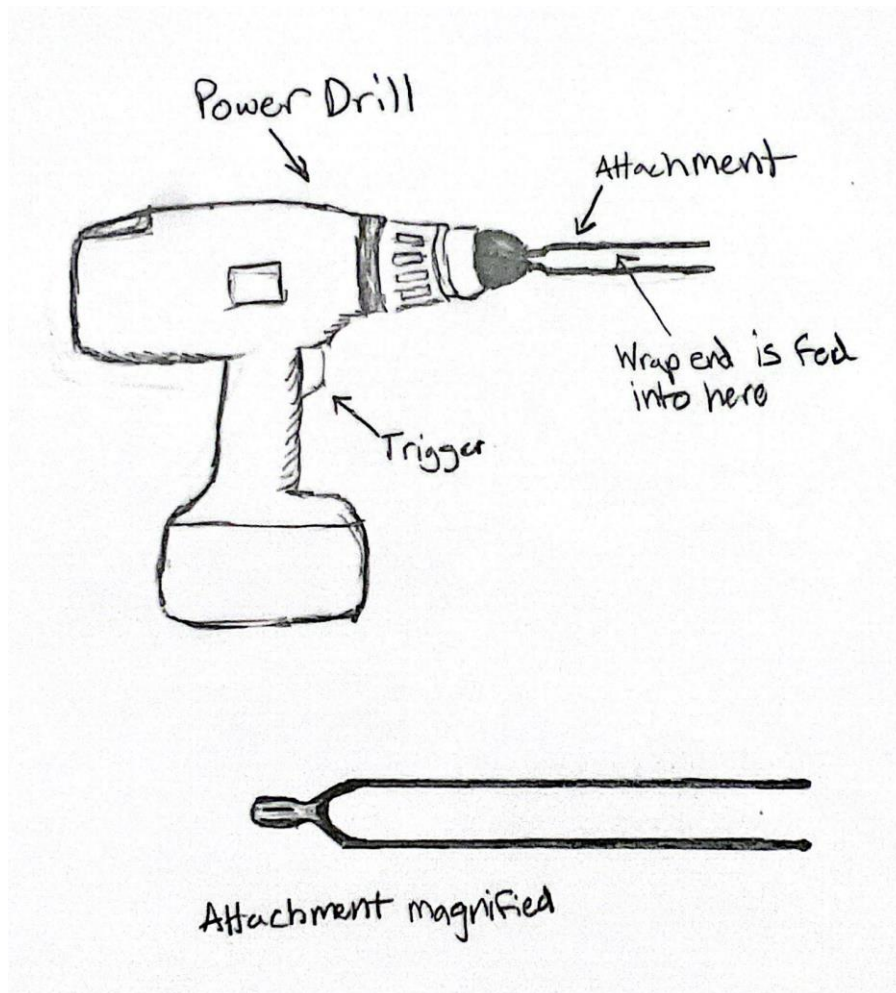


Figure 1- 7: Power Drill Rolling Solution

During research another area where this problem exists came to light in the form of leg wraps for horses. In “The What, Why and How of POLO WRAPS” the author states that many horses will have their legs wrapped to support tendons and ligaments [10]. The material of the polo wraps as they are often called is a stretchy synthetic pile with hook and loop fastening. On page 3, the steps to roll a leg wrap in preparation for application are laid out and as with many other examples, they do it manually by hand. As with hand wraps for martial arts, it is imperative that the leg wraps are properly applied to help protect the horse [10]. This is demonstrated in a YouTube video from Michigan State University “How to Wrap a Horse’s Leg” [11] where they begin with a loose wrap and proceed to roll it back up. This must be done correctly, so the Velcro ends up in the correct position as with other wraps. With the number of wraps required for horse activities, rolling them back up manually after washing leaves room for improvement. Figure 1- 8 is a sketch of the leg wrapping process for horses as demonstrated in the video.

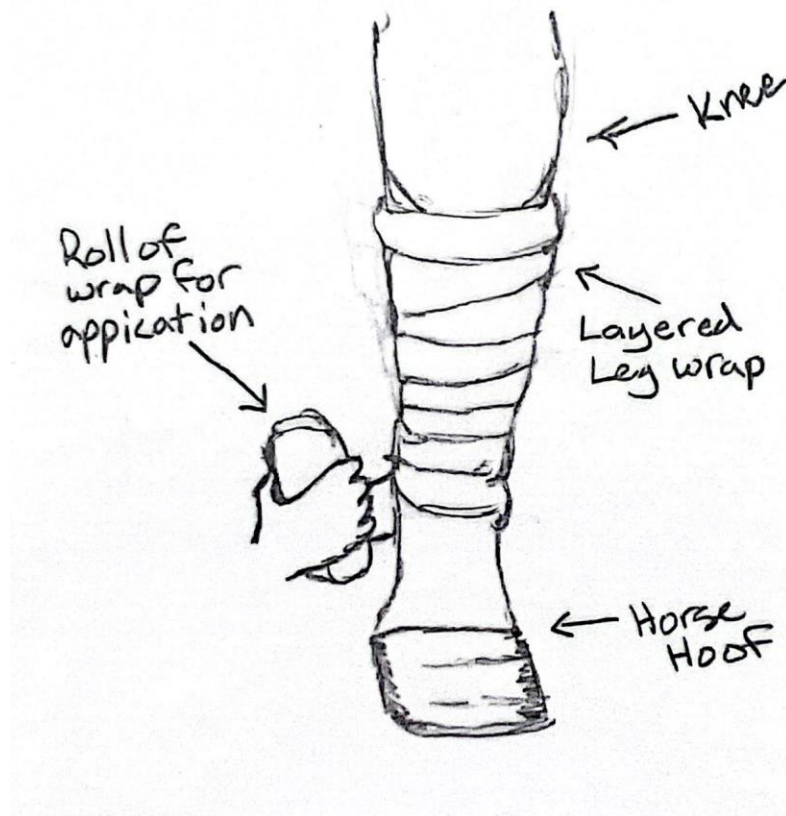


Figure 1- 8: Bandage Application for Horse

To round out the research focus was placed on traditional sports such as football and soccer where ankle injuries are very common. In “The influence of athletic taping and bracing on ankle sprain” it is stated that an estimated 28,000 ankle sprains occur every day in the US [11]. The use of taping and bracing as a preventative measure has been prevalent for years. In addition to those same methods being used to stabilize an injured ankle. The research for this problem has shown, however, that taping is the most common method and non-reusable. For this reason, it will not be relevant to the defined problem in this report.

1.3.4.2 Remaining Unknowns

We are in the early stages of research on this topic, so there remain many unknowns. One major unknown is the current solutions and how widespread their use is. In conjunction with that, what the appetite is for an automatic device to make the process easier will be important to know. Another unknown is how a rolling device would function for wraps of different sizes, materials and condition. One way to know how widespread the need for a solution is, would be to learn the statistics on wrap usage in the communities that we target.

1.3.4.3 Summary

In summary, the research performed for this section of the report has indicated that the problem of efficiently and properly rolling wraps exists well beyond the world of martial arts. Examples include weightlifting, equestrian, and every sport where ankle, knee, and wrist support could be needed. While

many of these wraps are often disposable tape, there are some instances where reusable joint bandages. When these are used in large quantities, such as with a barn full of horses, or regularly for athlete's training, the problem can become one that deserves a closer look.

The specific community that appears to have the greatest use for a solution to our problem, outside of our main area of focus, may be the Equestrian community. Many horse owners have multiple animals and go through the process of wrapping and rolling with every activity. Additional research would be needed to discover the true need within the community.

1.3.5 Daniel Benson

1.3.5.1 *Research*

A huge concern in the martial arts community regarding our problem being researched is the importance of hygiene and keeping everything clean to prevent any mold or bacteria from having a bad effect on the user. To learn more about the problem, two interviews were conducted with a boxer and a biologist, along with reviewing four academic papers.

In the first interview with the boxer, many different topics pertaining to hand wrap care were covered. The first being the need to store them in a mesh bag after practice unraveled to allow for the wraps to dry [14]. This bag is like the one shown in Figure 1- 9. Another thing mentioned was the interviewee would wait three weeks or nine training sessions in between cleaning his wraps [14]. This shows the need for our solution to have something to keep wraps dry to prevent any kind of mold from being grown. In addition to this the interviewee showed some of his old wraps, showing how the Velcro would fray and they would become dirty [14]. With this, our solution will need some kind of way to prevent the wear and tear and extend the life of the wraps themselves. Finally, the boxer mentioned that after a training session, they usually dry overnight and are completely dry by the morning, and while washing them, he puts them in the dryer, so they come out completely dry and ready to be rolled back up [14]. This shows the need for some sort of storage method that could be part of the system or works adjacent to it.



Figure 1- 9: Bag for Hand Wraps

My second interview was with Microbiology teacher Lynne Albert, who teaches at Red Rocks Community College. When asked about bacterial growth Professor Albert responded with there being a wide range of bacteria that can grow in any environment, the type depending on the humidity, oxygen concentration, if there is an open wound, and if there is more moisture in the environment, the more dangerous ones grow at 37 degrees Celsius [15]. Professor Albert also mentioned that mold spores are also an issue and can take special cream and plenty of time to heal. When asked how to get rid of bacteria, the professor responded that cotton and soap were good at removing bacteria, in addition to heating over 120 degrees Celsius, and putting the surface under 15 psi of pressure [15]. The solution created needs to include something that will be antimicrobial, to prevent bacteria growth on the product created, and the wrap itself. Lastly the professor told me that if your hands are cut wraps need to be washed often, and that antimicrobial fabric would be the best to prevent any bacterial growth [15]. From this the solution needs to have some antimicrobial effects to prevent bacteria such as staphylococcus shown in Figure 1- 10 as much as possible.

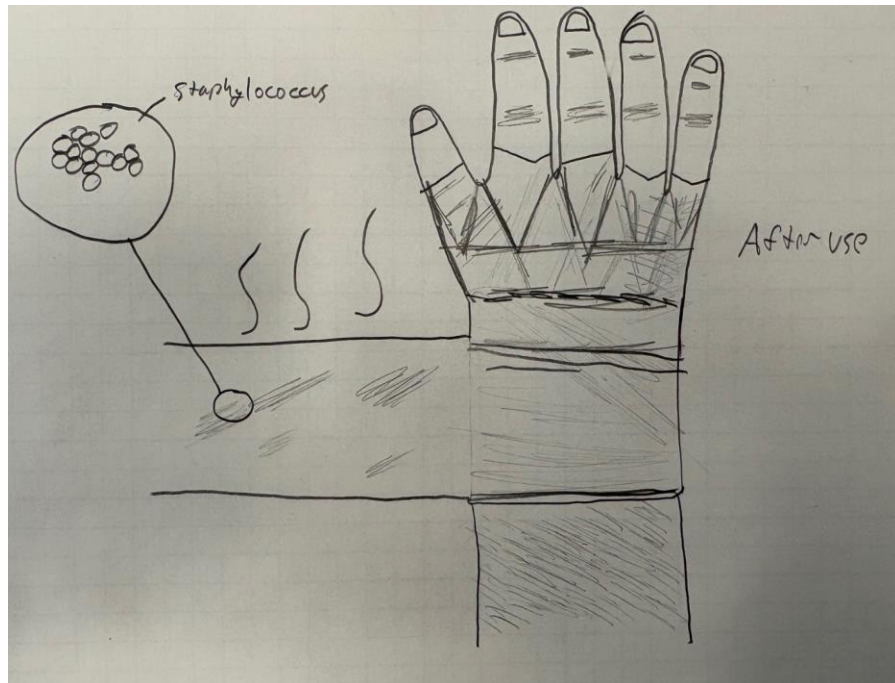


Figure 1- 10: Bacteria Growth on Hand Wraps

Bacterial colonization is a major issue in multiple fabric types, including those used in mixed martial arts and our hand wraps, which we are studying. Human sweat provides a variety of nutrients that promote the growth of bacteria on materials such as cotton and synthetic materials, with synthetic materials growing even more [17]. Amino acids and other nutrients in sweat often promote the creation of biofilms, which are tightly packed groupings of bacteria such as *E. coli*, and can lead to a bad smell [17]. Hand wraps are often composed of materials such as cotton or other nylon materials, making them breeding grounds for all sorts of bacteria and smells after use. This can create an additional danger in any area of skin that can be susceptible to an infection.

Moisture on skin can be trapped by clothes, creating an environment where bacteria can grow profusely and affect our skin's natural bacterial environment. The bacteria can often produce a negative effect for the wearer and the clothing itself, with the fabric deteriorating, and the bacteria causing issues such as allergies and infections in the wearer [18]. When a moist area is created on clothing from exercise or sweating, it can easily create a breeding ground for bacteria that poses both an adverse effect on the materials and the wearer, placing both in an unhealthy state. In addition to this, a moist area can often carry around one million bacteria per square centimeter compared to only one hundred bacteria per square centimeter, and this bacterium is able to transfer from clothing to skin and vice versa [16]. Bacteria and Fungi not only multiply in moist environments but also transfer from clothing, for example, the hand wraps can transfer bacteria to your hands as bacteria is transferred to them.

Certain materials can naturally help to regulate the amount of moisture in certain areas in addition to controlling the number of bacteria that can grow. Bioceramic fibers are included in one of these materials made of materials such as calcium, aluminum, and other metals that interact with your body's infrared radiation to help better distribute heat [18]. If these materials are used as a substitute for the old material, they could dissipate heat at a better rate and prevent the growth of bacteria. In fact, in testing the materials that used the bioceramic fibers had a CFU of only 171 as opposed to a CFU

of 215, CFU being colony forming units [18]. Overall, if Bioceramics are implemented into the design it will greatly reduce the cultures found in our hand wraps.

Another area of research was in different clothing of athletes and the bacteria that can provide adverse effects on the user. It was found that this fabric, when it was used without proper cleaning, would lead to many microorganisms growing over time [19]. In addition to this, the way that clothing is washed can also influence how many microorganisms can grow. The types of washing that can affect this include when something is washed at a lower temperature, when detergents are not used, and when something is air-dried, which allows bacteria to survive [19]. With this in mind, hand wraps clearly need to be washed at a higher temperature, with detergents, and to be air-dried to keep them from becoming a breeding ground for pathogens and bacteria.

1.3.5.2 *Remaining Unknowns*

Some of the remaining unknowns range from issues regarding athletes in our specific field of study all the way to the specific bacteria we must deal with and how to incorporate it into our product. For our athletes, I do not yet know the types of injuries that could lead to a bacterial infection or any other infection that would need to be treated. In addition to this an interview with someone who has had issues with injury or some sort of bacterial infection related to martial arts would be useful. In addition to this I was able to find information about Staph infections but had trouble finding anything related to other sorts of infections or what kind of bacteria would cause the smell. Finally, I want to find more about what kinds of applications of different antibacterial and health-related concerns have gone into the field of hand wraps and how we can include that into our own designs.

1.3.5.3 *Summary*

Throughout my interviews and articles, I gained a new insight into how hand wraps need to be treated and how to prevent any health concerns from them. I learned how to properly care for them, ranging from how to properly wash them and how often to replace them. I also learned what kinds of bacteria can grow on them and what dangers they can pose to the user. Finally, I found academic sources showing the bacterial growth and the best ways to mitigate the growth of those bacteria. Overall, I have a much better understanding of the issues cleanliness presents and more viewpoints I can use to go about developing our device.

In our own device, this research has truly guided my thinking in the things I might need to include in a solution to our problem. First is finding a good way to store the wraps to prevent any sort of bacteria growth and allow the moisture to dry as opposed to storing them dirty. Next, I would like to include something on the device itself that prevents any transfer from the device to the wrap, or vice versa. Finally, A good way to kill the bacteria that could be included with the wrap would be best to maintain the health of anyone who uses our product.

2. Module 2

2.1 Summary of Key Stakeholders and Existing Solutions

The main key stakeholders in our problem include students of martial arts, martial arts teachers, hobbyists, and Dojo owners. All four of them provide a key stake in the problem and could constantly be affected by it. In an interview with a student who partakes in boxing, he said he would train with the hand wraps three times a week, and he should wash all of them every week [14]. For that student he is

re-rolling these wraps around six times a week, and at 2 minutes a piece, this is 24 minutes spent every week trying to roll these up. The people in this group are constantly re-rolling these wraps every week, making them out main stakeholders for the issue.

Some of our lesser stakeholders fall into two groups which are the manufacturers, and the parents of the users. The manufacturers hold a big stake in the project since it depends on what they can produce, determines what is made in production, and whether the final product can recharge or has enough power. Parents are also a huge factor in production since they determine whether their child needs the product. While they themselves might not use the product, they are a huge determinant of the price and usability of the product.

As for existing solutions to our problem, there are around three different kinds of products that can be used. The first is a wall mounted roller that cranks manually, where there is a pin in the middle that the Velcro goes around, and the wrap feeds in, and once it is fully cranked, the pin is pulled out, and the wrapped wrap can be removed. Another one is a handheld crank system. It is built from a cylinder with a rod with a hole in the middle. The wrap is tied onto the pin and fed through the side while the crank is slowly rolling it up. Once rolled, the wrap can slide right out. The third solution is a drill, exactly like the one in Figure 2-24. The roll is hooked onto a split in the shaft on the drill and when the trigger is pulled, it spins. However, the Title product photographed was found to be very faulty according to its amazon reviews with multiple stakeholders discussing the motors lack of power, and

2.2 Requirements, Customer Needs, and Technical Specifications

The primary requirements for this device are ease of use, portability, affordability, and durability. These requirements reflect the needs identified by stakeholders and help guide the technical development of the design.

General Design Requirements

The device should be easy to use with minimal instruction, portable enough for one-handed operation, affordable compared to similar products, and durable enough to withstand accidental drops.

Stakeholder Needs

Stakeholders want a product that is simple to operate, easy to carry, reasonably priced, and capable of maintaining functionality during regular use. Since the device is handheld, users expect it to be compact and durable.

Technical Design Specifications

To meet these needs, the design should include clear operating instructions, maintain a compact form similar to manual wrap rollers, and be manufactured at a cost low enough to keep the selling price competitive. Internal components should be securely housed, and the outer structure should be made from a material capable of withstanding impact.

A summary of the design requirements, customer needs, and technical specifications is shown in Table 2-1.

Table 2-1

Requirements	Customer Needs	Technical Specifications
Easy to use	The user should need as little instruction as possible to use the device.	A short and concise instruction manual should be included, and preferably a video demonstrating use as well.
Portable	The product should be small enough to be held in one hand while in use.	The device should be similar in size to manual rolling tools so it can be operated with one hand while the other attends to the hand wrap.
Must be affordable	The product should be equal to or preferably less than other current electric roller prices. Potential users shouldn't be put in a position where they need to justify the higher cost of an item with the same general functionality as another.	The cost of production should be between 40 and 45 dollars or less to give a 10 - 20% profit margin. Ideally the cost of production would be below 20 dollars, to keep the price near manual rollers.
Drop Safe	The product should be able to withstand multiple drops onto various floor surfaces. Since this is a hand-held device, the expectation is that it will be dropped at some point.	Mechanical and electrical components must all be securely attached and connected in the housing, and the housing should be printed in a filament known to be able to withstand impacts.

2.3 Individual Looks-Like Prototype and Concept

2.3.1 Ryan Cherry

2.3.1.1 Individual Solution

This solution is a handheld electric hand wrap roller. It is unique compared to existing solutions because it combines the compact, portable form of manual wrap rollers with an electric drive system. This allows the device to remain small and easy to transport while reducing the time and effort required to reroll hand wraps.

The design includes a 3D-printed housing, an N-20 motor or similar compact motor, a power source, an activation button, and a set of rotating prongs that secure and roll the wrap, as shown in Figure 2- 1. These components work together to create a portable device capable of rerolling hand wraps in a consistent and efficient manner.

Figure 2- 2 shows a top view of the device separated into two pieces. The bottom assembly, shown on the left, contains the two prongs used to begin and guide the rolling process. One prong holds the wrap, while the second helps catch and roll it as the mechanism rotates. The top assembly, shown on the right, can be attached to enclose the system during operation.

Figure 2- 3 shows a side view of the device, including the opening through which the hand wrap is fed and the location of the rotating prongs. This view highlights the compact profile of the design.

Figure 2- 4 shows a bottom view of the device, with the motor exposed in the lower assembly. An additional component would be included in the complete design to house the battery and remaining electrical components.

Overall, this concept provides a compact and powered alternative to manual wrap rollers, with an emphasis on portability, simplicity, and ease of use.

2.3.1.2 Solution Field Sketches

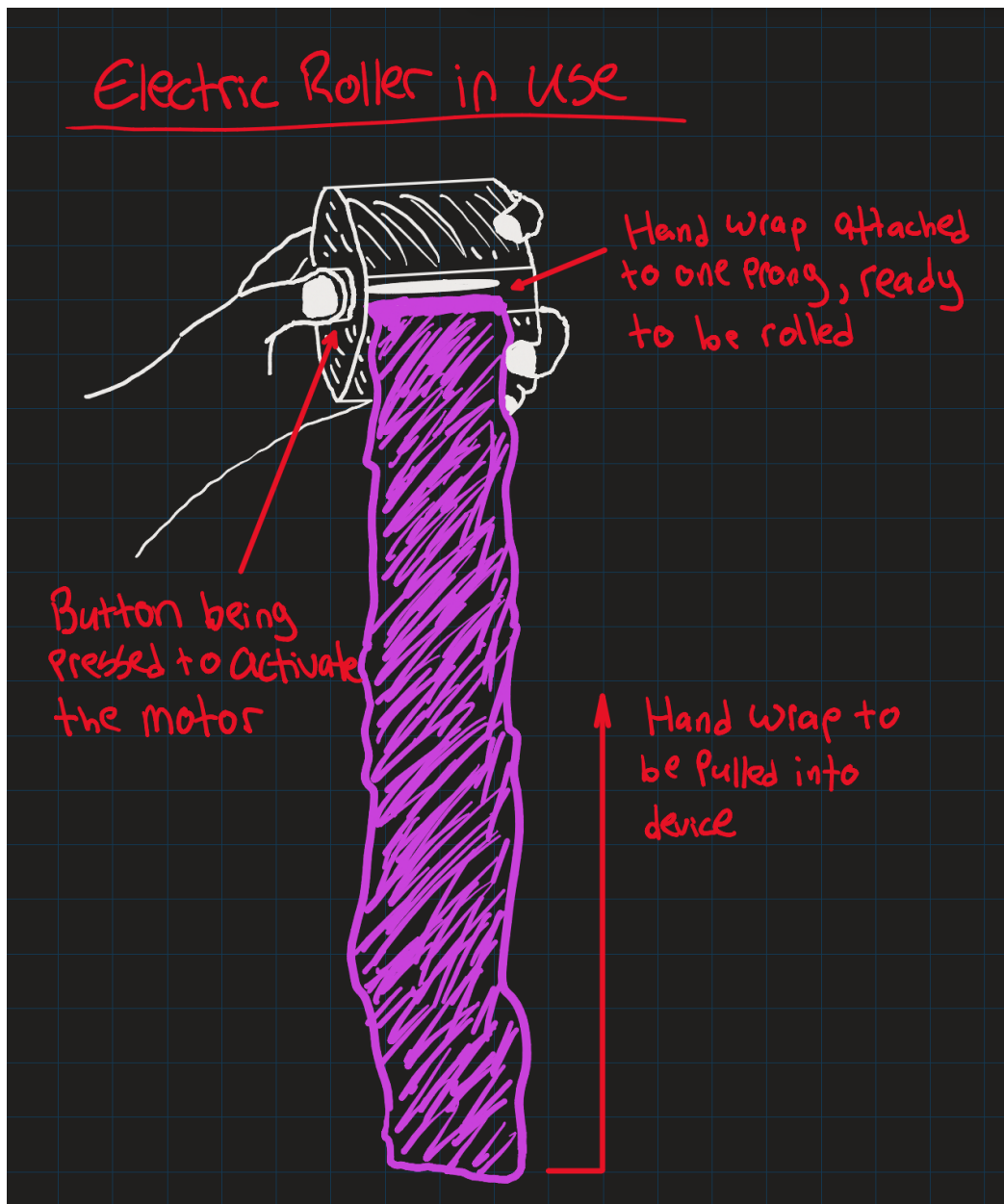


Figure 2- 1 Electric Roller Field Sketch

2.3.1.3 Photos of Solution



Figure 2- 2 Electric Roller Prototype Top-View



Figure 2- 3 Electric Roller Prototype Side-View

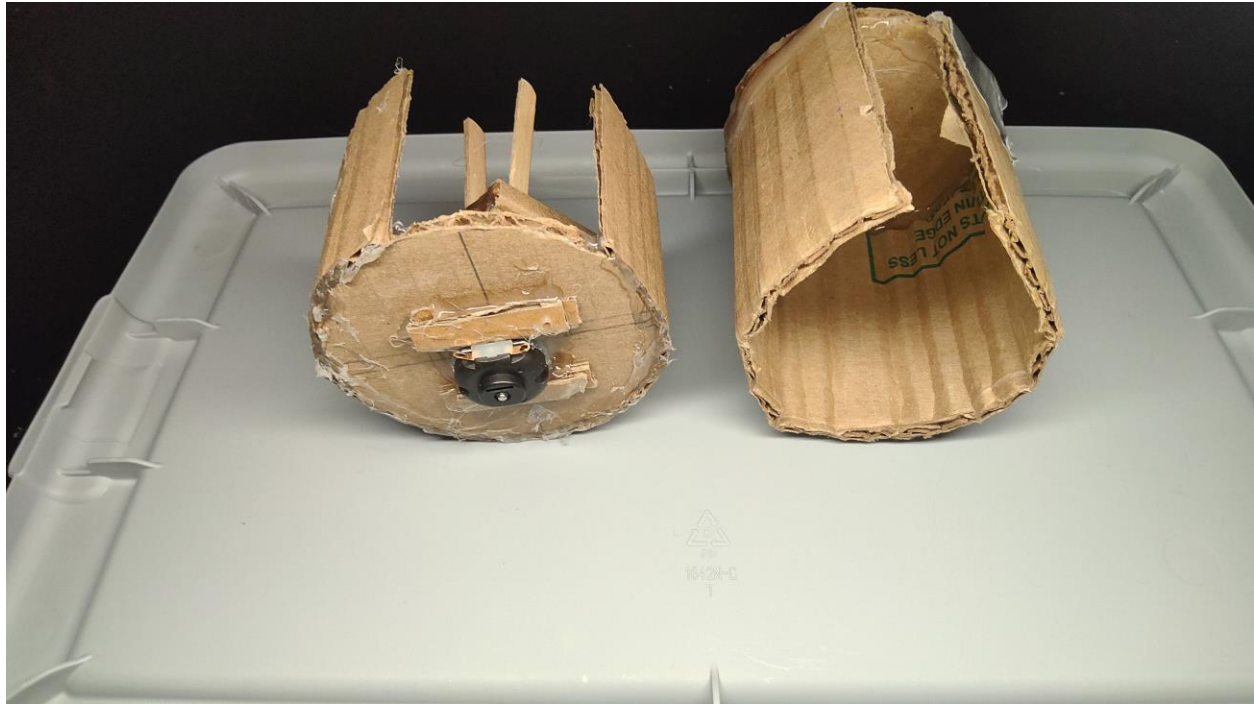


Figure 2- 4 Electric Roller Bottom View

2.3.1.4 Summary of How the Solution Addresses the Problem

This solution addresses the problem by reducing the time and effort required to re-roll martial arts hand wraps. The motorized design allows the user to quickly reroll wraps through simple button activation, providing a faster and more convenient alternative to hand rolling or fully manual devices.

Stakeholders consistently described frustration with the rerolling process and the inconvenience caused by unrolled wraps after washing. One stakeholder stated that “...does anyone also find that the whole rolling, re-rolling, and wrapping of hand wraps to be quite an ordeal? Used to just throw my washed wraps into my bag, but it makes wrapping hands during training to be insanely slow as compared to when it's rolled up...” [21]. This feedback shows a clear need for a solution that improves convenience and saves time.

Stakeholders have also developed their own improvised solutions to address this issue. One stakeholder explained, “I took a wire clothes hanger...and cut off a straight section 8 inches long, folded it in half so that both ends are touching, and then put it into a chuck of a wireless electric drill...What used to take me about 30 minutes now takes less than 5” [21]. This demonstrates that users value powered solutions and are willing to create their own tools to make rerolling easier. The proposed design builds on this demand by providing a compact, purpose-built electric device intended specifically for rerolling hand wraps.

2.3.2 Daniel Benson

2.3.2.1 Individual Solution

The solution takes the hand wrap roller from something that is handheld, to something that acts as a wall mounted hand wrapping station. My solution fixes two problems at once which include where to put the wraps after training to dry and finding a solution so they can dry without gathering bacteria. To create a storage solution, I used a box and added hooks onto the back so the box could be mounted onto a door to give the box some height and added hooks to the front of the box, see Figure 2-5, Figure 2-7, and Figure 2-8. The rolling part of the solution involved a 12V dc motor placed inside the box with a screw-in pin attached that rolls up the hand wraps as they are fed in through the side, and this system is activated with a switch, and powered by a 9V battery, see Figure 2-9. Once the wrap is rolled back up again, just unscrew the pin and allow the wrap to drop into the section below, see Figure 2-9, Figure 2-6.

In comparison to other products on the market, this includes two new features. The first is the ability to wrap the roll with just the flick of a switch. The second is being able to hang the wraps on the hooks to dry, and to hang the roller on a door. With all other solutions for wall mounts lacking hooks or power, this is a first of its kind. It fixes both the issues of storage of hand wraps and makes them easier to roll.

2.3.2.2 Solution Field Sketches

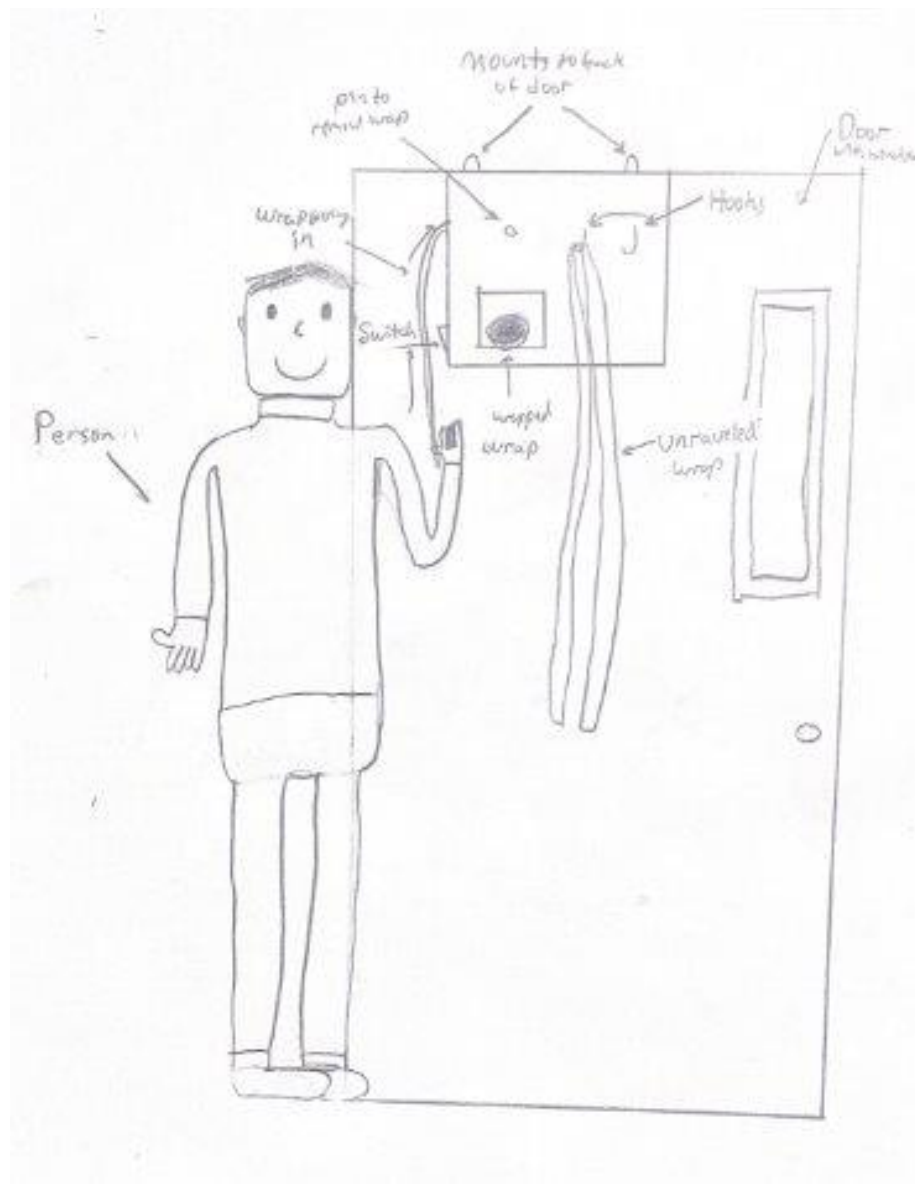


Figure 2-5



Figure 2-8



Figure 2-9

2.3.2.4 Summary of How the Solution Addresses the Problem

The solution addresses two of the main problems presented from the stakeholder's perspective. The first involving allowing them to dry as an interview with Jorge helped to put in perspective their need to be dried overnight and stored in an area where they can breathe [14]. The second issue involved finding a good place for the wraps to dry overnight since depending on the temperature that these wraps are stored since wraps can breed different kinds of bacteria and need to be cleaned with either soap or high temperatures [15]. The created solution solves both problems in two ways. The first is by including hooks on the design, so the wraps have a space where they can dry up in the air to allow them to breathe and properly dry out. The second was by adding the door mount so it can hang in the door of a laundry room where they can be washed, dried, and immediately rolled right out of the dryer.

The stakeholder had some input for the solution. The switch made the product easier to use, the hooks are fairly useful for drying, and the mounts need to be a bit lower [33]. In another iteration of the solution, the box needs to be lowered, so it is easier to reach for shorter people. The switch and the hooks were well liked, so those would need to be incorporated into another iteration. Overall, the stakeholder said the hooks and the door mount were good for making drying and storing the wraps easier but will need to be refined in another iteration.

2.3.3 Max Golterman

2.3.3.1 Individual Solution

This solution is a handheld electric drill roller. The prototype design was broken down into two distinct solutions. One, a modified battery powered drill, in this case using a Ryobi brand drill see Figure 2-12 Modified electric drill roller side view. Two, a purpose-built electric drill roller see Figure 2-11 Purpose built electric drill roller. Both solutions are unique compared to existing solutions because they include a system designed to straighten and unfold the wrap as it is fed into the roller. This system consists of two downward facing, curved metal prongs which unfold the wrinkled wrap and one metal loop that keeps the wrap straight during the roll (Figure 2-13 The straightening system).

The modified Ryobi power drill was designed as a cheaper wrap rolling alternative for individuals who already own a power drill. The forked bit would be inserted into the drill head like any other drill bit and holds the wrap. The wrap straightening system attaches to the drill handle see (Figure 2-10 Modified power drill roller).

The purpose-built power drill solution would be designed to perform the same function as the above solution. This would be an off-the-shelf version which does not require owning a drill. The looks-like prototype used the Ryobi power drill as a stand in for the power drill design and includes the same straightening system see Figure 2-13 The straightening system. The actual drill solution would be modeled after a power drill but in a compact size. The straightening system on this device would fold in between uses for portability and storage.

2.3.3.2 Solution Field Sketches

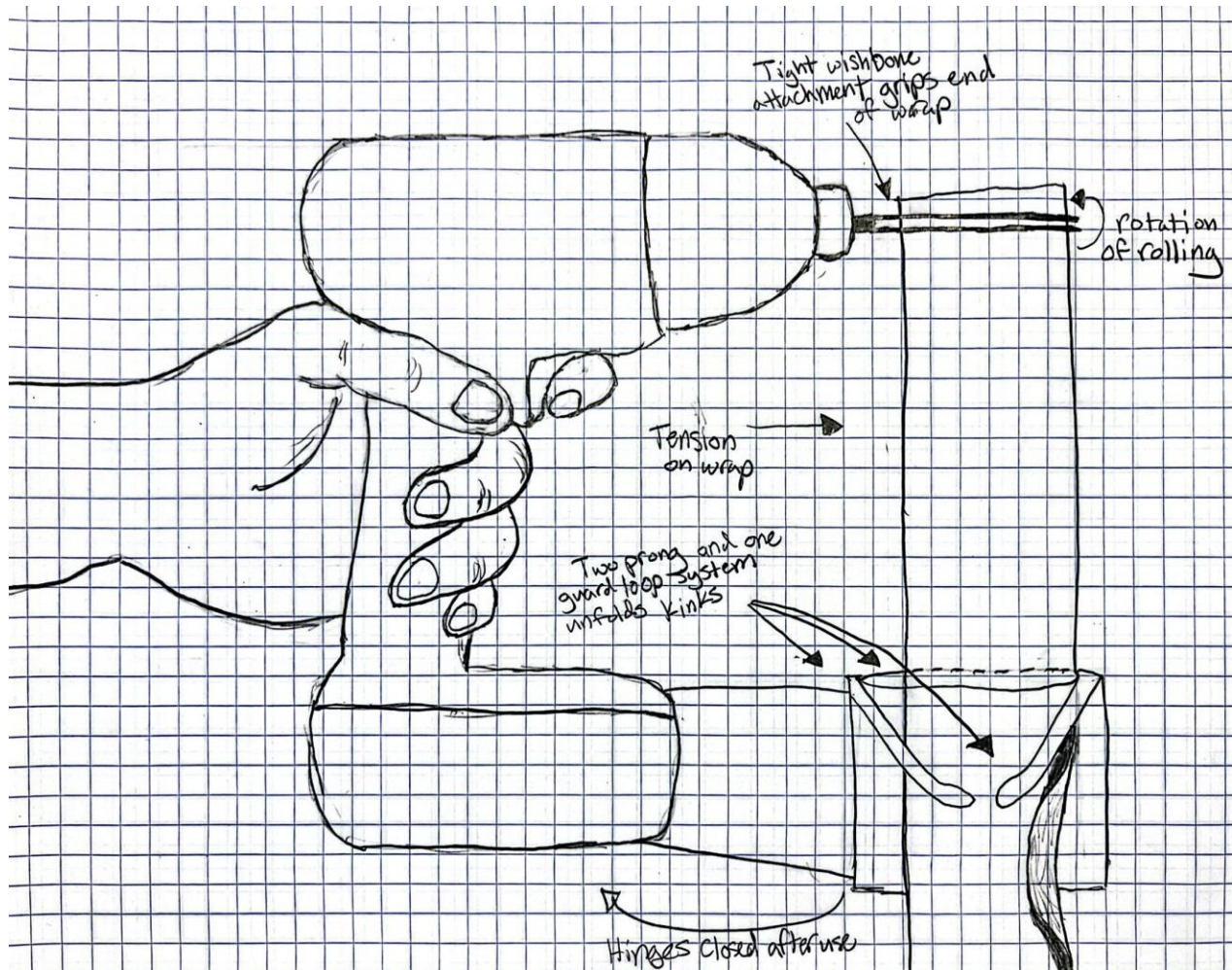


Figure 2-10 Modified power drill roller

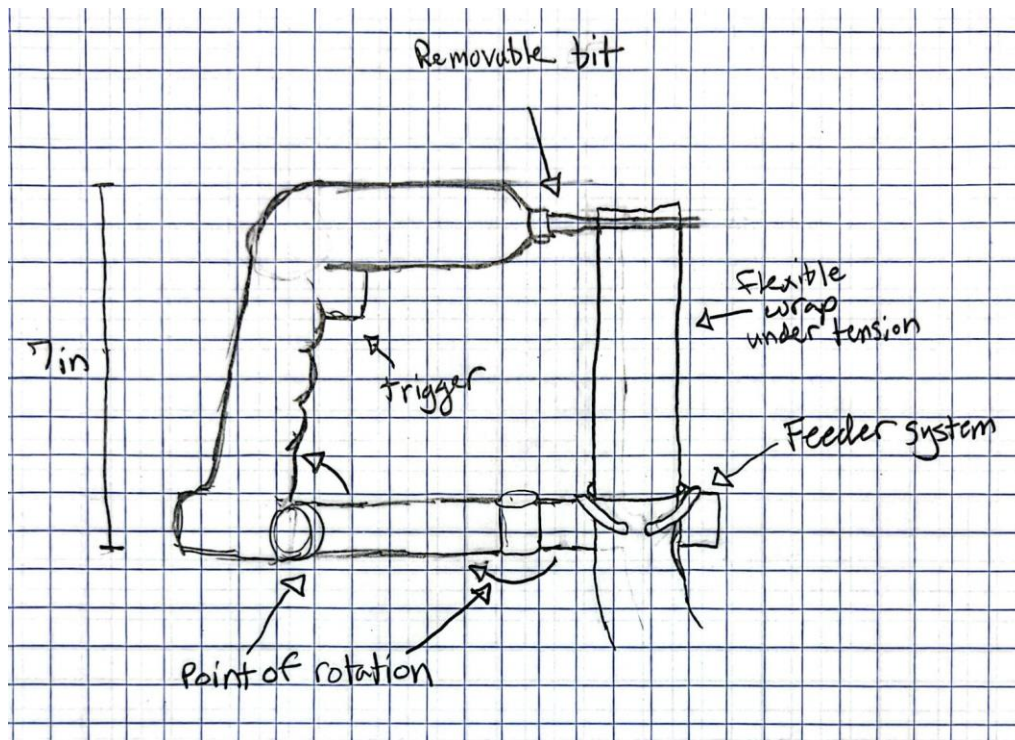


Figure 2-11 Purpose built electric drill roller

2.3.3.3 Photos of solution

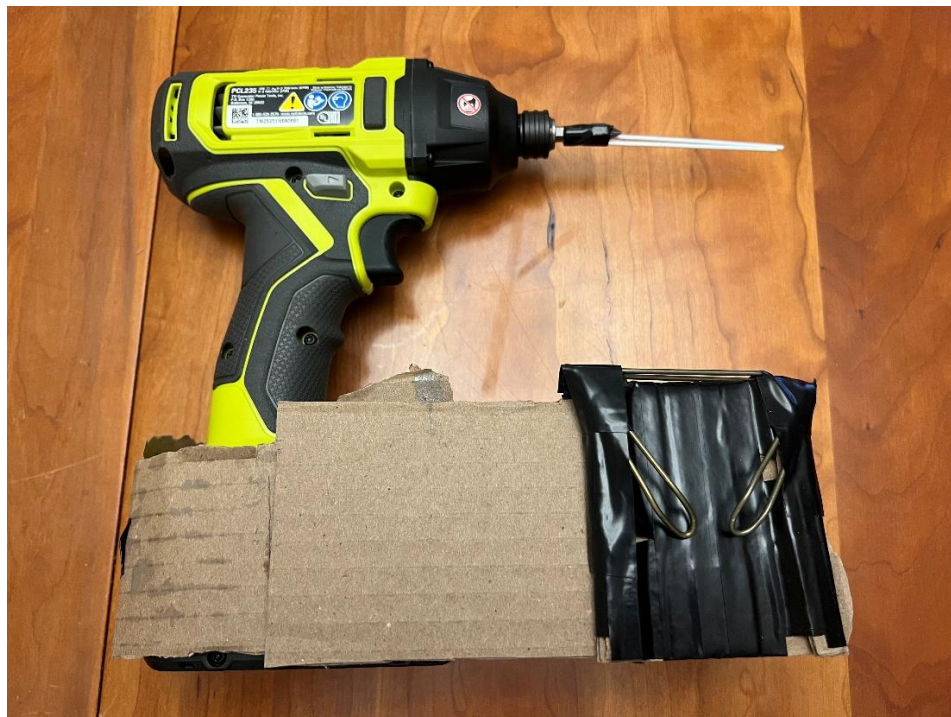


Figure 2-12 Modified electric drill roller side view



Figure 2-13 The straightening system up close

2.3.3.4 Summary of How the Solution Addresses the Problem

The solution addresses the problem by providing a simple motorized device which rolls hand wraps in an easy and efficient manner. The straightening system removes the need for a second hand providing tension to the rolling mechanism.

The benefit of the purpose built drill solution is its compact portable design that can be marketed as an all in one answer to the problem. The modified power drill solution provides drill owners with a low cost option which performs the same function as the purpose built prototype. Another key benefit of this design is giving the household drill which many people own another use without having to purchase another motorized device.

2.3.4 Revell Jenkins

2.3.4.1 Individual Solution

This concept is a portable device designed to automatically roll reusable hand wraps with minimal effort. The system uses a DC motor connected to an inline gear reduction assembly that rotates a forked spindle to wind the wrap into a compact roll.

The device is powered by a rechargeable lithium polymer (Li-Po) battery and enclosed in a protective housing that improves durability and portability. A push button allows simple one handed operation, while a removable top provides access for wrap insertion and removal.

Unlike existing manual crank rollers, this device is fully self contained and does not require any external tools. The geared motor provides controlled rotation to improve roll consistency and reduce folding or misalignment of the wrap during the rolling process.

2.3.4.2 Solution Field Sketches

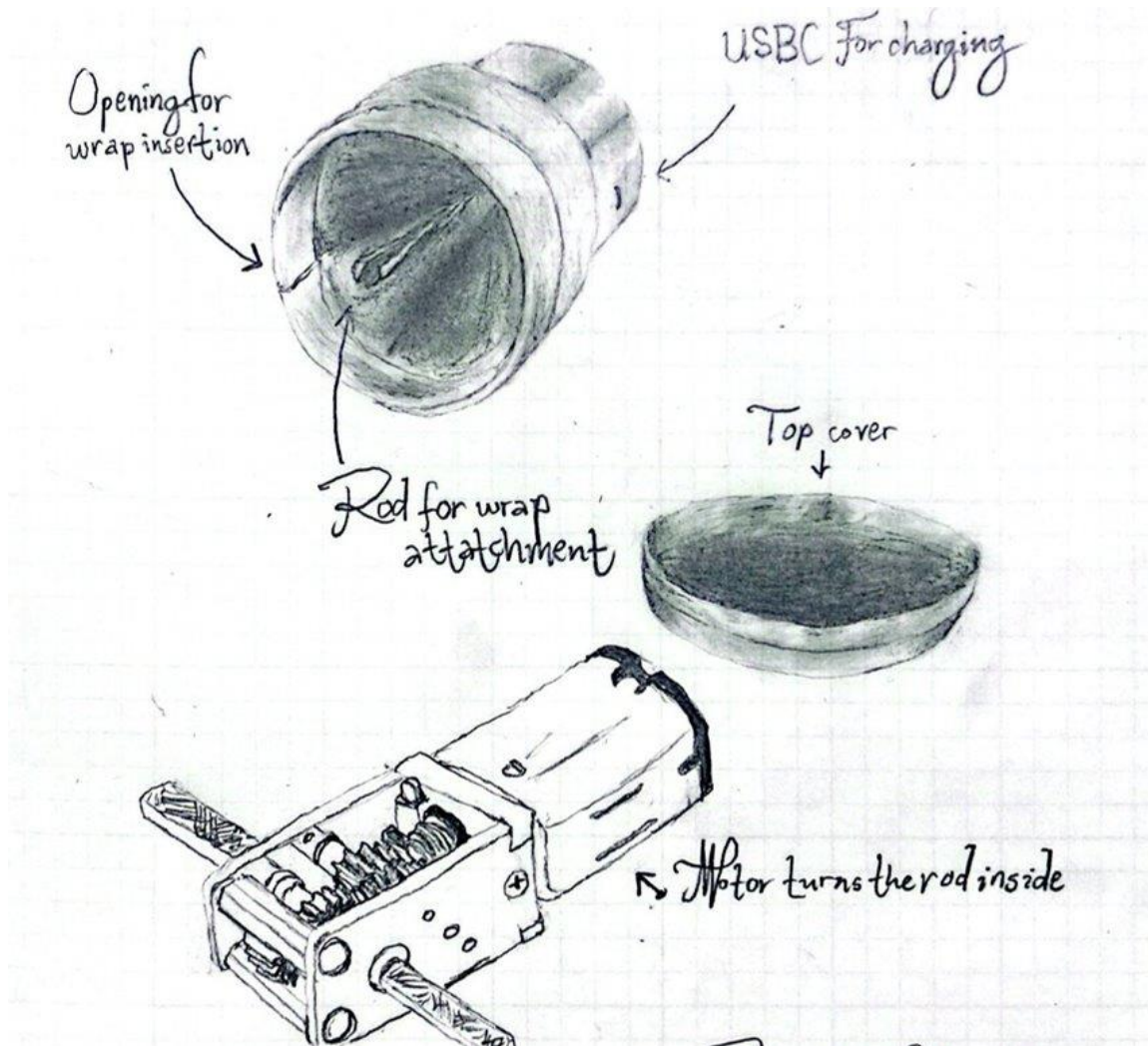


Figure 2-14

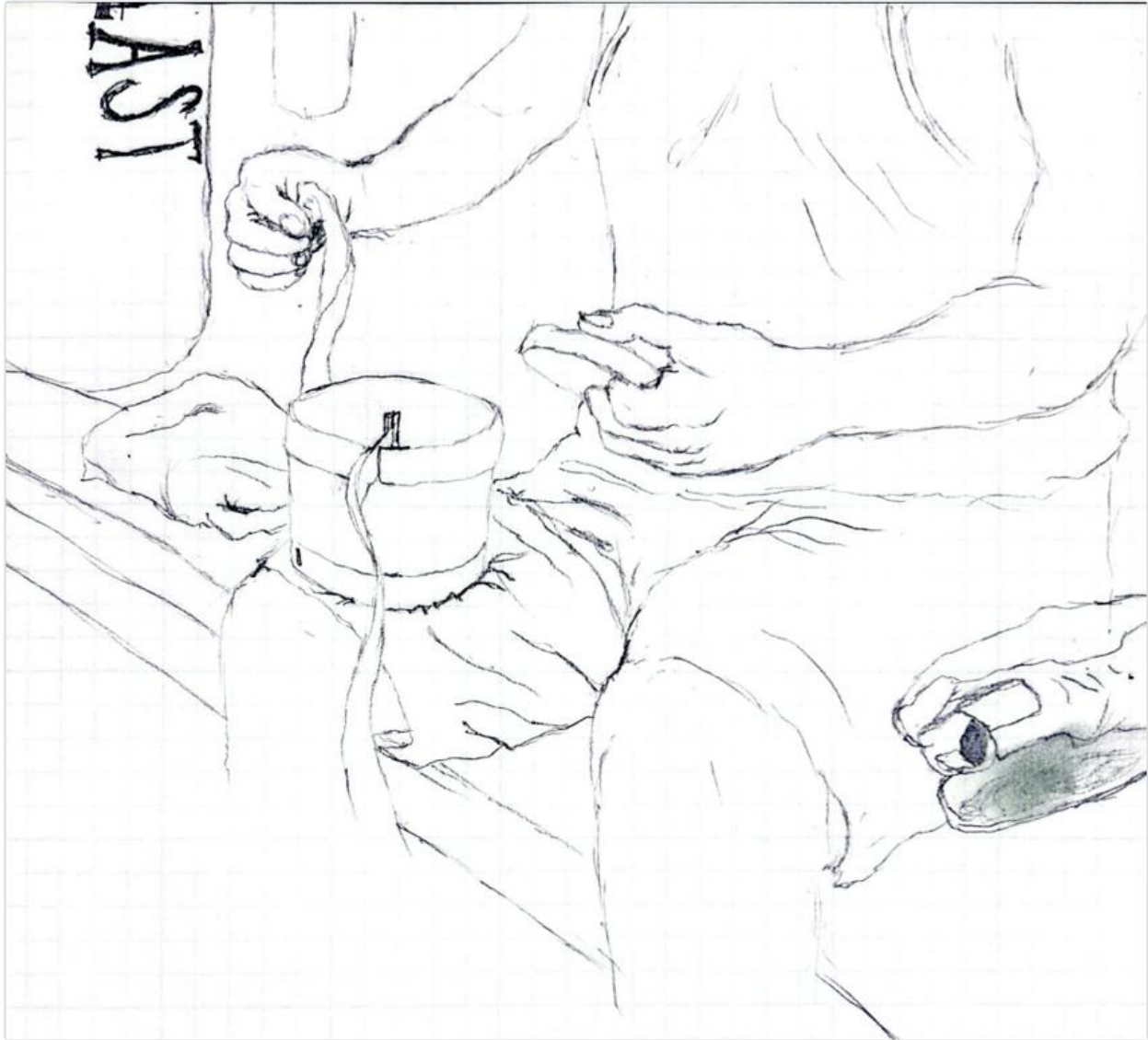


Figure 2-15

The field sketches illustrate the device being used in a realistic gym setting to demonstrate scale, ergonomics and user interaction. The user is shown seated on a bench while feeding the hand wrap into the device spindle and activating the motorized rolling mechanism. Additional detail sketches highlight key features, including the wrap insertion opening, removable top cover, internal motor assembly, and USB-C charging port. These sketches emphasize how the device would be operated in practice rather than focusing solely on mechanical structure.

2.3.4.3 Photos of Solution

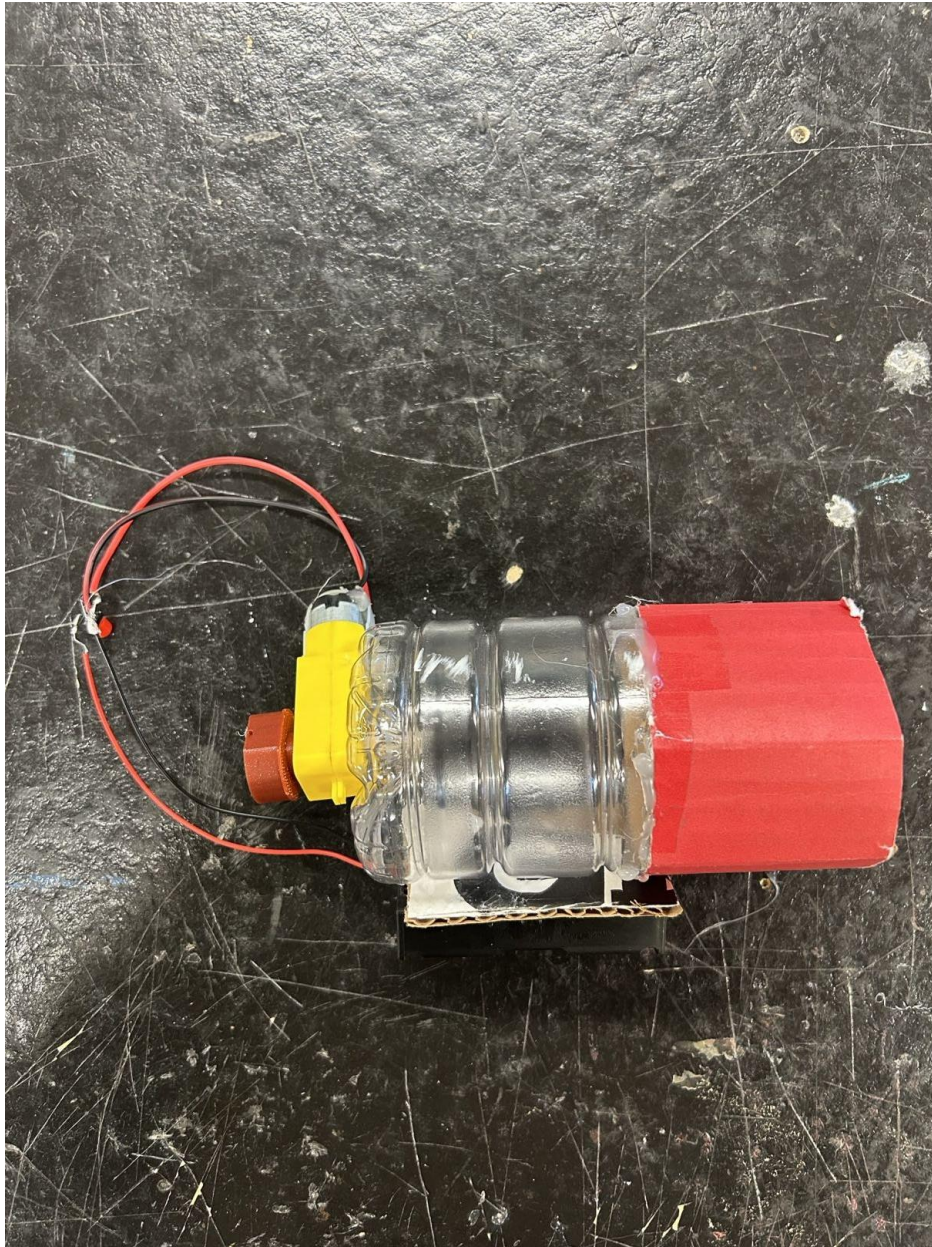


Figure 2-16 shows the front orthographic view of the prototype. In this view, the DC motor and spindle assembly are visible. The motor rotates the spindle to wind the hand wrap into a compact roll, while the wiring connects the motor to the power source and activation button.

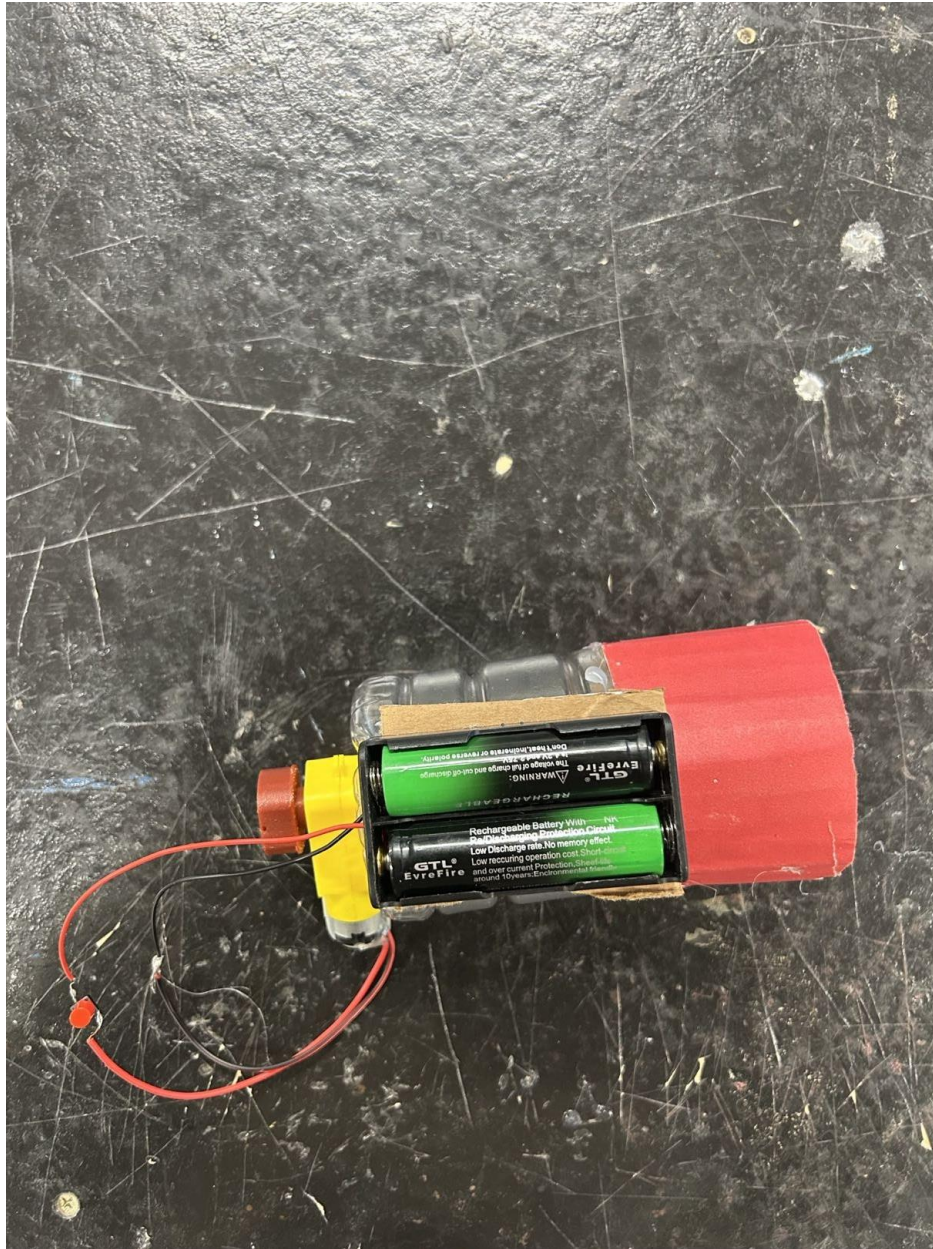


Figure 2-17 shows the top orthographic view of the prototype. This view highlights the placement of the battery pack, which supplies power to the motor. The battery holder is mounted above the housing to keep the electrical components accessible and balanced.

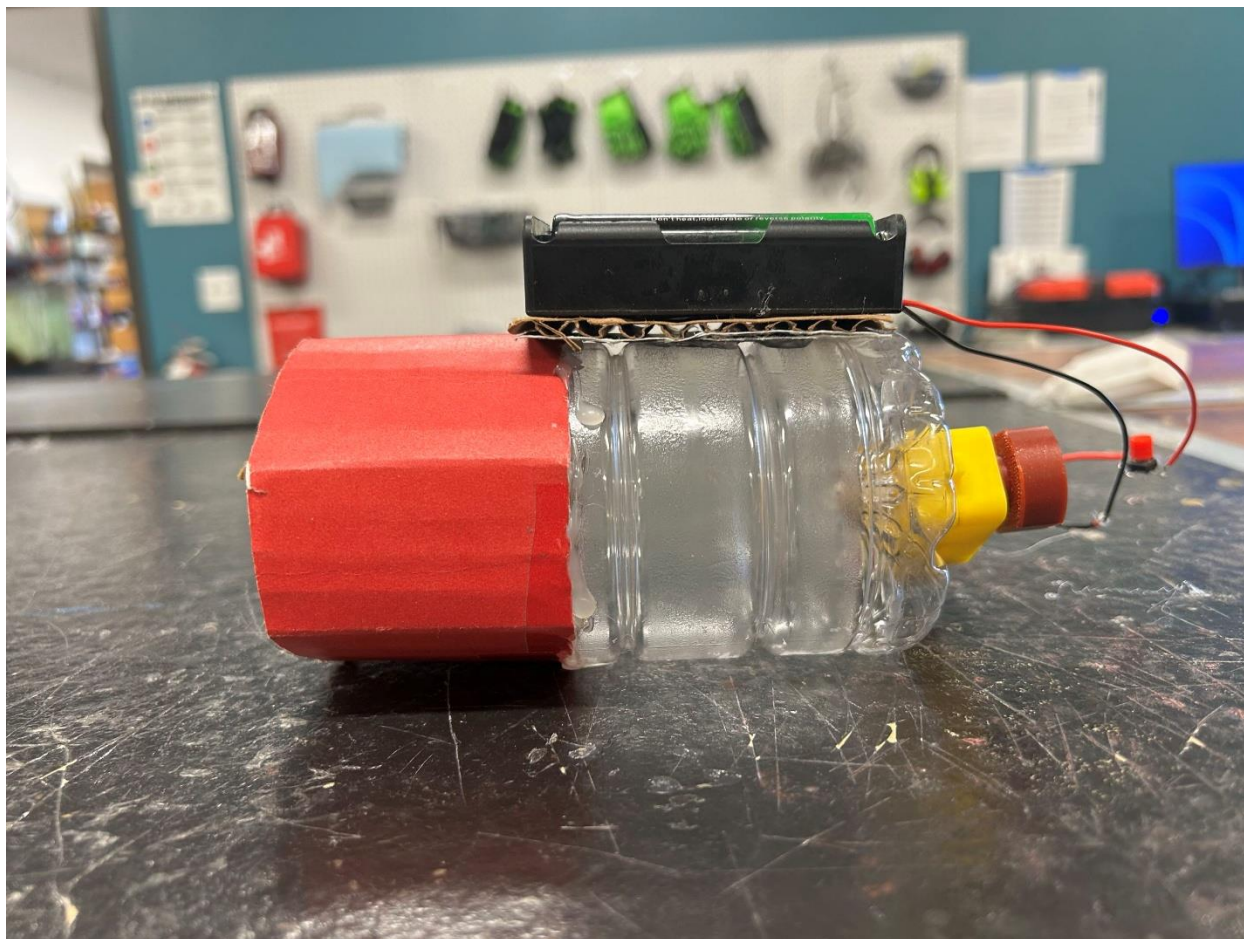


Figure 2-18 shows the side orthographic view of the prototype. From this angle, the cylindrical housing used to guide the wrap during the final rolling can be observed. The activation button and wiring are also visible, demonstrating how the user controls the device during operation.

2.3.4.4 Summary of How the Solution Addresses the Problem

This solution reduces the time and effort required to roll hand wraps by automating the process through motorized rotation. Manual rolling is slow, inconsistent, and physically repetitive. The push-button system simplifies the operation while the geared motor ensures consistent roll tightness and alignment

The compact portable design allows convenient use in gyms, training facilities, or home environments, improving preparation efficiency and readiness for stakeholders.

2.3.5 Terrence Micciche-Hall

2.3.5.1 Individual Solution

This solution is the gun-shaped electronic wrap roller. It aims to address the general problem of rolling wraps in both the sports fighting and medical field, and does so by utilizing one motor, a detachable claw, one power supply, and one normally open button. These internal components are depicted as a part of the device (Figure 2-21). What makes this solution unique is its ability to detach its

claw for use on other wraps, such as bandages. An example of how the claws look is shown in Figure 2-22 and Figure 2-23. This device can be used in any environment, specifically after the wraps have been washed. Figure 2-19 depicts the stakeholder using the device within their home.

2.3.5.2 Solution Field Sketches

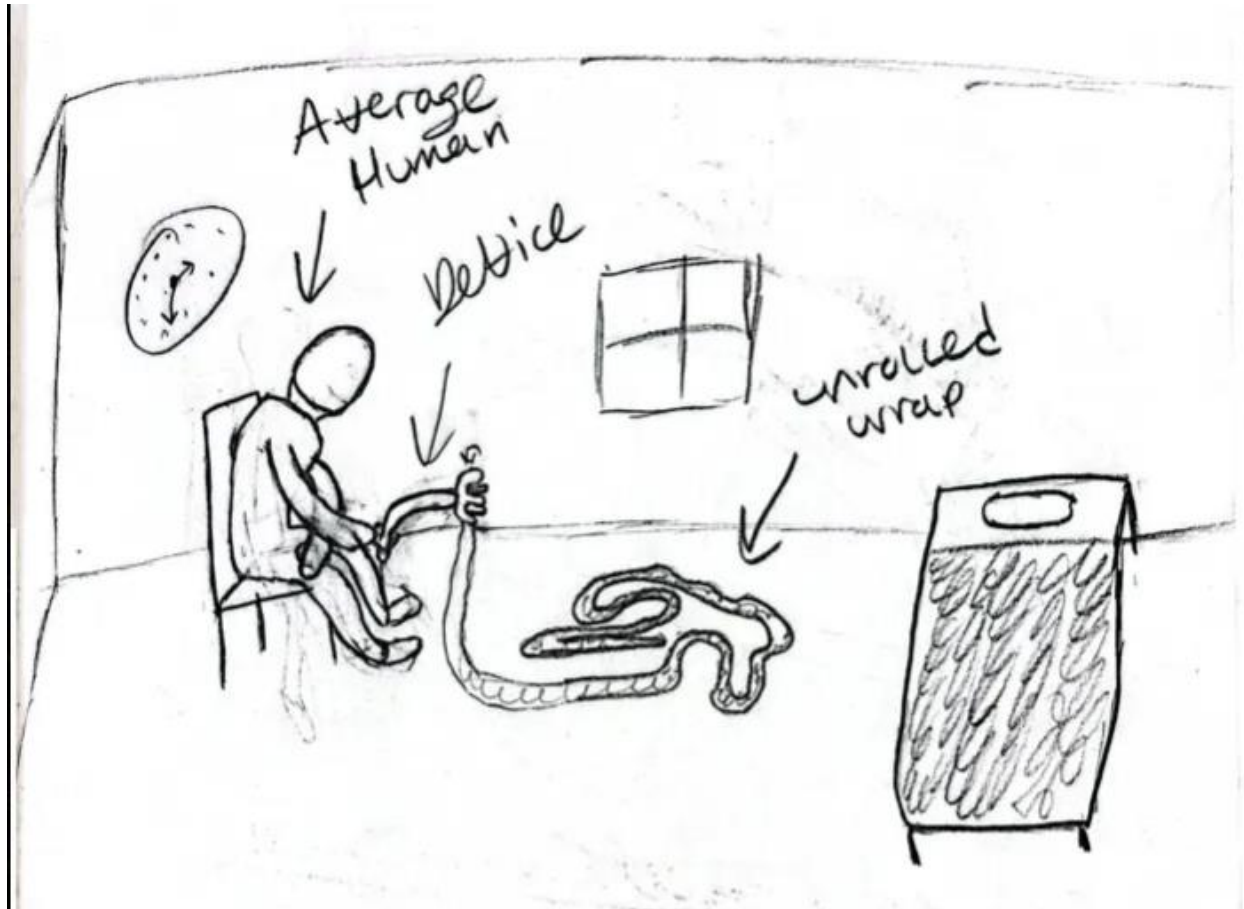


Figure 2-19: Stakeholder Utilizing the Device in Their Home

2.3.5.3 Photos of Solution

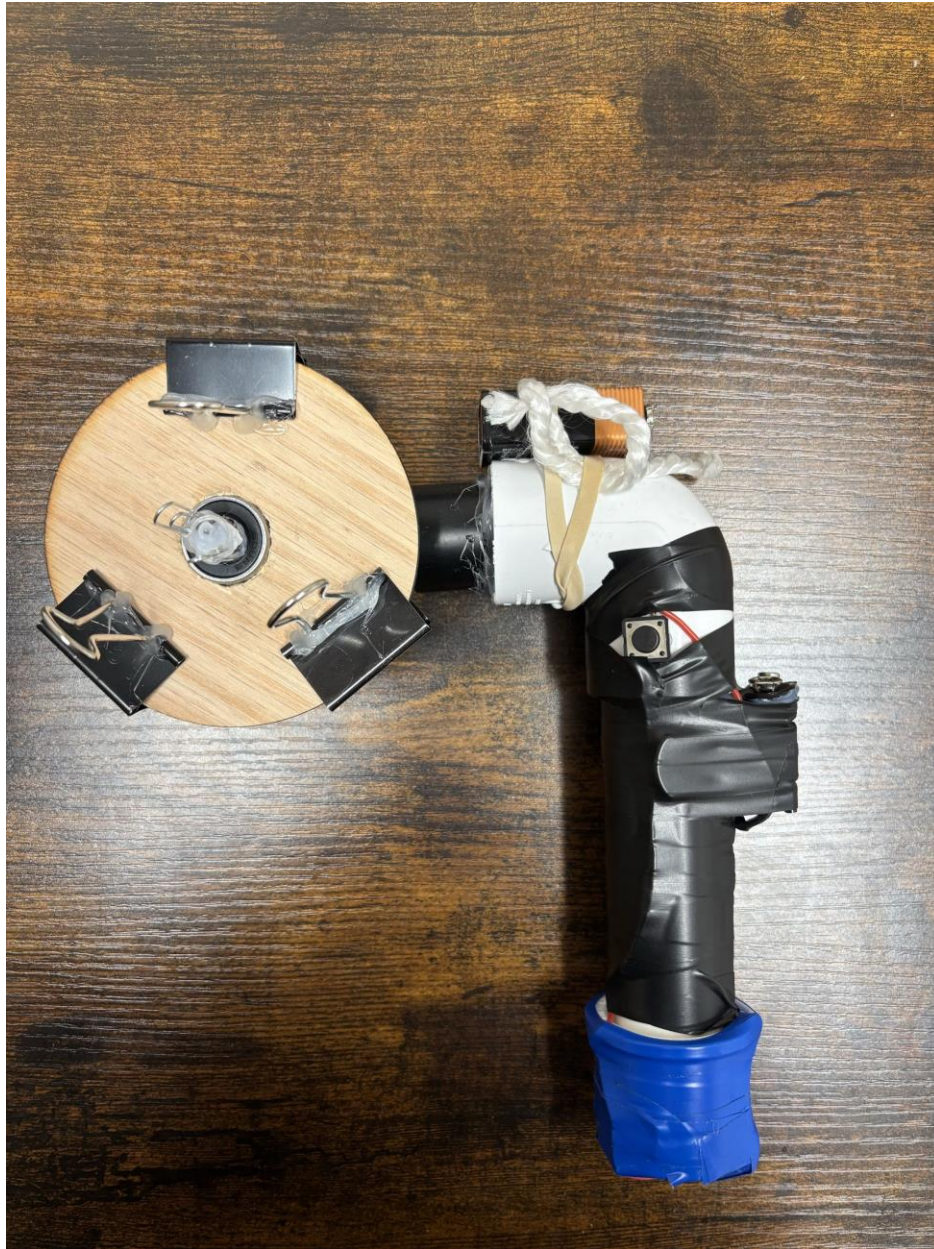


Figure 2-20: Side Profile of Gun Shaped Roller Prototype

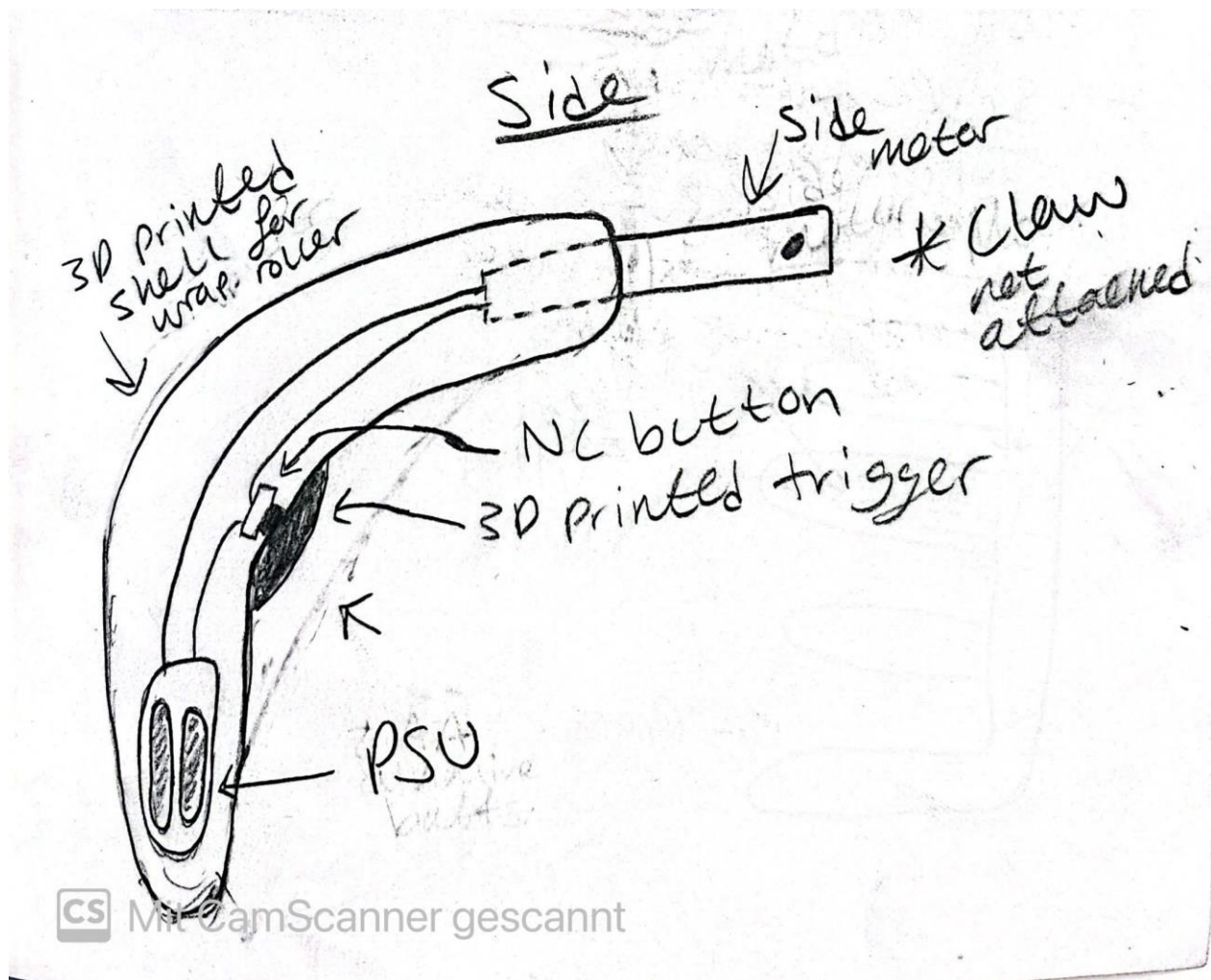


Figure 2-21: Sketch of Side Profile of Gun Shaped Roller Prototype

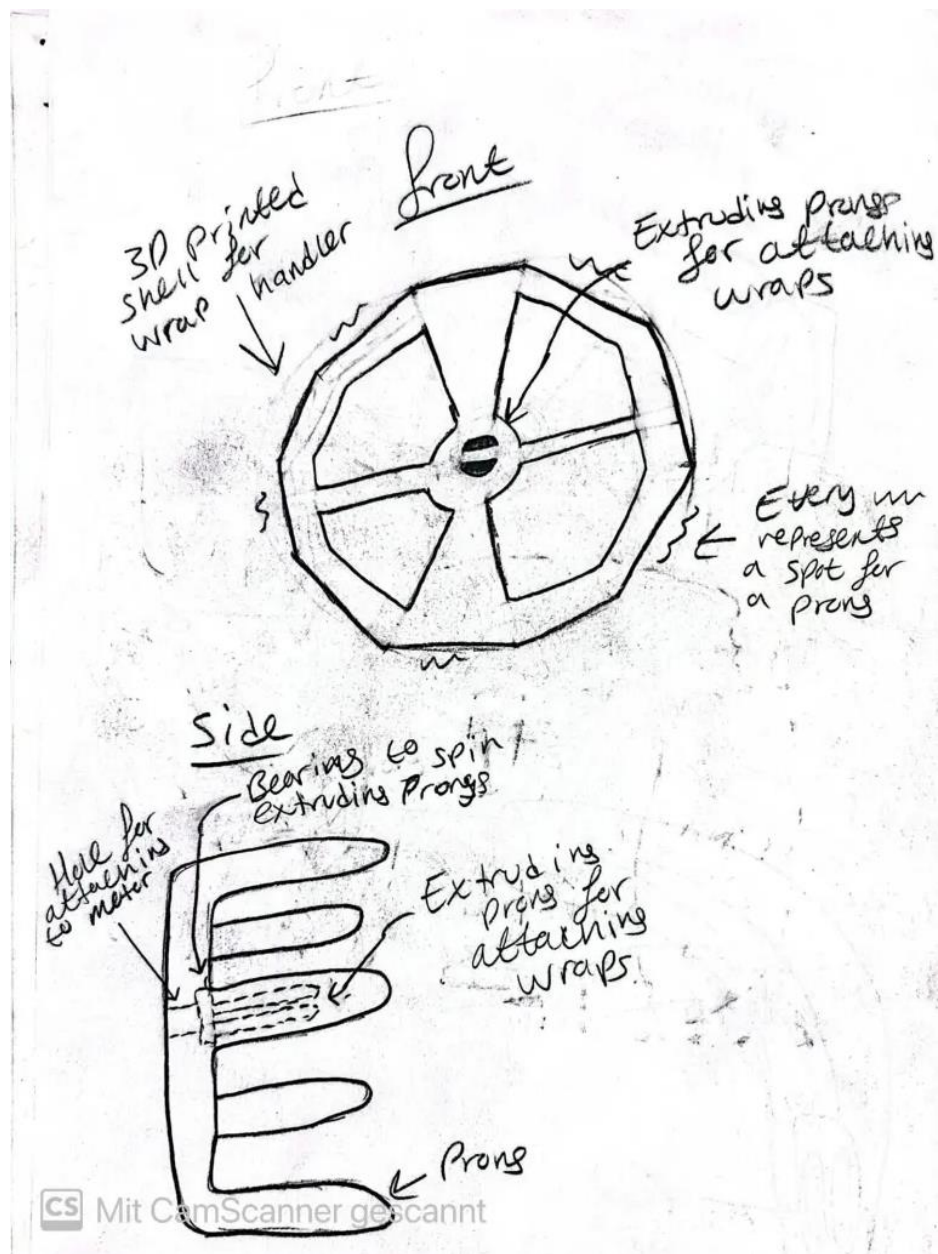


Figure 2-22: Sketch of Side and Front Profile of Detachable Claw



Figure 2-23: Front Profile of Gun Shaped Roller Prototype

2.3.5.4 Summary of How the Solution Addresses the Problem

The electronic wrap roller solves the problem of the slow rolling time it takes for an unexperienced roller to roll up their wraps. When recreating the problem, it has been recorded to take 2.5 minutes to roll up one single wrap [30]. Assuming averages and no lost time, ideally it should take 5 minutes to roll two wraps. In Figure 2-20, a side profile of the looks-alike prototype is displayed. At the top left is a motor which is rated for 40rpms at $2.2\text{kg}/\text{cm}^3$. It takes about 50 rolls to wrap one hand wrap of length 120 inches, as per the problem recreation, so using the motor to wrap up the hand wraps should take about 1.25 minutes. This cuts the time for the inexperienced roller to roll up their wraps in half.

Figure 2-21 reveals the sketch behind the design, which depicts the internal components of the device. There is plenty of empty room, and further consideration about how the device will be opened has not been had.

2.3.5.5 Solution Field Sketches

Figure 2-19 is a field sketch of the final device. This field sketch depicts the user utilizing the device in their room. Another purpose of this device was portability, as many of the other existing solutions were primarily stationary devices. It is worth noting that this device should not be used until after the wraps have been cleaned, as wraps act as a perfect breeding ground for bacteria growth after use [31].

Figure 2-22 and Figure 2-23 show the claw design that is used on the look-alike prototype. This design utilizes a tried-and-true claw design, used within the Meister Omega Portable device by Meister [32], which cuts slits in the walls of the bounding cylinders as well as prongs in the middle. The split wall allows for the placement of the wrap onto the prongs in the middle, allowing for a smooth user experience. The objective with this prototypes claw design is to allow for the claw to detach and be replaced by other claws when applicable.

2.4 Summary of Concepts Considered for Final

Before landing on a final solution to our proposed solution, 5 looks-like prototypes were created. Each of the prototypes provided vital information regarding the cost, user-appeal, long-term impact, and buildability. A decision matrix, observed in Table 2-2, was made to aid in the consideration of how much of each prototype will be integrated into the final solution. The overall score, paired with research regarding stakeholder and user needs, served as the final deciding factor.

Table 2-2 Decision Matrix

Decision Matrix					
Design Objectives					
Design Alternative	Viability (Cost)	Desirability (User Appeal)	Sustainability (Long-Term Impact)	Feasibility (Can You Build It?)	Overall Score
Option 1: Circular Handheld Roller (Revell)	4	3	3	5	15
Option 2: Circular Handheld Roller (Ryan)	4	3	3	5	15
Option 3: Door Mounted Roller (Daniel)	4	1	3	4	12
Option 4: Drill Shaped Roller with Prongs (Max)	5	3	4	5	17
Option 5: Gun Shaped Roller (TJ)	4	3	2	3	12

Scale: 1 = Worst, 5 = Best

In Table 2-2 Decision Matrix, there are 4 distinct options: Circular Handheld Roller, Door Mounted Roller, Drill Shaped Roller with Prongs, and Gun Shaped Roller. Each of these options solves the solution in unique ways, though they are prone to their own downsides.

2.4.1 Circular Handheld Roller

What made this prototype unique is its minimal design. While each of the prototypes are fairly cheap, this option only needs enough material to barely cover an entire wrapped hand-wrap as well as a motor, power supply, and a switch to power the motor. Furthermore, this design is very buildable using SolidWorks software. However, this design lacks a solution to a specific stakeholder's concern of ensuring a neat and tightly rolled wrap [8]. The minimal design will be implemented into final solution in order to ensure it is cheap and buildable.

2.4.2 Door Mounted Roller

This solution is unique because of its ability to mount the hand-wrap roller onto a door. The pros of this design is that it allows for the users to mount such roller in a doorway for easy, out of the way, access. The caveat with this, however, is that there exists a lack of stakeholder input which provides the grounding to follow through with such a technical ability. In addition to this, there are stakeholder suggestions for a desire of portability of the device. This is found as a reviewer of an existing solution had said, “This is an okay portable hand roller. I like the portable aspect to it,” [28]. Overall, this prototype is cheap and buildable, but not practical. From this solution, the addition of a slide switch to toggle the PSU (Power Supply) will be implemented into the final solution.

2.4.3 Drill Shaped Roller with Prongs

This solution scored the highest within the decision matrix in Table 2-2 Decision Matrix. This is because of the high practicality and buildability of its unique approach to the solution: Prongs. Because stakeholder input exists which justifies the need for a tight and compact roll when rolling hand-wraps [8], this design is very attractive. Furthermore, the prongs hold the wrap down when rolling in order to reduce wrinkles and overlapping.

Regarding the cost of this solution, there is question about the availability of hand-drills within the targeted stakeholders. Assuming the audience has a hand-drill, this solution may be very accessible and cheap. However, assuming otherwise drives the cost of this solution into the hundreds; immediately nullifying any practicality of the solution. Given the focus on only the prongs, this solution is very buildable as it only requires a metal rod and the ability to bend such rod. Because of this, our final solution will include prongs.

2.4.4 Gun Shaped Roller

The gun-shaped roller boasts the ability to detach the wrapping claw and replace it with another. Furthermore, the desired intention of such a design is to allow the user to determine whether they want to roll bandage wraps or Muay Thai wraps. The issue with this solution is that it strays too far from the problem at hand, and it attempts to address issues which are not being considered by the targeted stakeholders. Because of this, paired with its great similarity to existing products on the market [29], this solution will only contribute its research into how wraps are rolled for the final solution. Figure 2-24 depicts the existing solution that this prototype is modeled after.



Figure 2-24: Title Brand Existing Solution

2.4.5 Team Brainstorming

The team had initially come up with over fifty ideas. Each idea was written on a sticky note and presented. Not one idea was shamed, but rather each idea was uplifted. The next step for the team was to limit the number of ideas to five. This rounded to one idea per person. Each of these ideas are described in sections 2.4.1 to 2.4.4. From there, a decision matrix was made to land on one main solution to draw inspiration from. In addition to this, stakeholders were asked about their concerns and desires for a final solution. One stakeholder revealed their concern about the pricing of the device [27]. When using the decision matrix to vote on the inspiration for the final device, cost was greatly considered. Regarding the final device, the vote on the decision matrix did not determine what device

would be made. Rather, the overall score of each solution determined how much of it will find its way into the final device. For example, the circular handheld roller was incorporated into the final device for its high feasibility.

2.5 Final Proposed Design

Visual Aids and a table of design requirements

The team's proposed solution is a compact handheld electronic device designed to automatically roll reusable hand wraps quickly and consistently. The device contains an internal motor powered by a compact power supply and controlled using a slide switch for simple operation.

The housing follows a circular handheld form that minimizes material use while allowing sufficient internal space for mechanical and electrical components. The device is designed to be placed on a bench or flat surface during use.

A pronged spindle mechanism is incorporated to guide the wrap during rotation. These prongs maintain alignment, reducing folding and ensuring a tight and uniform roll. The enclosed housing protects internal components and improves durability while keeping moving parts safe during operation.

The design emphasizes portability, ease of use, and consistent rolling performance for athletes and trainers. The motorized mechanism reduces manual effort and preparation time, while the pronged spindle improves roll consistency. The compact housing enhances portability and allows convenient storage. Enclosed mechanical components improve safety and durability during repeated operation.

2.6 Summary of Solution Generation and Project Selection

Solution generation for this problem fell into four main categories; research, team brainstorming, look-alike prototypes and a decision matrix. We began with research and stakeholder interviews to determine the scope of the problem as well as the key issues. The scope of the problem included millions of boxers and Muay Thai fighters in the US [25][26]. Our research showed that the primary stakeholder concerns were time, cost, and the neatness of the roll [24].

The brainstorming process started with a broad scope of ideas. These ideas were narrowed down for reasons including feasibility, cost and fit with stakeholder needs. Eventually the field was narrowed to five ideas. To showcase what was unique about each of the five, look-alike prototypes were created.

The prototypes all fit the primary stakeholder need of rolling hand wraps more quickly than being done by hand, but got there in different ways. Amongst the five, there were two circular rollers, two drill style rollers and a door mounted one. The prototypes all had pros and cons so we utilized a decision matrix to come to a final proposed design.

The decision matrix allowed us to discuss the aspects of each design and how they fit with stakeholder needs in Table 2-2. While the process was important to the final design a decision was made to ignore the overall scores in the decision matrix. In the end we decided to use it as a reference and take the best aspects from each prototype and include them in the final design.

The final design proposed is a circular handheld electronic rolling device which is intended to address key stakeholder concerns. The device will be compact, portable, easy to use, and will reduce the time and energy needed to roll hand wraps. The design will be a success once a working prototype is able to consistently turn wrinkled hand wraps into neat rolls, ready for the next use.

3. Module 3

3.1 Subsystem Description – Electrical Logic, Ryan Cherry

The electrical logic subsystem is responsible for storing, controlling, and delivering electrical power to the hand-wrap roller so that the device operates reliably and with minimal user effort. In the full-scale design, this subsystem is integrated into the handheld housing and manages battery power, charging, user controls, and motor activation. Its primary purpose is to route electrical energy through the device in a controlled way so that the motor only runs when intended, and the system can be recharged for repeated use.

3.1.1 Scaled Up Solution

3.1.1.1 Overall Objective of the Subsystem

The overall objective of the electrical logic subsystem is to convert stored electrical energy into controlled motor operation for the wrap-rolling mechanism. This subsystem must provide the proper voltage and current to the selected motor while remaining compact enough for a portable handheld design. It must also ensure that the device can be powered on, activated, and recharged through a simple user interface.

The key components of this subsystem are a rechargeable 9.6 V battery pack, a boost converter, a USB-C charging port, a charging and protection module, a slide switch used as the main power switch, a normally open push button used to activate the motor, electrical wiring and terminals, and the selected 12 V DC worm gear motor.

3.1.1.2 How the Subsystem Works

The subsystem works by storing electrical energy in the battery pack and routing that energy through a simple control sequence before it reaches the motor, outlined in Figure 3- 1. First, the slide switch acts as the main power switch for the device. When the slide switch is turned on, the circuit is enabled. Second, the normally open push button acts as the activation control. The motor only runs while that button is pressed, which gives the user direct and momentary control over operation.

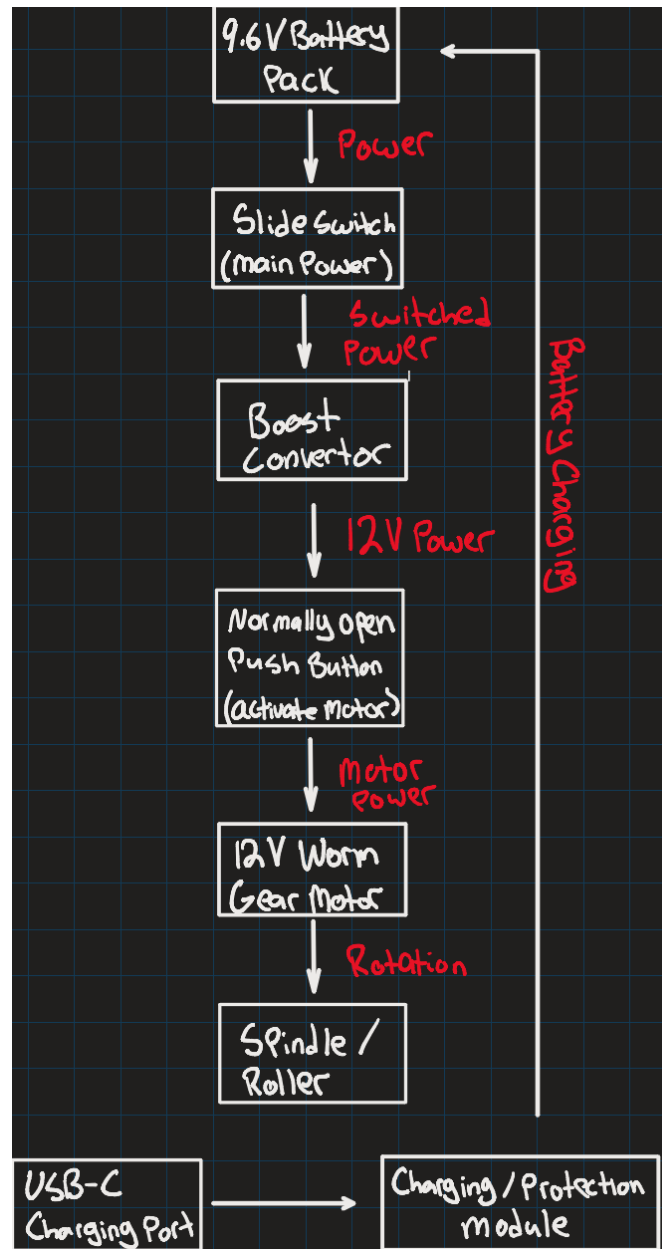


Figure 3-1: Simplified Electrical Diagram for Hardwired Control System

When the user turns on the slide switch and presses the normally open push button, current flows from the 9.6 V battery pack to the worm gear motor. The motor then converts electrical energy into rotational motion, which drives the spindle that rolls the wrap. The USB-C charging port is not intended to power the motor directly during operation. Instead, it supplies input to the charging and protection module, which recharges the battery pack.

This control arrangement keeps the electrical logic simple while still giving the user intentional control over the device. The slide switch prevents the device from remaining active during storage, while

the normally open push button prevents continuous motor operation unless the user is actively pressing it.

3.1.1.3 Subsystem Inputs

The subsystem requires stored electrical energy from a rechargeable 9.6 V battery pack, user input through the slide switch, user input through the normally open push button, charging input through the USB-C port, and physical and electrical connection to the motor. Because the selected motor is rated for 12 V operation, the subsystem also requires a boost converter to raise the battery output to the motor operating voltage.

3.1.1.4 Key Subsystem Components

Each major component serves a distinct electrical function. The 9.6 V battery pack stores portable electrical energy for device operation. The boost converter raises the battery output to the 12 V level required by the selected worm gear motor. The USB-C charging port allows the battery to be recharged without removing it from the housing. The charging and protection module regulates charging and helps protect the battery from improper charging conditions. The slide switch functions as the main power control, enabling or disabling the circuit. The normally open push button gives the user direct momentary motor activation. The wiring and terminals complete the electrical path between the power source, controls, boost converter, and motor. The motor serves as the final electrically driven component, using the delivered power to produce spindle rotation.

3.1.1.5 Purchased Components

Boost Converter

The electrical logic subsystem uses the DC-DC boost converter shown in Figure 3-2 to raise the battery output from 9.6 V to 12 V for the worm gear motor. This component is necessary because the selected motor is rated for 12 V operation, while the battery pack operates below that voltage. Testing showed that full 12 V provided the best combination of speed and torque for the hand-wrap roller, making it the preferred operating condition for the subsystem. The boost converter allows the design to retain a rechargeable battery while still delivering the motor voltage needed for effective performance.

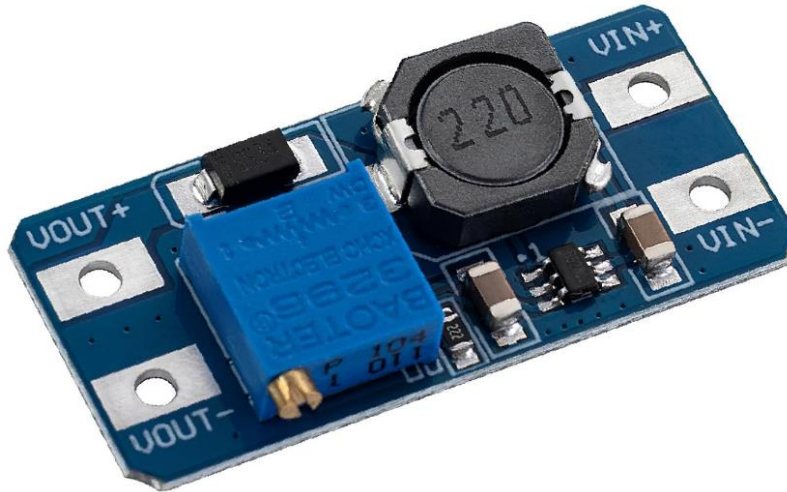


Figure 3-2: Boost Converter

Normally Open Push Button

The momentary normally open push button pictured in Figure 3-3 is used as the motor activation control. The button closes the circuit only while pressed, which prevents the motor from running continuously unless the user is actively operating the device. This improves intentional control and reduces accidental activation.

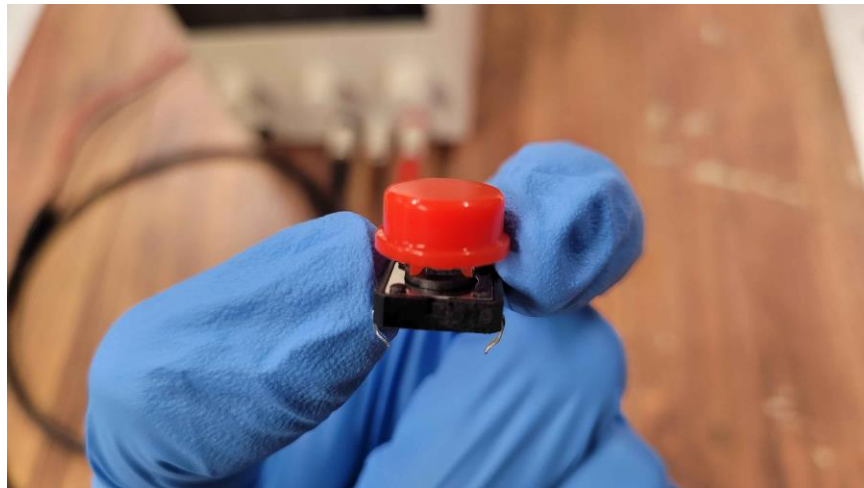


Figure 3-3: Normally Open Push Button

The slide switch in Figure 3-4 is used as the main power switch for the subsystem. This switch enables or disables the primary electrical circuit and allows the user to turn the device fully off during storage or transport. This helps reduce unintended battery drain and improves usability.

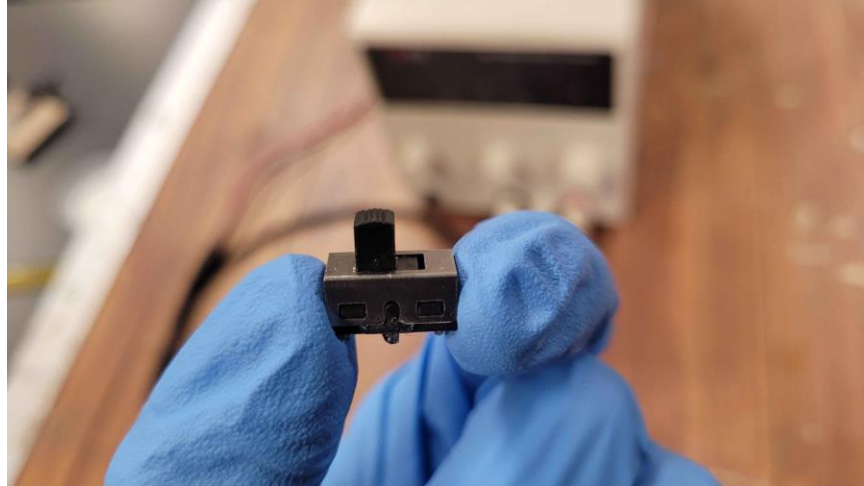


Figure 3-4: Slide Switch

3.1.2 Physical Properties of Components

The electrical logic subsystem must fit within a compact handheld housing and must be arranged so that the battery, charging components, controls, and motor do not interfere with the spindle or user grip. Since the device is intended to be portable, the subsystem must balance electrical performance with space and weight limitations. The USB-C charging port, slide switch, and push button must remain accessible from the exterior of the housing, while the battery, charging module, wiring, and motor connections must remain protected inside the device.

Table 3-1: Component Properties

Component	Function	Type / Material	Key Specification	Device Integration	Dimensions (Inches) LxWxH	Weight (kg)
9.6 V battery pack	Stores portable electrical energy	Rechargeable battery pack	Boosted 12 V target output	Mounted inside housing	4.09 x 0.63 x 2.32	0.204kg
USB-C charging port	External charging interface	Panel-mount USB-C port	Recharges battery through charger board	Mounted on outer housing wall	0.66 x 0.3 x 1.10	~0.003kg

Slide switch	Main power control	SPST slide switch	On/off device enable	Mounted on outer housing	0.5 x 0.26 x 0.73	0.0015kg
Normally open push button	Motor activation	Momentary NO push button	Motor runs only while pressed	Mounted near user grip	0.47 x 0.47 x 0.28	0.0014kg
Worm gear motor	Drives spindle rotation	12 V geared DC motor	12 V, 160 RPM no-load, 6.86 Nm rated torque	Mounted inside housing	4.73 x 1.57 x 1.93	0.38kg
Wiring	Transfers current	22 AWG Insulated copper wire	Sized above 1.6 A motor current requirement	Routed internally to prevent snagging	N/A	N/A
Boost Converter	Raises battery voltage for motor use	DC-DC boost converter	Steps 9.6 V to 12 V	Mounted internally near battery and motor	1.44 x 0.68 x 0.56	0.0038kg

Evaluation of Alternative Ideas

Several similar wire gauges were considered for the electrical logic subsystem, including 22 AWG (American Wire Gauge), 20 AWG, and 18 AWG insulated copper wire. The decision matrix shown below Table 3-2, was used to compare the alternatives based on current capacity, flexibility, ease of routing, durability, space efficiency, and cost. Current-capacity values were based on standard AWG reference data [40]. Although 18 AWG provided the greatest current capacity, it was less desirable because of its larger size and reduced routing flexibility in the compact handheld housing. 22 AWG received the highest weighted score because it provided the best balance of flexibility, space efficiency, and sufficient current capacity for the subsystem. As a result, 22 AWG insulated copper wire was selected for the electrical logic subsystem.

Table 3-2: Wire Gauge Decision Matrix

Wire Gauge	Current Capacity	Ease of Routing	Durability	Cost	Flexibility	Total
22 AWG	3	5	3	5	5	21

20 AWG	4	4	4	4	4	20
18 AWG	5	3	5	3	3	17

Two control strategies were considered for the electrical logic subsystem: a simple hardwired control system and a microcontroller-based control system Figure 3-5. The hardwired design uses a slide switch and a normally open push button to control motor operation, while the microcontroller-based option would allow added features such as adjustable motor speed. These alternatives were compared using Table 3-3 to determine which control method best fit the current design needs.

Table 3-3: Device Control Decision Matrix

Control Option	Simplicity	Ease of Build	User Control	Cost	Expandability	Total
Slide Switch + Button	5	5	3	5	2	20
Microcontroller	2	2	5	2	5	16

The hardwired control system was selected for the current design because it is simpler, easier to build, and less expensive while still meeting the functional needs of the subsystem. The microcontroller-based option remains a possible future improvement if greater control over motor speed becomes necessary.

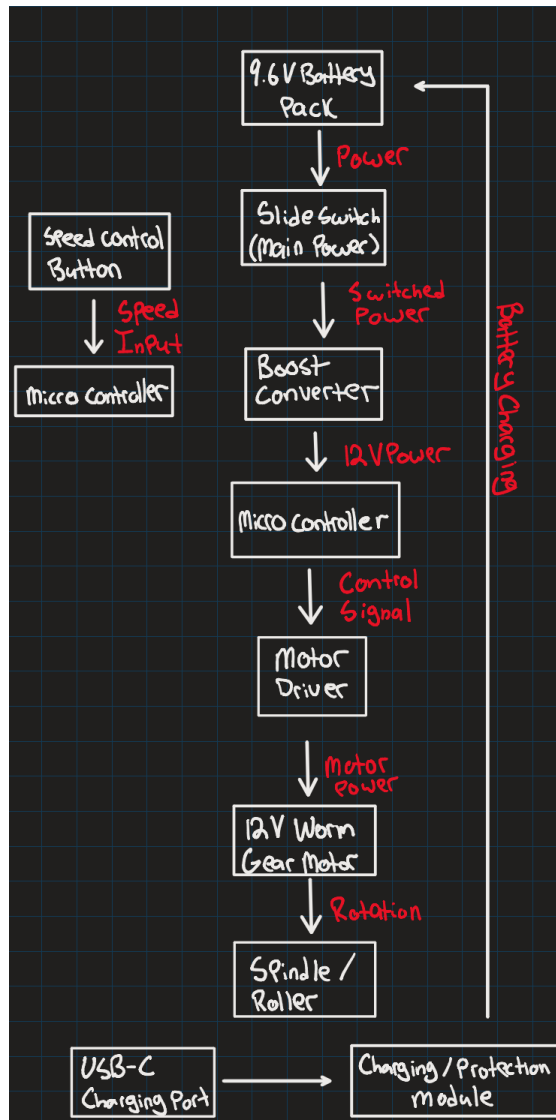


Figure 3-5: Simplified Electrical Diagram for Microcontroller System

Data Considered

3.1.2.1 Test Results

Testing for the electrical logic subsystem was conducted using the selected 12 V DC worm gear motor and a variable DC power supply pictured in Figure 3-6. The purpose of this testing was to determine the most appropriate operating voltage for the subsystem after the design changed from the earlier 3.7 V N20 motor configuration to the current 12 V motor configuration. By using a variable power supply rather than immediately selecting a final battery, the subsystem could be evaluated across a range of voltages before finalizing the portable power system.

The test procedure consisted of connecting the 12 V motor to the variable DC power supply and operating the motor at multiple voltages between 3.7 V and 12 V. This range was chosen because the previous subsystem iteration used a 3.7 V LiPo battery, while the updated motor is rated for 12 V operation. The testing was intended to determine whether a lower-voltage approach would still provide acceptable motor behavior or whether the subsystem should be designed around the full rated voltage of the selected motor.

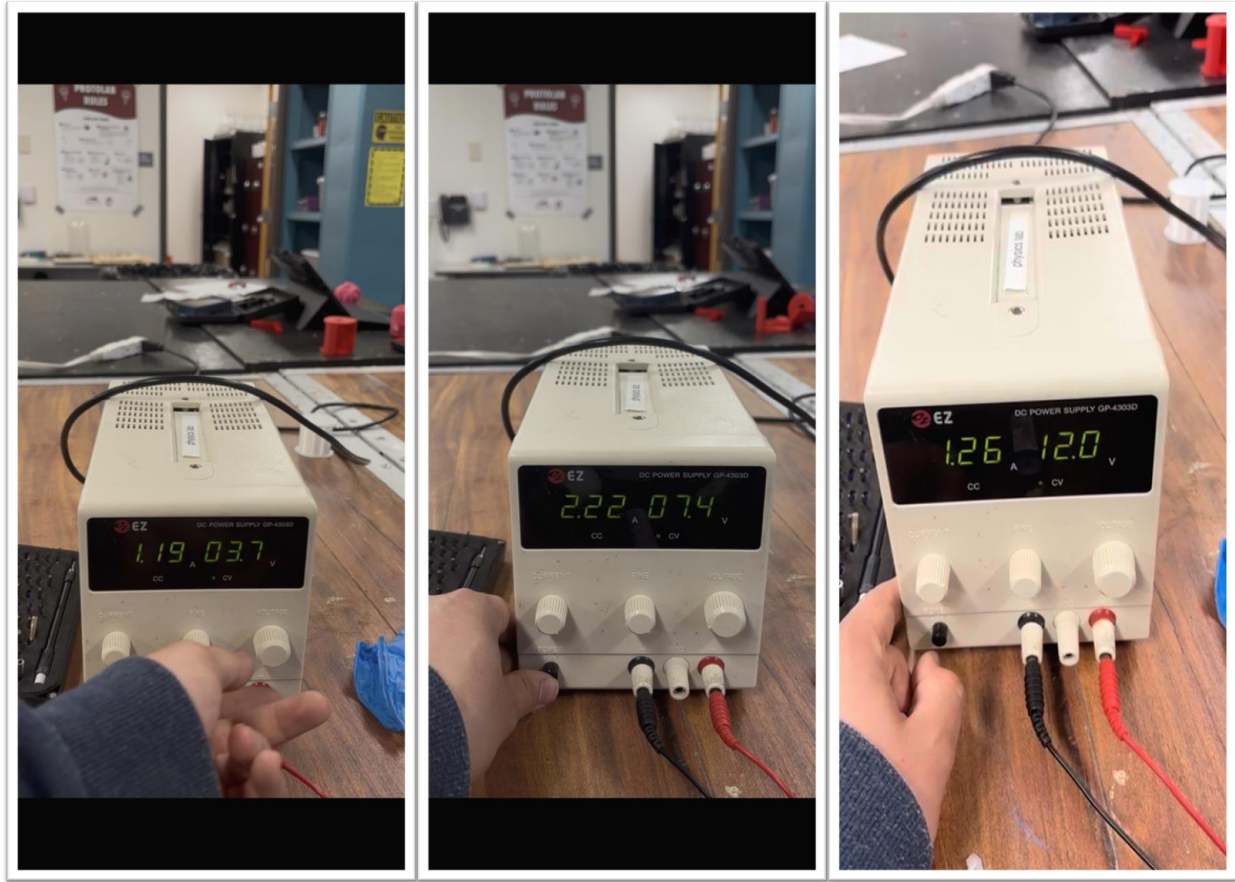


Figure 3-6: Testing Voltages with a Variable DC Power Supply

From this testing, it was determined that the electrical logic subsystem should target full 12 V operation. Lower voltages were considered during testing, but 12 V was selected because the torque and speed at that voltage best matched the needs of the hand-wrap roller. This was important because the subsystem needed a balance between sufficient torque to rotate the wrap reliably and sufficient speed to roll the wrap efficiently. The testing also confirmed that the previous 3.7 V electrical setup was no longer appropriate once the design pivoted away from the N20 motor. At this stage, battery testing has not yet been completed, but the variable power supply tests established the intended operating voltage for the subsystem and guided the selection of the charging and control architecture.

3.1.2.2 Analysis and Calculations

The electrical logic subsystem was designed around the requirements of the selected 12 V worm gear motor. Because the motor is rated for 12 V operation, the subsystem should provide a power source capable of delivering that voltage rather than relying on the earlier 3.7 V system used with the N20 motor. Designing the subsystem around a lower voltage would reduce the motor's ability to reach its intended operating condition and would no longer match the selected component.

The variable-voltage testing also showed that 12 V was the most appropriate operating target because it provided the best combination of torque and speed for the application. The hand-wrap roller requires enough torque to rotate the spindle while handling the resistance of the wrap, but it also requires enough speed to improve efficiency over rolling by hand. Based on testing, 12 V provided the best balance of these two needs.

The selected motor has a rated current of approximately 1.6 A. Based on this requirement, the subsystem wiring must safely carry current above that value without being unnecessarily large for the compact handheld housing. For this reason, 22 AWG insulated copper wire was selected. Current-capacity reference data shows that 22 AWG wire is capable of carrying current greater than the motor's rated demand, making it sufficient for the low-voltage DC circuit while still remaining compact and easy to route within the enclosure [40]. Compared with larger wire gauges, 22 AWG also improves flexibility and space efficiency, both of which are important in a handheld device.

The switch logic was also analyzed as part of subsystem design. A single on/off control would have been simpler, but it would also allow the motor to continue running whenever the circuit was energized. Instead, the subsystem uses two control stages: a slide switch as the main power switch and a normally open push button as the activation control. This arrangement allows the user to enable the device with the slide switch while keeping the motor off until the push button is pressed. As a result, the subsystem provides more intentional control over motor operation and reduces the chance of unwanted continuous running.

The charging path was separated from the motor power path as well. The USB-C port is intended to provide charging input to the battery through a charging and protection module rather than powering the motor directly during operation. This separation simplifies subsystem logic and supports repeated portable use of the device.

3.1.2.3 Secondary Research

Secondary research supported the selection of the major electrical components used in this subsystem. The selected worm gear motor is rated for 12 V operation, with a no-load speed of 160 RPM and a rated torque of 6.86 Nm, making it appropriate for a compact wrap-rolling device that requires controlled rotational output rather than high-speed free spinning. Standard American Wire Gauge reference data was also used to compare current capacity and physical size of 22 AWG, 20 AWG, and 18 AWG copper wire. This comparison supported the selection of 22 AWG wire because it provides sufficient current capacity for the motor while remaining flexible and compact enough for routing inside the handheld housing. The need for a boost converter is also supported by the mismatch between the 9.6 V battery pack and the 12 V motor requirement. Together, these sources support the feasibility of the selected electrical logic architecture and justify the chosen subsystem components.

3.2 Subsystem Description – Wrap Holding Mechanism, Daniel Benson

3.2.1 Scaled Up Solution

The overall objective of my subsystem is to pull the wrap with the power of the motor and spin the unraveled hand wrap into a neat roll. The spindle must be attached to the motor and be able to spin with it. It also must be smooth to make it easy to take the wrap off, and it must be strong enough to not break while spinning. Through attaching to the motor and being smooth, the spindle will be able to pull the wrap onto itself and once fully wrapped, the hand wrap can easily be slid off the spindle as seen in Figure 3-7.

3.2.1.1 How the Subsystem Works

My subsystem works through being the sole connection between the motor and the wrap harnessing the motors torque to re-roll the wrap. My spindle has a hollow bottom designed to fit directly onto the motors shaft that creates a stable and strong connection between the two. The bottom is designed with a cutout to fit directly on the motor to ensure the spindle is straight and will not strip on the motor. The top of the motor is made up of two prongs with gaps between the two designed to be the exact size of the wrap. The wrap can be placed onto one of the prongs on the spindle, and as the motor turns, the spindle turns pulling the wrap along with it and wrapping the wrap into a tight and straight bundle as shown in Figure 3-7.

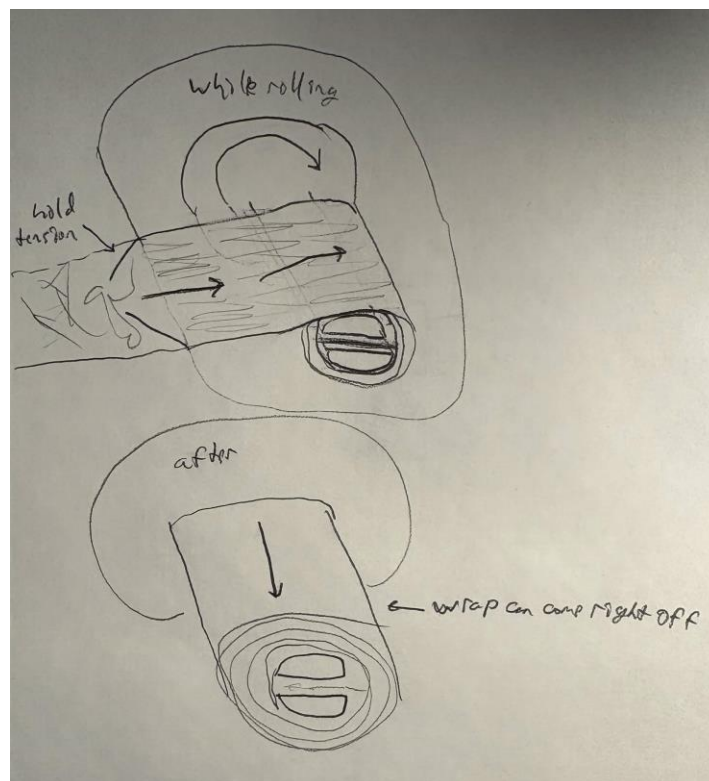


Figure 3-7: Process of Rolling Wrap and Taking Off

3.2.1.2 Subsystem Inputs

Two main inputs are needed to make this subsystem work properly and they are both shown in Figure 3-8. The first is the torque from the motor, and the second is the wrap that was just straightened by the straightening mechanism. The torque from the motor moves the spindle around to wrap the hand wrap. The wrap is actively getting straightened before going onto the spindle and then wrapped by the spindle to create the output, wrapped wraps. Through both the torque of the motor and the wrap itself, we can get our outputs of heat and a perfectly wrapped wrap.

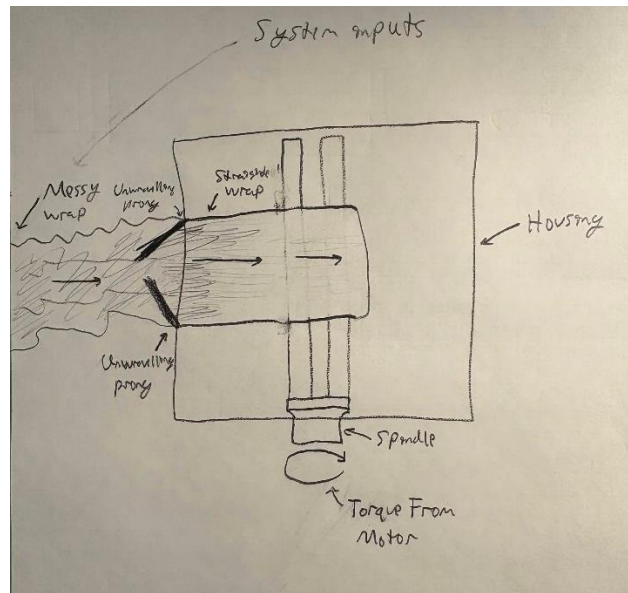


Figure 3-8: System Inputs

3.2.1.3 Key Subsystem Components

In the wrap holding mechanism, the key component is the spindle that the wrap will wrap onto. The spindle has two major features to it which are the hole on the bottom and the two prongs on the top. The prongs are designed to hold the wrap between them as the spindle is spun to wrap the wrap. The hole on the bottom is there so the spindle can have a tight mount into the motor. Both smaller systems of the spindle work together to have a straight, tight wrap.

3.2.1.4 Purchased Components

The shaft was designed in fusion 360 and printed with a Bambu Lab printer. Below is Table 3-4 with all the materials used to build, and Table 3-5 with the amounts of materials used for each component in the system.

Table 3-4: Purchased Materials

Product	Price	Size
1.75mm Bambu Lab PLA	\$32.70	1.0 kg

Table 3-5: Built Components

Printed Item	Price per unit	Mass
Spindle	$(\$32.70/1\text{kg}) \times .114.41\text{kg} = \3.74	.114.41kg



Figure 3-9: Filament

3.2.2 Physical Properties of Components

The dimensions of the holding subsystem component, the spindle, can be shown below in Figure 3-10 with dimensions. The material used to build this is PLA and its weight is around 114.41 grams. Its construction involves a fully filled 3D print with the spindle having two prongs as shown Figure 3-11, and a motor mount in the back as shown in Figure 3-12. It is designed specifically to fit into the shaft of the 12V motor also shown in Figure 3-12 and not too tall for our housing to keep the wrap straight and tight.

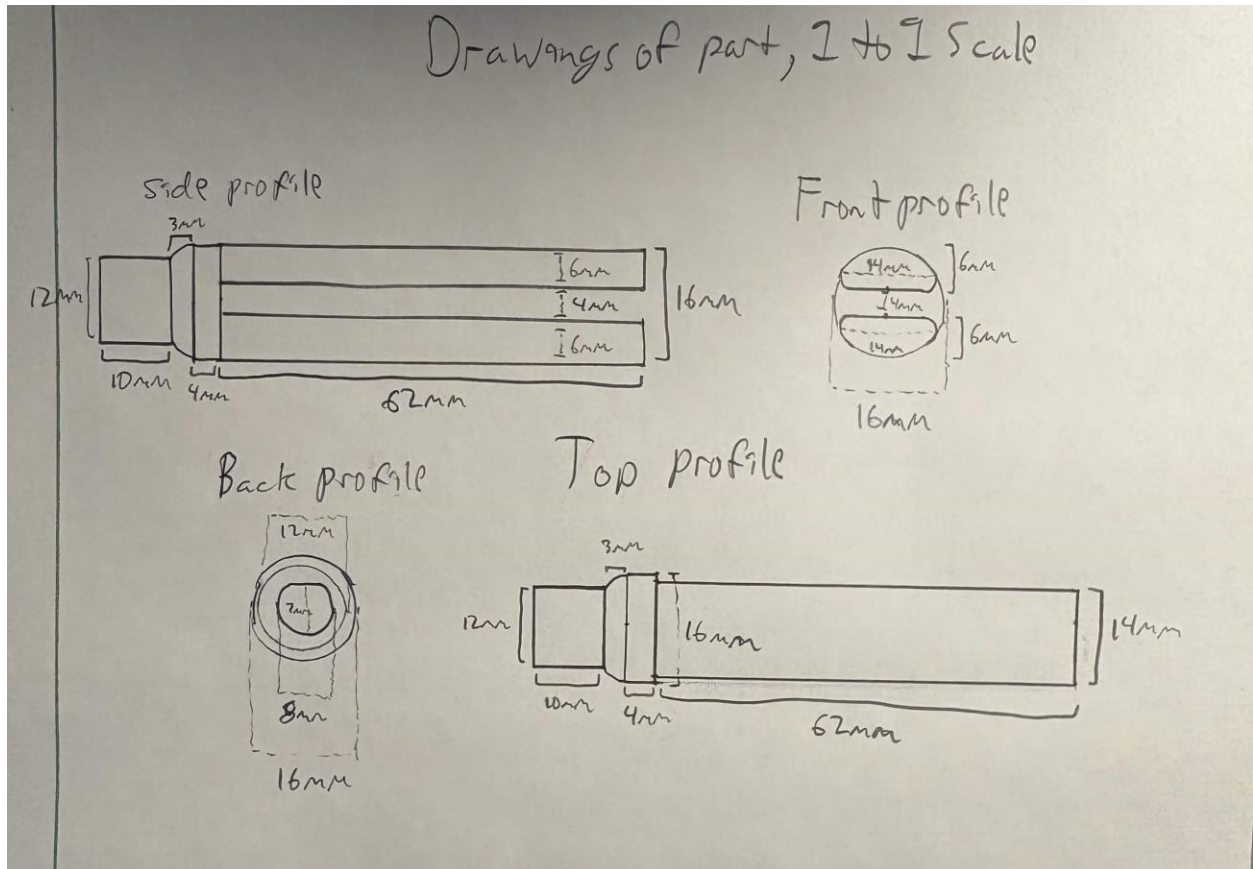


Figure 3-10: Scale Drawing

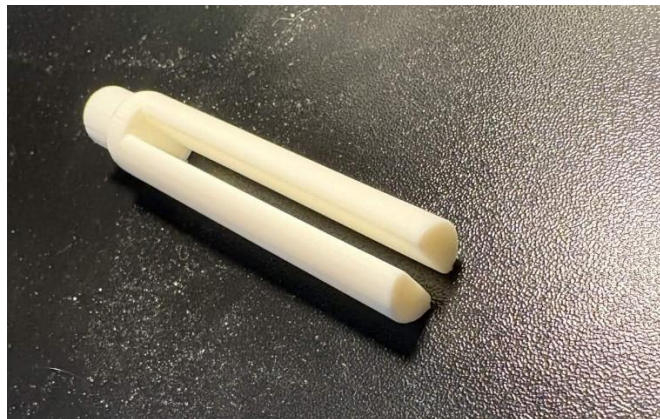


Figure 3-11: Front of Spindle

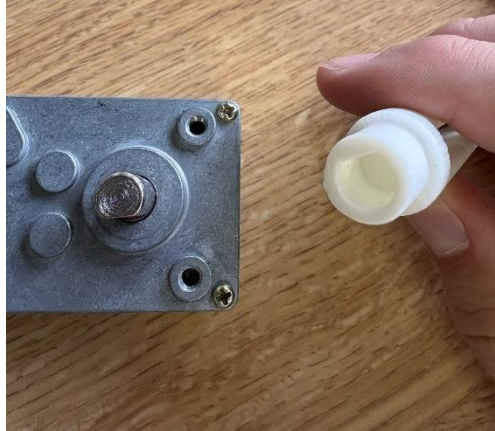


Figure 3-12: Back of Spindle and Motor

3.2.3 Evaluation of Alternative Ideas

Alternatives considered for the wrap holding mechanism involved a spindle made from resin, and a hollow spindle. Below in Table 3-6 is the decision matrix laying out how we decided which qualities were most important and how well each prototype matched those. In the table while the resin was strong and fir well, it weighed too much and the hollow shaft was light but was not strong enough. The conclusion was that the Solid PLA had most of the qualities we wanted in the final design.

Table 3-6: Decision Matrix

Material	Fitment on Motor	Strength	Weight	Finish	Total
PLA Hollow	4	2	5	3	14
PLA Solid	3	5	3	4	15
Resin	4	4	1	5	14

3.2.4 Data Considered

3.2.4.1 Test Results

Testing involved taking the various prototypes and testing the amount of strain they face under high amounts of force to simulate how much force the spindle will be under. The two tested were the hollow and solid PLA printed prototypes.

To test, the hollow and solid prints were put into vice grips with the ends with prongs facing out and having one prong facing downward. A piece of electrical tape was wrapped in a circle creating a loop between the end of the prong and a digital force meter. The digital force meter was then pulled down, and the peak was read for calculating the maximum force the spindle can handle. The spindle was then rotated at one hundred and eighty degrees, and the process was repeated on the other prong. Below is a model of the testing procedure used in Figure 3-13.

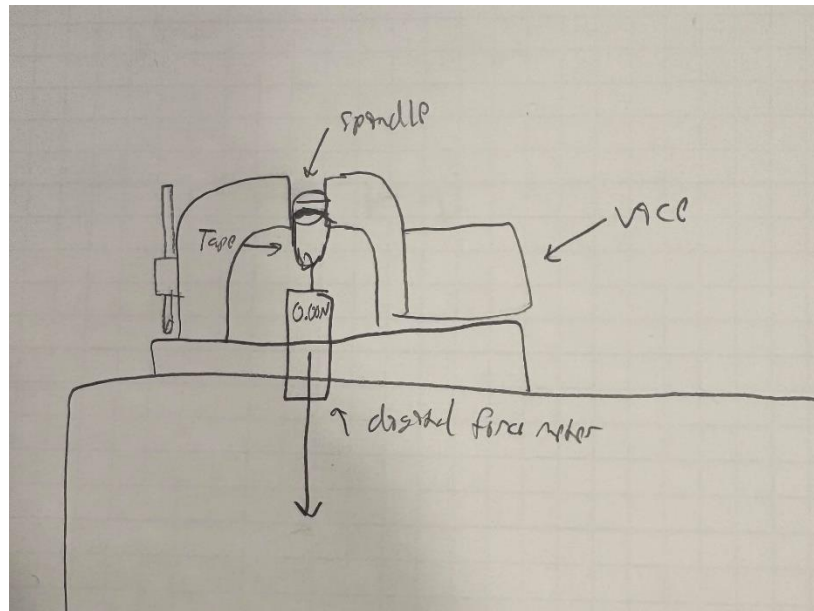


Figure 3-13: Testing Procedure

Below is Table 3-7 for the hollow spindle test, The spindles were breaking just before they got to forty newtons. As a result, despite weight saving, it concerns how quickly the spindles bend.

Table 3-7: Hollow Test

Test	Force	Notes
Test 1	37.2 N	Felt strain at 30 and that continued until it broke at 37.2.
Test 2	38.7 N	Started necking at 30 and broke at 38

Below is Table 3-8 for the solid spindle test, The spindle is completely solid PLA and was proven to be much more resistant to the force on the spindles. At the maximum force I was able to produce, the spindles did not break and did not even get any necking on them. The vice was unable to hold as the base mount gave out due to how it was mounted in the vice to keep still.

Table 3-8: Solid Test

Test	Force	Notes
Test 1	80 N	80 N is the maximum value I was able to produce
Test 2	80 N	Maxed out at 80 N meaning above isn't an outlier

3.2.4.2 Analysis and Calculations

Of the two materials being tested, PLA has a scientific strength advantage over the resin, making it a superior material for use in the subsystem. While being tested against resin, PLA is proven to have a greater tensile strength and can handle more stress [36]. Of the materials, PLA has a clear overall strength advantage to handle the tension of the wrap. In addition, PLA also happens to have the best strength against cyclical stress over resin [36]. This proving PLA superior for our prototype since it is the best at a load being added and removed multiple times as will happen with the wrap holding system.

3.2.4.3 Stakeholder and Expert Feedback

The final design prototype for the holding system was shown to a stakeholder who partakes in martial arts. The feedback given was mixed with good and bad things to say about the prototype. The positives included the stakeholder enjoying the overall look and feel of the prototype saying that it was smooth, and the stakeholder enjoyed the fitment into the motor. Some of the stakeholder's recommendations for improvement included trying a resin print or making the structure more rigid to prevent bending to the wrap can easily be taken in and out [35].

Given this advice, the current prototype is good, but more research needs to be done into attempting a resin print for better rigidity of the final product. In addition to this, a test with the final housing needs to be done to see if the stakeholders' fear of the spindle bending is true.

3.3 Subsystem Description – Wrap Straightening Mechanism, Max Golterman

3.3.1 Scaled Up Solution

3.3.1.1 Overall Objective of the Subsystem

The objective of the straightening mechanism is twofold. First, unfold the wrinkled and twisted wrap. Second, to provide tension to the rolling mechanism ensuring a tight roll. The subsystem includes two metal prongs that curve from the outside into the middle and downward to get underneath folds and spread them outward Figure 3-17. Additionally raised sidewalls, and a metal loop ensure the wrap does not pull sideways and stays within the system Figure 3-19.

3.3.1.2 How the Subsystem Works

The subsystem unfolds the wrap using the two prongs which curve inward getting underneath creases and forcing them toward to outside. The wrap remains underneath the prongs due to the raised sidewalls and metal loop which prevents lateral movement. The prongs fit snugly against the flat surface beneath them, which holds the wrap firmly enough to provide tension on the rolling mechanism, without overpowering it.

3.3.1.3 Subsystem Inputs

The only input to the straightening mechanism is the loose wrap. Typically, this wrap would be freshly washed and in a loose bundle with many kinks and folds.

3.3.1.4 Key Subsystem Components

The key components include two metal prongs which point diagonally inward and press down firmly upon the track to provide tension on the rolling mechanism. Raised sidewalls constructed from cardboard and vinyl tape guide the wrap to the metal stability loop. From there the wrap extends into the rolling mechanism.

3.3.1.5 Purchased Components

Purchase components of this subsystem include:

- Steel wire from a standard coat hanger see Figure 3-14. The wire measures 2mm in diameter and typically cost between \$0.20 and \$1.00 depending on quantity purchase [37].
- Scotch Super 33 Vinyl Electrical Tape in black. Approximately 2-3 feet was utilized. Cost from Walmart.com \$4.53 per unit which is $\frac{3}{4}$ inch x 52 feet. This gives the approximate cost of tape for prototype at \$0.26.

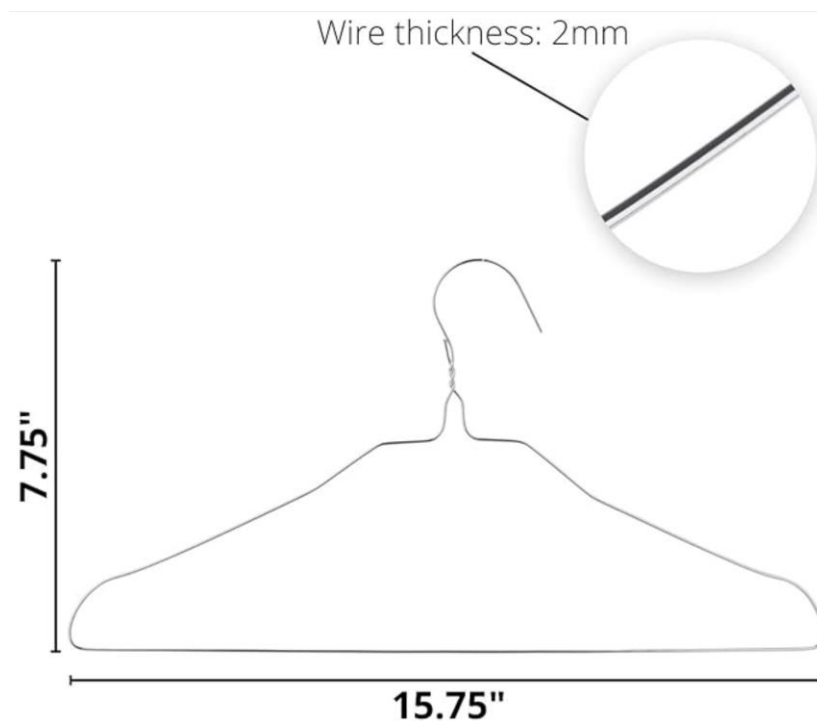


Figure 3-13: Coat hanger

3.3.2 Physical Properties of Components

As shown in Figure 3-15 the components in Table 3-9 are integrated into the outside of the rolling cylinder. Figure 3-16 provides additional context to placement of system on the cylindrical housing.

Table 3-9: Physical properties of components

Component	Material	Dimensions (cm)	Key Properties
Tension prongs	Alloy steel wire	5.7 x 1 each	Strength and flexibility
Stability loop	Alloy steel wire	6 x 0.7	Strength and flexibility
Raised sidewalls	Vinyl tape and cardboard	6 x 5 x 0.5	Low friction and cost per volume

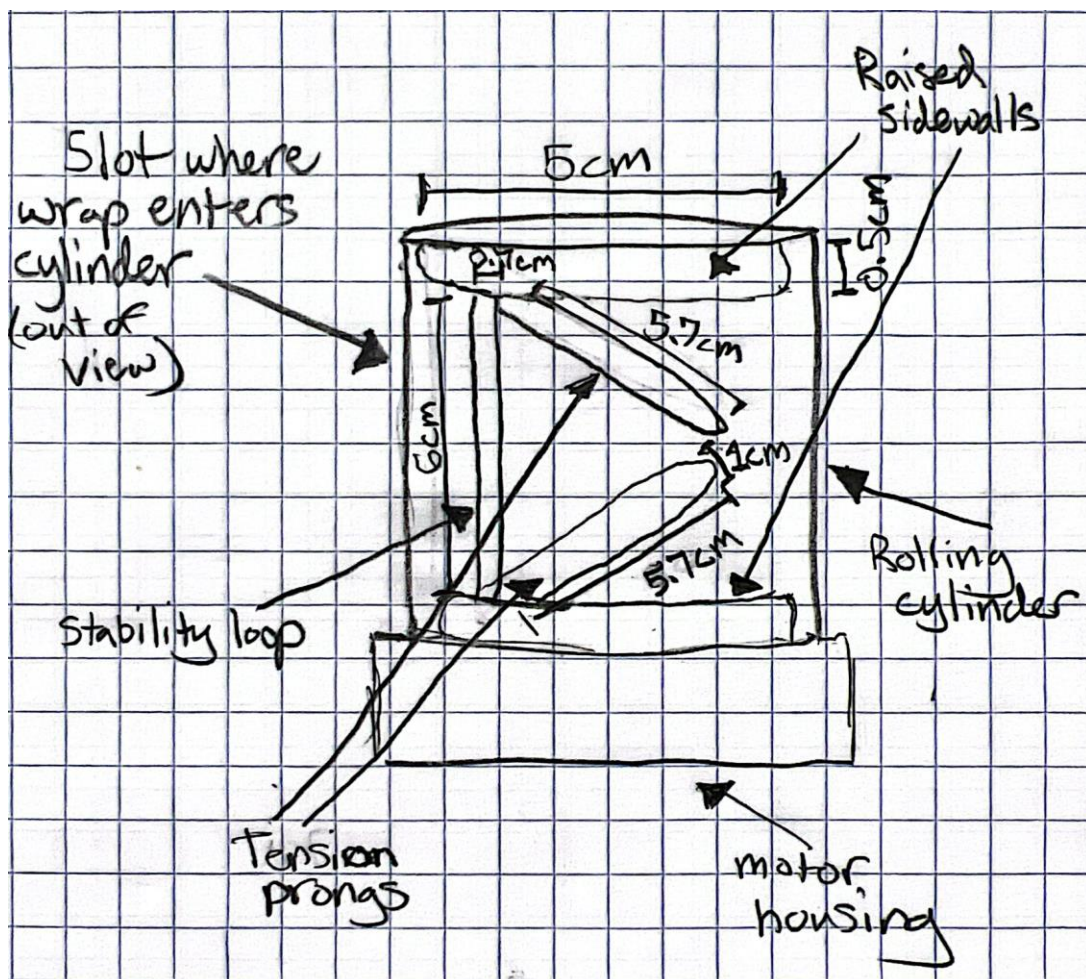


Figure 3-14: sketch of integrated system (side view)

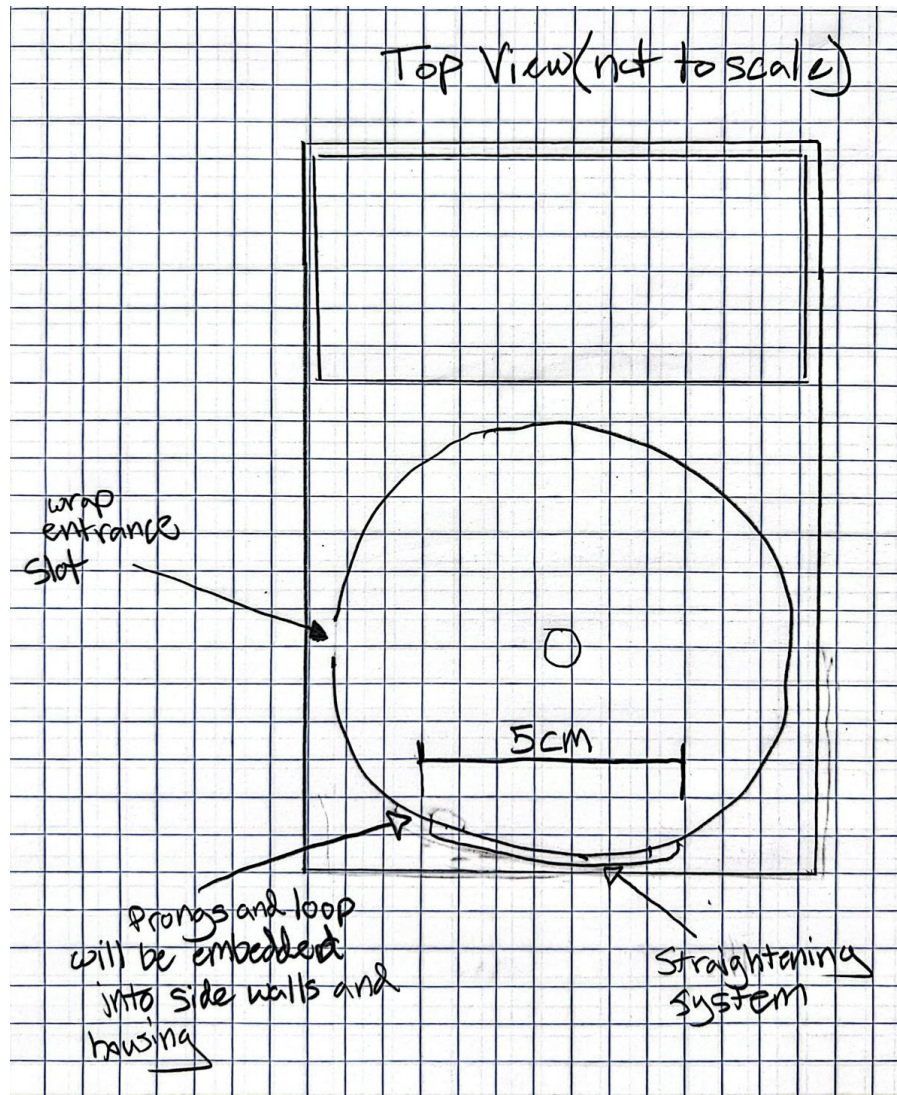


Figure 3-15: sketch of integrated system (Top view)

3.3.3 Evaluation of Alternative Ideas

As discussed later in this section, three prototypes were created during the design of this subsystem. The first prototype consisted of just two unfolding prongs. That proved to be unsuccessful as lateral stability was necessary to keep the wrap stable during the rolling process. The second prototype built upon the first with an added stability loop and raised sidewalls which prevented sideways slippage. Finally the third prototype was designed on stakeholder feedback to reduce load time utilizing hinges and a rotating loop to provide tension and lateral stability.

The primary tools used during the design process were testing and stakeholder feedback. With each prototype testing was done to determine viability and areas for improvement. As demonstrated in 3.3.4.1 Test Results, both prototypes 2 and 3 successfully completed the intended goal of unfolding the wrinkled wrap and providing tension on the motor which ensures a tight roll.

3.3.4 Data Considered

3.3.4.1 Test Results

The original prototype design for the straightening system was created to remove the need for human input on the wrap as it is fed into the roller. The initial system consisted of two downward curving prongs for unfolding and raised sidewalls for lateral stability see Figure 3-17. For initial setup the wrap is inserted underneath the prongs. Testing was done on the initial prototype see Table 3-10.

Table 3-10: Prototype 1 testing

Average load time	10.5 seconds
Test 1:	Unsuccessful, wrap pulled out from underneath one prong see Figure 3-18
Test 2:	Unsuccessful, the wrap pulls to one side. Determined additional lateral stability is required.



Figure 3-16: Prototype 1 loaded



Figure 3-17: Prototype 1 slipping

The second iteration of the straightening system includes an added metal loop for increased stability see Figure 3-19. The location of the loop on the roller side of the system provides lateral stability preventing the wrap from slipping out from underneath the unfolding prongs. For initial setup the wrap is carefully fed under the prongs and through the stability loop. Testing results from prototype 2 are shown in Table 3-11.

Table 3-11: Prototype 2 testing

Average load time:	9.67 seconds
Test 1:	Hand pulled wrap through the updated mechanism at a slow and steady rate. Wrap was successfully straightened as it would be fed into the device. The test did not require any human intervention below the straightening system.

Test 2:	This test the pile of wrap was turned over several times to simulate the twisted mess after it is washed. Same results. Wrap stayed within the system and no issues with fold or kinks.
----------------	---



Figure 3-18: Prototype 2

Even with the success of the second prototype a third prototype was created to test the viability of a system with improved ease of loading. In this prototype flat metal hinges are used in place of the curved prongs to provide tension to the roller and unfold the wrap as it is fed into the roller as shown in Figure 3-20. Testing was done as shown in Table 3-12.

Table 3-12: Prototype 3 testing

Average load time:	4.57 seconds
Test 1:	Wrap immediately came out from underneath hinges. Moved hinges closer together for next test.
Test 2:	Wrap was approximately halfway through before coming out from underneath the hinges. Lateral stability required.

Test 3:	Hinges held wrap too tightly, tension would be greater than motor can handle.
Test 4:	Improvements to ensure lateral stability include sidewalls under the hinges and a metal loop which rotates for loading and process see Figure 3-21 and Figure 3-22. This test went smoothly with the wrap stable laterally throughout the length of the wrap. The tension was not excessive to overload the motor.



Figure 3-19: Prototype 3 loading



Figure 3-20 Prototype 3 open



Figure 3-21: Prototype 3 loaded

3.3.4.2 Analysis and Calculations

An analysis was done to determine the best materials to use for the straightening system. During testing alloy steel wire from a standard metal coat hanger was utilized for the unfolding prongs and the stability loop. This specific material is favorable due to its strength and flexibility which was shown in testing. According to the strength-stiffness chart steels have a better strength vs stiffness ratio than most as shown in Figure 3-23 [34].

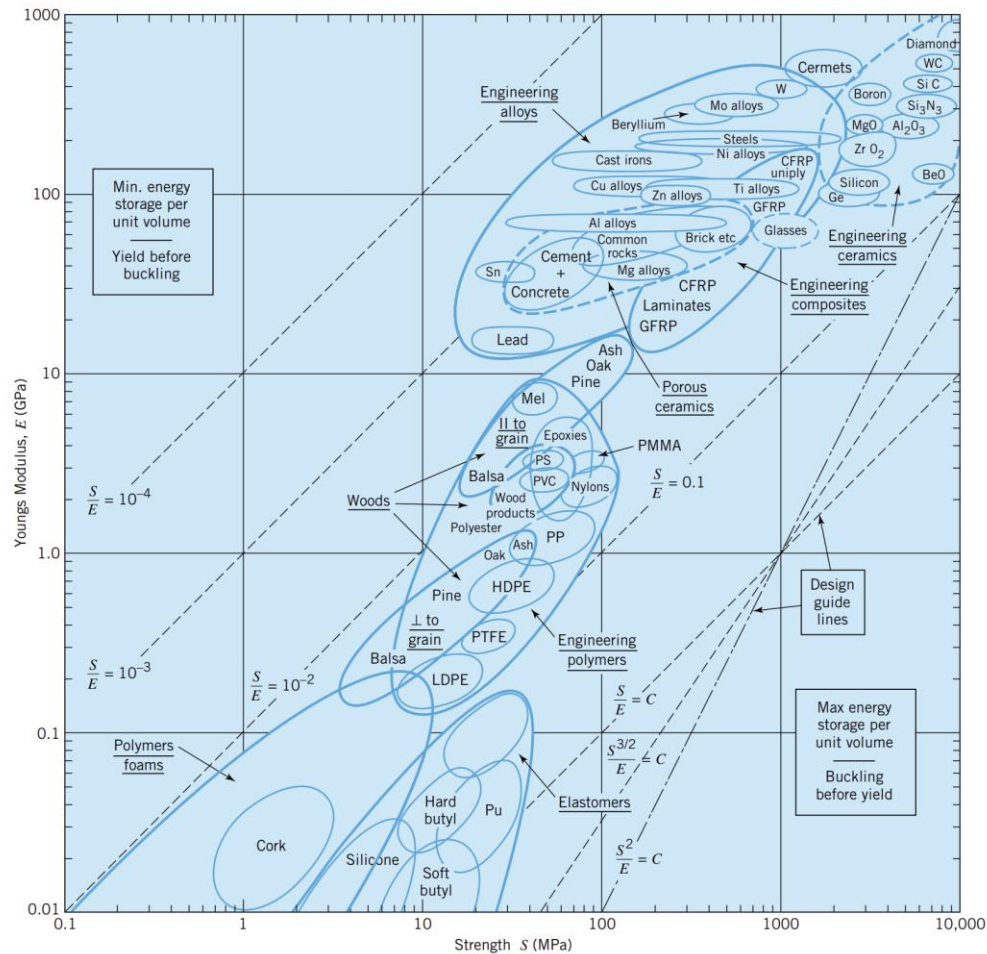


Figure 3-22: Material selection chart Strength vs Youngs Modulus (Stiffness) [34]

In addition to performance, alloy steel also has the added benefit of being cost efficient. This is represented in Figure 3-24 which shows alloy steel has one of the lowest cost's per unit volume [34].

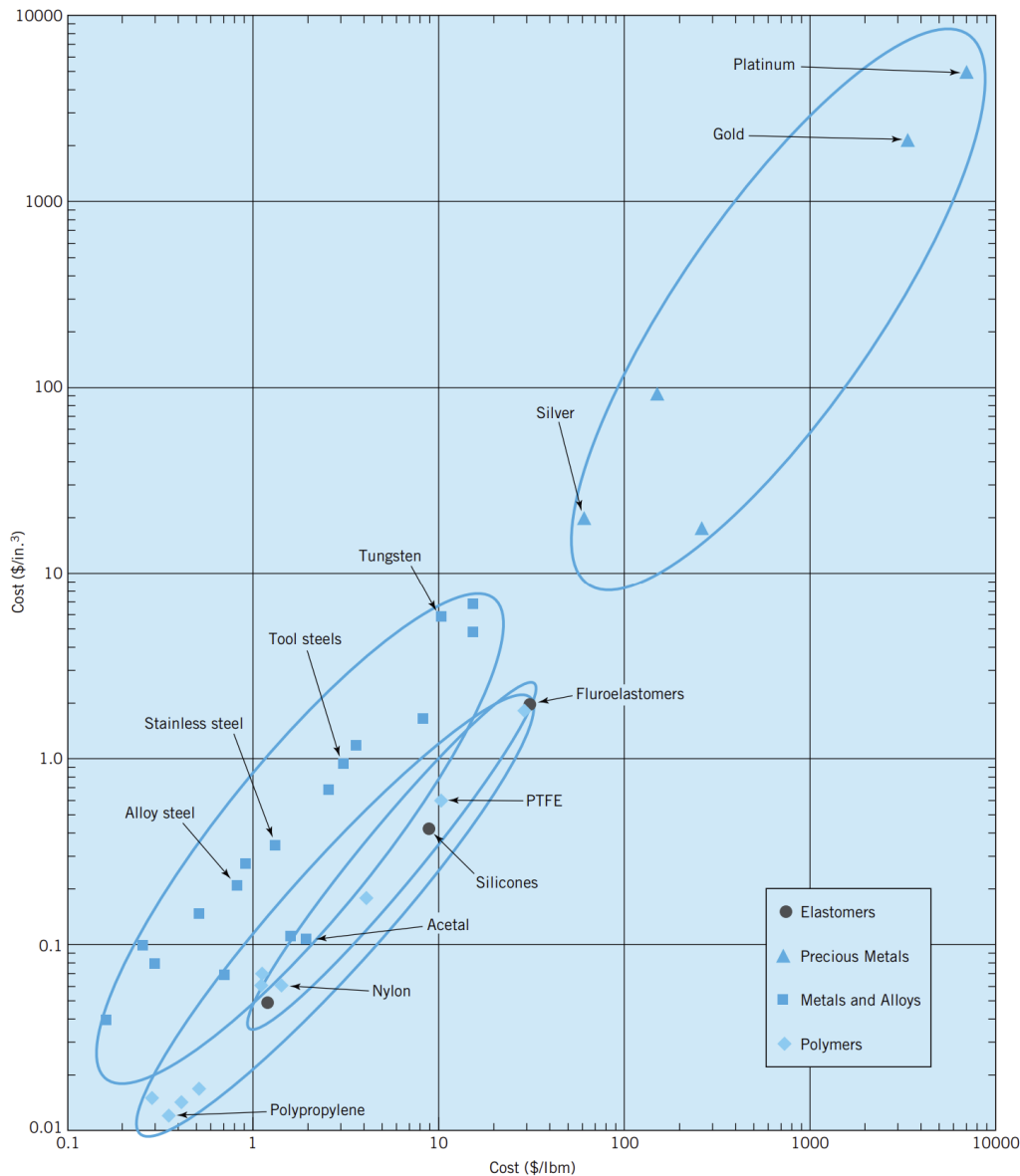


Figure 3-23: Material selection chart cost per unit volume

3.3.4.3 Stakeholder and Expert Feedback

After prototype 2 of the straightening system was successful, a demonstration was performed for stakeholders [41]. The stakeholders interviewed appreciated the hands free nature of the system reducing the need for human input into the device. Additionally, they saw recognized how the system would provide the tension necessary to ensure the roll is tight.

The stakeholder provided feedback on the aspect of loading the wrap into the system. Because the two prong and one loop design is fixed, the wrap must be carefully slid underneath the prongs and through the loop. This feedback provided the idea for the design of prototype 3 which features unique updates for ease of loading. As shown in Figure 3-21 the hinges open and the lateral stability loop lifts, allowing the wrap to be easily placed beneath them. Once the hinges and loop are closed see Figure 3-

22 the system is ready for activation. The hinge design improved load time by 5.10 seconds on average as shown in Table 3-11 and Table 3-12.

3.4 Subsystem Description – Electrical Energy, Revell Jenkins

3.4.1 Scaled Up Solution

3.4.1.1 Overall Objective of the Subsystem

The objective of the electrical energy subsystem is to store and supply sufficient electrical energy to power the DC motor while maintaining portability, safety, and rechargeability. This subsystem is responsible for delivering consistent voltage and current to support motor operation during the wrap rolling process.

The system is designed to provide adequate energy capacity for repeated use while remaining compact enough to fit within the device enclosure. A 9.6 V battery is used to better match the voltage requirements of the motor, improving overall performance compared to lower-voltage configurations. The subsystem must also allow for safe and efficient recharging to support long-term usability.

3.4.1.2 How the Subsystem Works

The subsystem operates through a direct electrical circuit powered by a 9.6 V battery, which stores electrical energy and supplies it to the system when the circuit is completed.

When the system is activated, the battery provides a voltage potential across the circuit, allowing current to flow to the motor. The amount of current drawn depends on the mechanical load of the motor, with higher resistance resulting in lower current and lower resistance resulting in higher current draw. This electrical energy is converted into rotational motion by the motor to drive the wrap rolling mechanism.

The battery maintains a relatively stable voltage output during operation, ensuring consistent motor performance throughout most of the discharge cycle, as illustrated in Figure 3-26.

3.4.1.3 Subsystem Inputs

The electrical energy subsystem requires the following inputs to function:

- Stored chemical energy within the 9.6 V NiMH battery pack, which is converted into electrical energy during operation
- Load demand from the DC motor, which determines the amount of current drawn from the battery

These inputs define how energy is stored, transferred, and consumed within the system, ensuring that sufficient power is available to operate the motor under varying load conditions.

3.4.1.4 Key Subsystem Components

The electrical energy subsystem consists of the following key components:

- **9.6 V NiMH Battery Pack**

Serves as the primary energy storage device, supplying sufficient voltage and current to drive the DC motor at improved performance levels compared to lower-voltage configurations.

3.4.1.5 Purchased Components

The electrical energy subsystem includes both the initially selected battery and the updated battery used in the final design.

- **3.7 V LiPo Battery (2000 mAh)**

The original design used a single-cell lithium polymer battery due to its compact size and high energy density. While this battery provided sufficient portability and rechargeability, it was ultimately determined to be inadequate for the system's power requirements. The voltage output was too low to effectively drive the DC motor at the desired speed and torque, resulting in reduced performance during testing. As shown in Figure 3-25, the original 3.7 V LiPo battery was compact and lightweight but did not provide sufficient voltage for the system's requirements.

- **9.6 V NiMH Battery Pack**

The updated design uses a 9.6 V nickel-metal hydride battery pack to better match the voltage requirements of the motor. This higher voltage allows the motor to operate more efficiently and produce sufficient torque to complete the wrap rolling process. Although the NiMH battery is larger and heavier than the original LiPo battery, it provides improved performance and reliability for the system. As shown in Figure 3-25, the 9.6 V NiMH battery pack provides a higher voltage output, allowing the motor to operate more effectively and produce sufficient torque.

- **USB-C Charging Module**

A charging module is included to allow the battery to be recharged from an external power source. This supports repeated use of the device and eliminates the need for disposable batteries.

Battery Model	104245
Product size	10*42*47mm
Sstandard	3.7v
Nominal Capacity	2000mAh
Charging Voltage	4.2V
Discharge Voltage	2.75V
Recharging Current	Standard 1A max 2A
Recharging Temperature	0°C–45°C
Discharge Current	Standard 1A max 2A
Store Temperature	–20°C–60°C
Certification	MSDS



Figure 3-24: 3.7 V 2000 mAh LiPo Battery with JST connector



Figure 3-25: 9.6 V NiMH Battery Pack

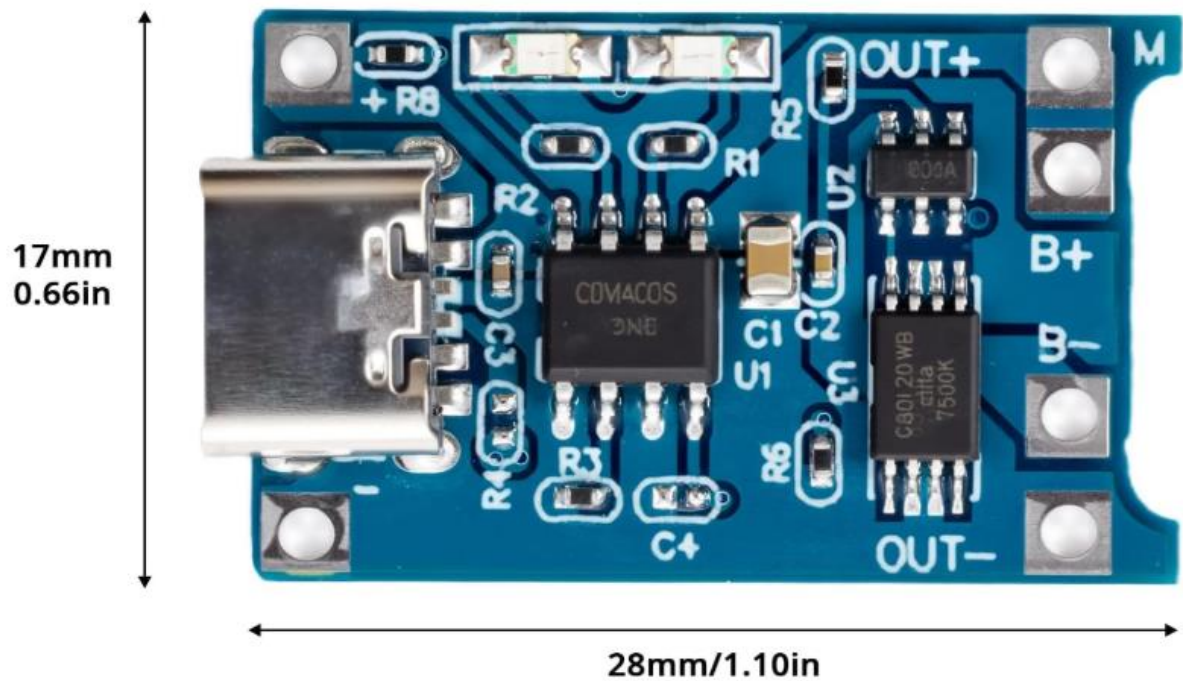


Figure 3-26: USB-C LiPo charging module

The transition to a higher voltage battery was a critical design decision that directly improved system performance.

3.4.2 Physical Properties of Components

The electrical energy subsystem components are designed to fit within the lower enclosure of the device while maintaining portability and proper weight distribution. The updated battery selection introduces changes in size and weight that impact the overall design. The physical size and configuration of the battery are shown in Figure 3-26 and summarized in Table 3-13.

Table 3-13: Component Properties

Component	Dimensions (mm)	Weight	Key Properties
9.6 V NiMH Battery Pack	135 × 47 × 25	~204 g	Higher voltage output, rechargeable

The 9.6 V NiMH battery pack is significantly larger and heavier than the originally considered LiPo battery, requiring adjustments to the internal layout of the device. However, this increase in size and weight is justified by the improved voltage output, allowing the motor to operate more effectively and produce sufficient torque.

The battery is positioned within the lower housing to maintain stability and minimize interference with the mechanical subsystem. While this reduces overall compactness, it improves system performance and reliability.

3.4.3 Evaluation of Alternative Ideas

Several alternative approaches for powering the system were considered during the design process. As shown in Table 3-14, multiple battery options were evaluated based on performance, size, and system complexity.

- 3.7 V LiPo battery (single-cell)
- Higher capacity LiPo battery
- Multiple LiPo cells in series
- Alkaline battery packs (AA/AAA)

Table 3-14: Alternative Considerations

Option	Advantages	Disadvantages
3.7 V LiPo	Compact, lightweight, rechargeable	Insufficient voltage for motor performance
Larger LiPo	Longer runtime	Still limited by low voltage unless combined in series
LiPo in Series	Higher voltage output	Increased complexity and safety concerns
Alkaline Batteries	Easy to replace	Not rechargeable, bulky, inconsistent output

Selected Approach Justification

The 9.6 V NiMH battery pack was selected because it provides a voltage level that better matches the requirements of the DC motor, resulting in improved speed and torque during operation. The initial 3.7 V LiPo battery was determined to be insufficient for driving the motor effectively, which limited system performance.

Although the NiMH battery pack increases the size and weight of the device, it simplifies the design by eliminating the need for additional voltage regulation components. This allows for a more reliable and straightforward system while still maintaining rechargeability.

The selected approach provides a balance between performance, simplicity, and reliability, making it suitable for the current prototype while allowing for future optimization.

3.4.4 Data Considered

3.4.4.1 Test Results

Initial testing with the 3.7 V LiPo battery showed that the motor was unable to operate at the desired speed and torque. The system experienced reduced performance, and the motor struggled to maintain consistent rotation under load. As shown in Figure 3-28 and Figure 3-29, testing was conducted using a power supply to simulate different voltage levels. Lower voltage conditions resulted in reduced motor performance, while higher voltage levels improved speed and torque, supporting the selection of a higher voltage battery.



Figure 3-27

After switching to the 9.6 V NiMH battery pack, the system demonstrated a noticeable improvement in performance. The motor operated more consistently and produced sufficient rotational speed to effectively roll the hand wrap. The increased voltage allowed the motor to overcome load resistance more efficiently, reducing the likelihood of stalling during operation.



Figure 3-28

Testing also confirmed that the system could be activated repeatedly without significant performance degradation over short periods of use. The battery maintained stable operation throughout testing, supporting reliable system functionality. As shown in Figure 3-30, increasing current results in a proportional increase in power output, improving motor performance.

3.4.4.2 Analysis and Calculations

Battery capacity:

- 2000 mAh = 2 Ah

Estimated motor current draw:

- Typical load: 0.2 – 0.5 A
- Peak (stall): ~3 A

Estimated Runtime

Runtime = Battery Capacity / Current Draw

- At 0.2 A -> ~10 hours
- At 0.5 A -> ~4 hours

Since the device is used intermittently, actual runtime is expected to exceed these continuous-use estimates.

Power Output

$$P = V \times I$$

- At 9.6 V and 0.5 A -> ~4.8 W

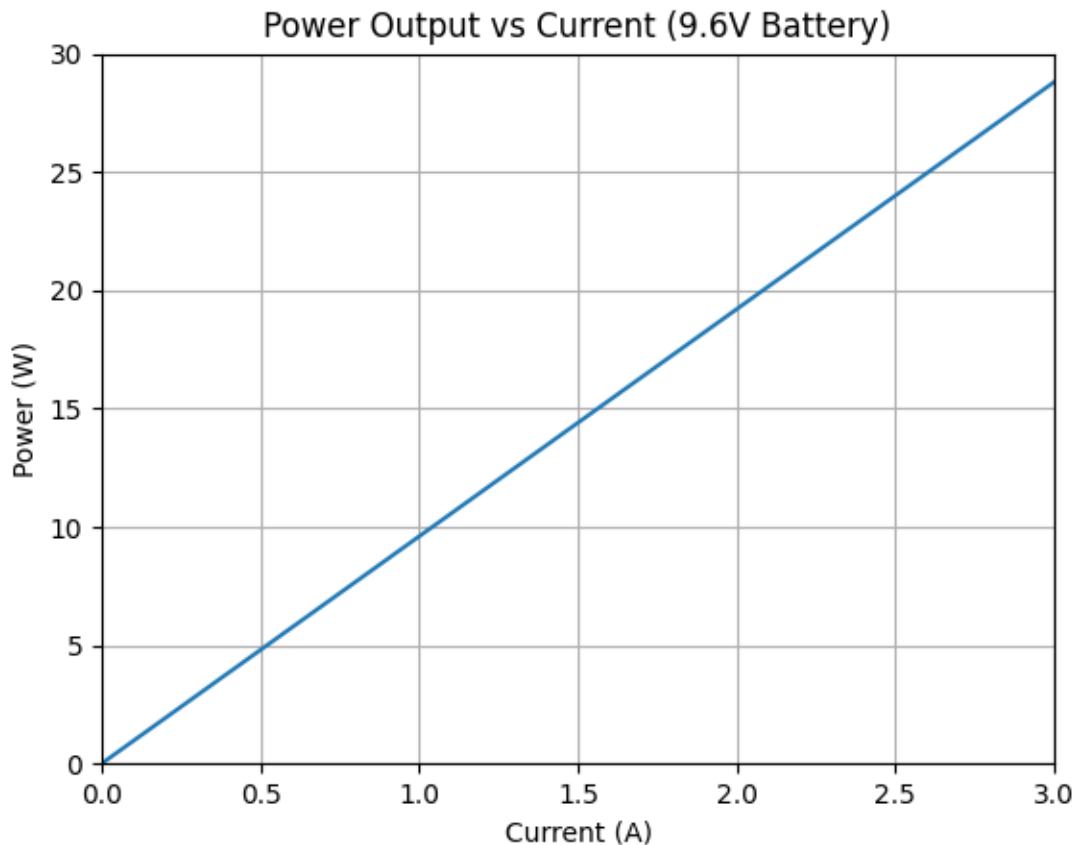


Figure 3-29

This is a substantial increase compared to the previous 3.7 V system and allows the motor to operate at a higher speed and torque, improving the effectiveness of the wrap rolling process.

Performance Justification

The increase in voltage from 3.7 V to 9.6 V directly improves motor performance by increasing the available power output. This results in more reliable operation and reduces the likelihood of stalling under load. Although higher power output increases energy consumption, the selected battery capacity still provides sufficient runtime for repeated use, making the system both effective and practical.

3.4.4.3 Secondary Research

Nickel-metal hydride (NiMH) batteries are commonly used in portable electronic systems due to their reliability, rechargeability, and ability to deliver higher voltages through multi-cell configurations. Research shows that NiMH batteries provide stable voltage output under moderate loads and are capable of supplying sufficient current for motor-driven applications.

Compared to single-cell lithium polymer batteries, multi-cell NiMH battery packs offer higher total voltage without requiring additional voltage regulation components. This makes them suitable for applications where increased motor performance is required.

Additionally, NiMH batteries are less sensitive to charging conditions than lithium-based batteries and are widely used in consumer electronics, power tools, and hobby systems. This supports their selection as a practical and reliable energy source for the current prototype.

3.4.4.4 Stakeholder and Expert Feedback

Feedback from users and peers emphasized ease of use, portability, and consistent performance. Initial evaluation showed that the 3.7 V battery did not provide sufficient power to drive the motor effectively, which led to the decision to switch to a 9.6 V NiMH battery pack.

This change improved motor performance by increasing torque and speed, resulting in more reliable operation. Additional feedback supported maintaining a simple design, so unnecessary components such as voltage regulation were avoided in the initial prototype.

Overall, stakeholder and expert feedback directly influenced the selection of the battery system, leading to improved performance while maintaining simplicity and usability. As shown in Table 3-14, the initial battery selection did not meet performance requirements, which led to the transition to a higher voltage system.

3.5 Subsystem Description – Rotational Force, Terrence Micciche

3.5.1 Scaled Up Solution

3.5.1.1 Overall Objective of the Subsystem

The overall objective of this subsystem is to create a rotational force that is efficient and effective. The key component of this subsystem is a motor. The chosen motor, which will be further evaluated in the section of Data Consideration, will demonstrate the success of this subsystem. An inefficient motor would best be described as a motor whose RPM (rotations per minute) are less than or equal to 40RPM, which is the speed of manually rolling the wrap [30]. An inefficient motor is also described as a motor whose torque, measured in Nm (newton-meters), is not strong enough to roll the wrap.

3.5.1.2 How the Subsystem Works

This subsystem works by utilizing a motor to create rotational force. This force, when paired with an input of an unrolled wrap, will result in the output of a rolled wrap. A custom motor will not be created for this project, as it is far out of the scope of the team problem. Rather, multiple different motors will be tested and compared to determine which is the most effective for the problem at hand. Below is Figure 3-30, which is a field sketch of an individual in their home observing the output of a rolled wrap. The output is tightly rolled and uniform.

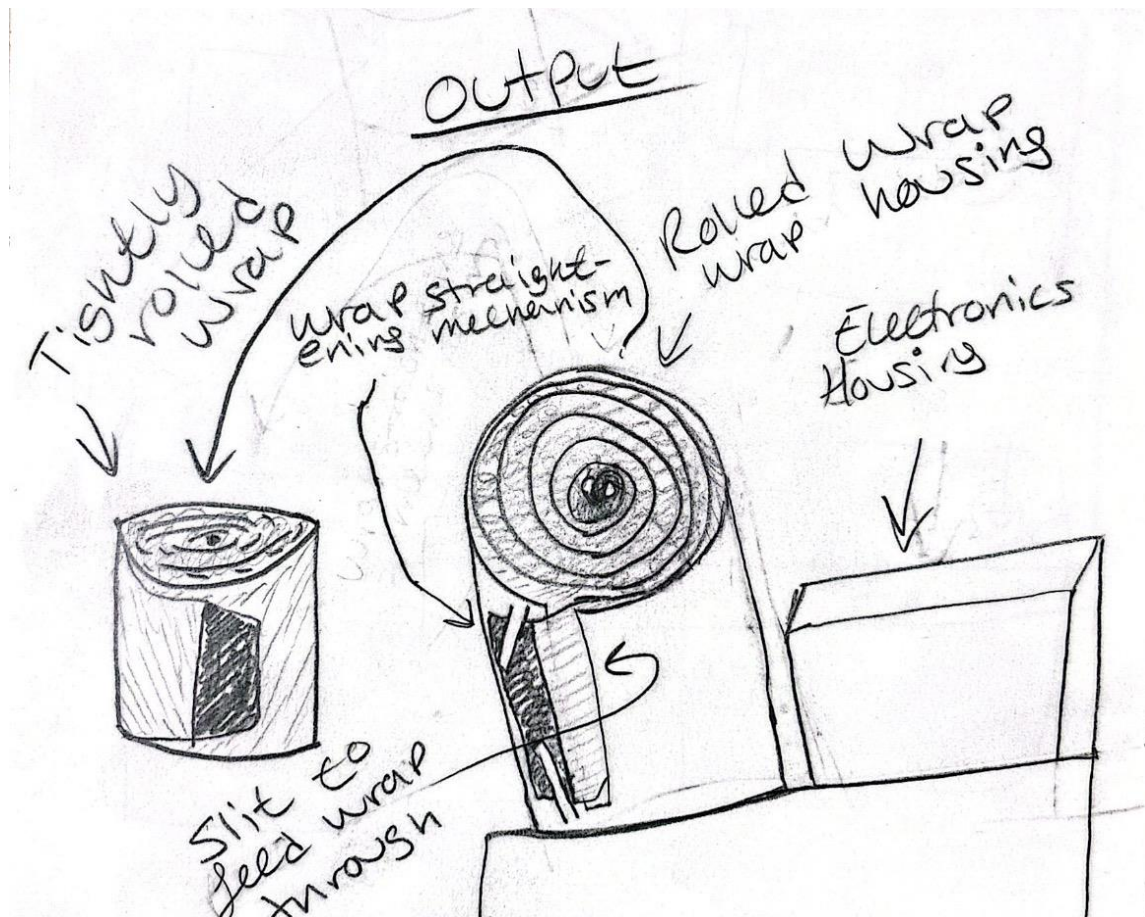


Figure 3-30: Shows the rolled wrap (output) being discharged from the final solution

3.5.1.3 Subsystem Inputs

The input required to make this subsystem work is the unrolled hand wrap. The point of this subsystem, with respect to the interface with other subsystems as well as the overall team solution, is to cause a rotational force that has enough RPMs and torque to cause a straightened hand wrap to roll into a neat roll in an efficient time. Below is Figure 3-31, which is a field sketch of the input, being an unraveled hand wrap, placed into the final solution.

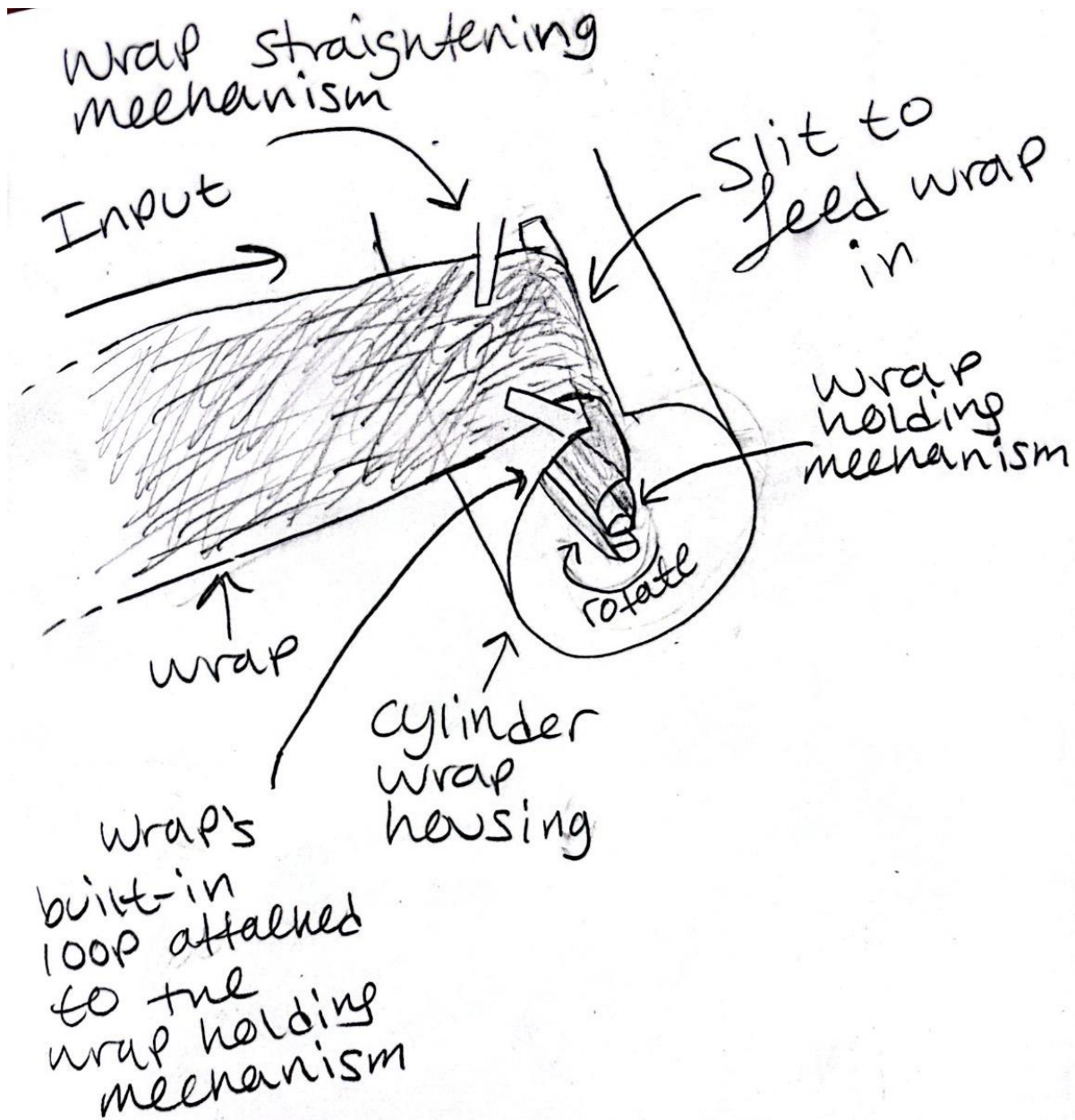


Figure 3-31: Shows the unrolled wrap (input) being fed into the final solution

3.5.1.4 Key Subsystem Components

The key subsystem components needed to accomplish the desired outputs are a motor of stall torque greater than about 0.216 Nm and a speed of greater than 40 RPM. For this subsystem, the evaluation of the motor alone will provide the exact input needed to move forward with the interfacing of other components. As a side note, the scope of the team's problem does not involve creating a unique motor. Motors will be purchased for testing and utilization in the final solution.

3.5.1.5 Purchased Components

With the completion of the first working prototype of the final solution, only one component has been purchased. This component is a 1218GE-N20 Reduction Mini DC Motor, and it is observed in the Figure 3-32 below.

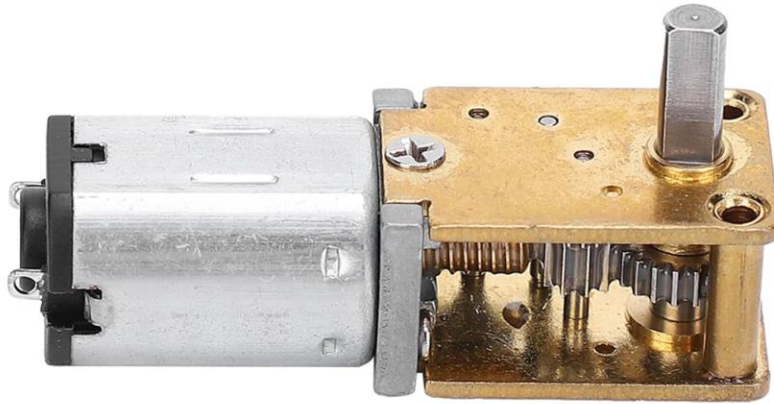


Figure 3-32: Shows the N20 motor [42]

This motor is rated for a maximum of 6V at 61RPM with a block load of about 0.216Nm. This purchase was extremely insightful, as it allowed the team to understand that a torque of 0.216Nm at stall is too little to support the effective rolling of a wrap. Moving forward, the team will investigate utilizing motors that have block load torques that are greater than 0.216Nm.

3.5.2 Physical Properties of Components

With the consideration of the N20 motor, the final motor that will be used will be large. The initial motor is very small, and its dimensions are shown in Figure 3-33 below.

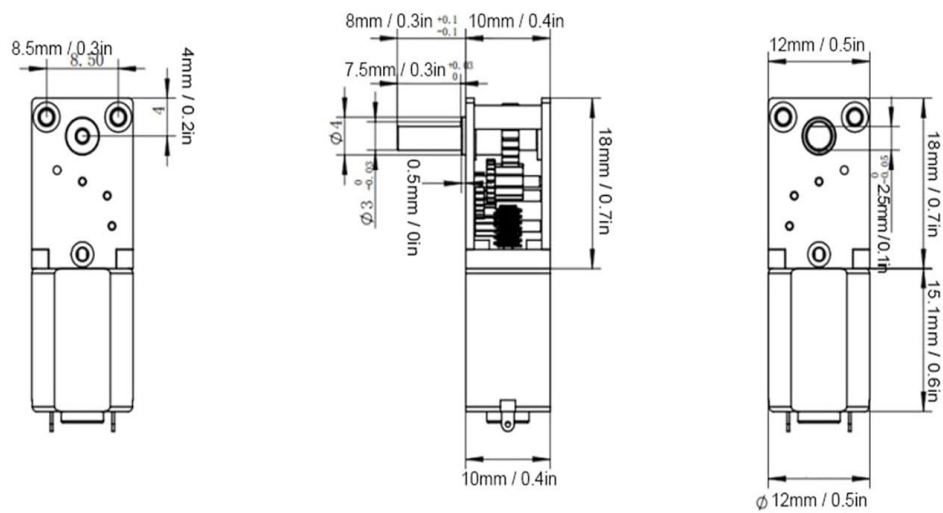


Figure 3-33: Shows the dimensions of the N20 motor [42]

The size, and weight of only 14 grams, of the initial motor is perfect for the team to satisfy the goal of making the final solution portable. However, due to its lack of strength, it no longer is viable to make the final solution portable. This is supported by the dimensions of the next available motor, which is more than triple the initial volume. The next motor is a 12V 220RPM DC worm gear motor, weighing 380 grams. This motor is optimal for its affordable price and smaller size compared to other models of the same torque. The physical dimensions of the motor are displayed in Figure 3-34.

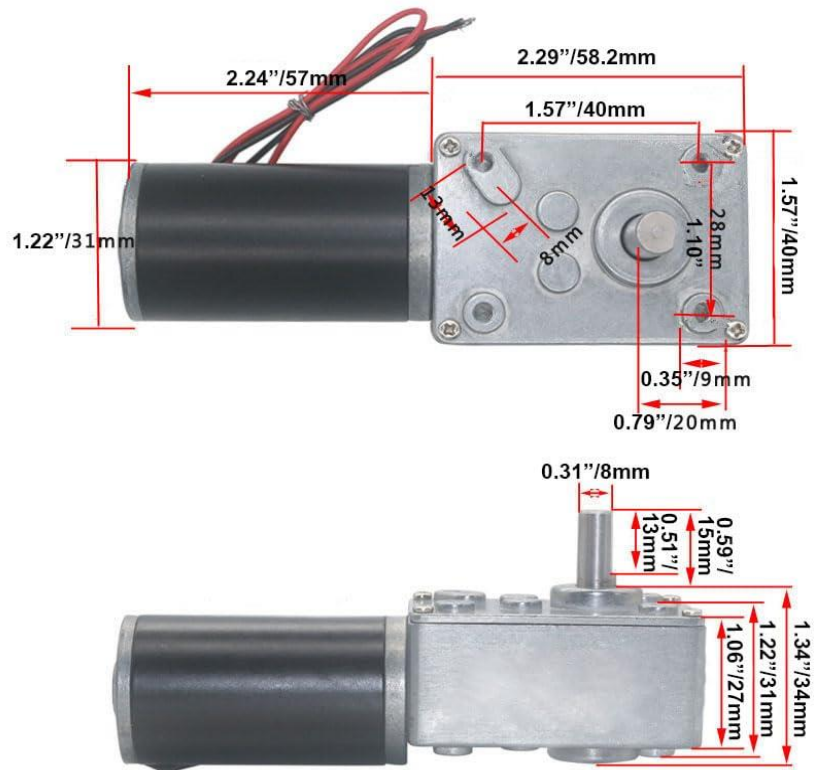


Figure 3-34: Shows the dimensions of the new motor [43]

3.5.3 Evaluation of Alternative Ideas

The ideas that were generated to help the team reach their final solution all related to the N20 motor that was used in the first working prototype. This motor was not strong enough to roll the wrap effectively, so 2 ideas were considered by the team to help subdue the issue. The first idea was to put more than 6V into the motor, which was its rated voltage. Doing so resulted in a noticeable gain of torque. However, this idea was unusable, as using more than 6V is only reasonable for short bursts of time. The next idea regarded the team investing in a larger motor, which is the same motor depicted in Figure 3-34. This would be done to replace the smaller motor, sacrificing the portability of the device. But due to the danger of the first idea, the second idea was selected as the most practical choice for the team. This was supported by a vote of 4 to 1, with the one vote being for the sympathy of maintaining portability.

3.5.4 Data Considered

3.5.4.1 Test Results: Initial Motor

The first working prototype of the device was printed, assembled, and tested with ease. However, the results were not what the team was hoping for. More specifically, the motor which was used did not have enough strength to effectively roll the testing wrap because it could not maintain light pulling of the wrap, which is necessary to keep the wrap from folding onto itself. The motor used was a 1218GE-N20, such as the one depicted in Figure 3-32 above. Below in Table 3-15 the performance parameter table of the N20 motor. Based upon the below figure, the team can determine what stall torque is too low to roll the wrap. Table 3-15, Shows the stall torque, RPM, and Current (amps) of the N20 motor given different voltages

Table 3-15: Stall Torque, RPM, and Current of N20 Motor [42]

Voltage	Reduction ratio	No-load		Load			Block load	
		Rotate speed	Current	Rotate speed	Current	Torsion	Current	Torsion
3V	1:236	16RPM	≤30MA	13RPM	≤40MA	120g.cm	≤0.4A	2.6KG.CM
3V	1:236	30RPM	≤30MA	25RPM	≤40MA	160g.cm	≤0.4A	2.2KG.CM
6V	1:236	35RPM	≤50MA	28RPM	≤40MA	220g.cm	≤0.4A	1.5KG.CM
6V	1:236	70RPM	≤60MA	61RPM	≤40MA	320g.cm	≤0.4A	2.2KG.CM

From the above table, it is concluded that a torque of over 0.216Nm (or 2.2 kilogram-force-centimeter as seen in the above table) is necessary to wrap rolls effectively. This will be considered in the search for another motor.

3.5.4.2 Analysis and Calculations

The conclusion of the test results suggests that a stronger motor is needed, and this is supported by the moment of force equation, observed in Equation 3- 1.

$$\tau = Tr$$

Equation 3- 1: The equation for torque (Tao) given tension and radius

This equation shows that the torque of the motor is equal to the tension of the wrap being multiplied by the radius of the wrap. The team is only interested in the strongest motor possible, so the radius will be set to the maximum radius of the rolled wrap, which is about 0.4m. Moreover, the tension needed to keep the wrap from folding onto itself will be estimated at a value of 15 Newtons. Given these values, the resulting torque required to roll the wrap comes out to ~0.588Nm. This is just under double the stall torque used in the first working prototype, which was 0.216Nm. As 0.588Nm is the minimum stall torque required to theoretically roll the wrap, the team concludes that the next motor should be ~0.980Ncm, which accounts for any error of calculations.

3.5.4.3 Secondary Research

Further research validates the conclusion that a motor with a stall torque of 0.980Nm is necessary to roll a wrap. This research investigates the reasonability of using 0.4m as the wrap's final rolled radius. To test the reality of such a radius, a stakeholder had allowed the team to borrow their

hand wrap and measure it. The radius of the hand wrap was measured to be about 0.45m. This hand-wrap was thicker than most casual hand-wraps, as it's thickness was about 3mm. This is outside of the optimal range of wrap thicknesses [44]. This secondary research has confirmed that a motor with the torque strength of ~0.980Nm would be optimal, as even a thick hand-wrap is only about 0.45m in radius when rolled. Such a radius still yields a required torque under the current 0.980Nm threshold.

3.5.4.4 Test Results: New Motor

Given the testing for the new motor, 2 hypotheses were evaluated:

- At 12V the motor will exceed 1 Nm of torque
- At 12V the motor can roll a wrap in less than 30 seconds

Each hypothesis aims to determine the effectiveness of the motor in rolling hand wraps in a controlled and expeditious manner. Regarding the torque test, a force gauge sensor was connected to a rod attached to the wrap holding mechanism on the motor. The distance from the axial point to the connection of the sensor and the rod was measured. The distance was 0.37m. The motor was then started, and stalled by the sensor, and a reading of N (newtons) was produced by the sensors. The average of all trials was calculated, and such result was multiplied by the distance to yield torque. Given three trials, the results are observed in Table 3-16 below.

Table 3-16: Shows the measured torque of the motor at different voltages

	Trial 1	Trial 2	Trial 3	Average Torque
12V	3.441Nm	3.404Nm	3.552Nm	3.465Nm
7.4V	3.145Nm	3.145Nm	3.108Nm	3.132Nm
3.7V	2.664Nm	2.812Nm	2.775Nm	2.750Nm

From this table, the motor is well above the required 1Nm of torque at all tested voltages. This means that the first hypothesis is validated. However, the next hypothesis will reveal the drawback of using lower voltages.

For the second hypothesis, the wrap was unraveled and straightened. Then, it was attached to the wrap holding mechanism and the motor was started. The time, in seconds, was measured to determine how long it takes to wrap the wrap. Below in Table 3-17, the results are observed.

Table 3-17: Shows the time required in seconds to roll a wrap using the motor at different voltages

	Trial 1	Trial 2	Trial 3	Average Time
12V	14.11s	12.81s	13.81s	13.58s
7.4V	19.32s	N/A	N/A	19.32s
3.7V	47.90s	N/A	N/A	47.90s

These results show that the voltages of 7.4V and 12V pass the hypothesis, with their wrapping times being well below the required 30 seconds. However, 3.7V did not pass the hypothesis, as the

average time was over 30 seconds. It is also worth noting that the second and third trials were not recorded for 7.4V and 3.7V, as the wrap holding mechanism had broken. Considering the two hypotheses, the tested motor will be more than suffice to roll the wraps in a controlled and expeditious manner.

3.6 Subsystem Interfaces

3.6.1 Interface Descriptions

The final solution starts with the electrical logic. The electrical logic will transfer current from the battery to the entire system, which consists of a slide switch, a normally open button, a USB-C plug, and two 3.7 LiPo batteries. To connect each component, 22 AWG (American Wire Gauge) wires will be used. The main power source of the solution is two 3.7v 2000mAh LiPo batteries. These batteries have been selected due to their simplicity, compact size, and ability to support rapid prototyping. The power source connects directly to the rotational force, which is a 12V 3.465Nm 160rpm worm-gear DC motor. The rotational force will allow for a spinning motion, which will be able to overcome the weight of the unrolled wrap as well as the wrap straightening mechanism. The straightening mechanism will be two prongs attached to the side of the solution, which will allow for the unraveled wrap to be fed under them for straightness and tightness. For the wrap to roll, it must first be connected to the wrap holding mechanism. This mechanism is attached directly to the motor and will take the shape of a 3D printed spindle. Once the wrap is finished being rolled, it may be discharged from the device and stowed for later use. The interfaces between each subsection are designed to adhere to Table 3-18, which depicts the updated requirements of the final solution based on stakeholder suggestions and the team problem.

Table 3-18: Updated Requirements and Specifications

Requirement / Criteria	Design Feature that Addresses Requirement
Reduce time required to roll hand wraps	Motorized spindle automatically rolls the wrap
Produce tight and uniform rolls	Pronged spindle maintains tension and alignment
Portable for gym or home use	Compact circular handheld housing
Simple and intuitive operation	Slide switch power control
Safe for users	Enclosed housing protects moving components
Compatible with standard hand wrap dimensions	Spindle sized to fit typical wrap diameter and length
Durable for repeated use	Rigid protective casing and internal component support
Rechargeable power source	Integrated power supply with USB-C charging port

3.6.2 Subsystem Diagrams

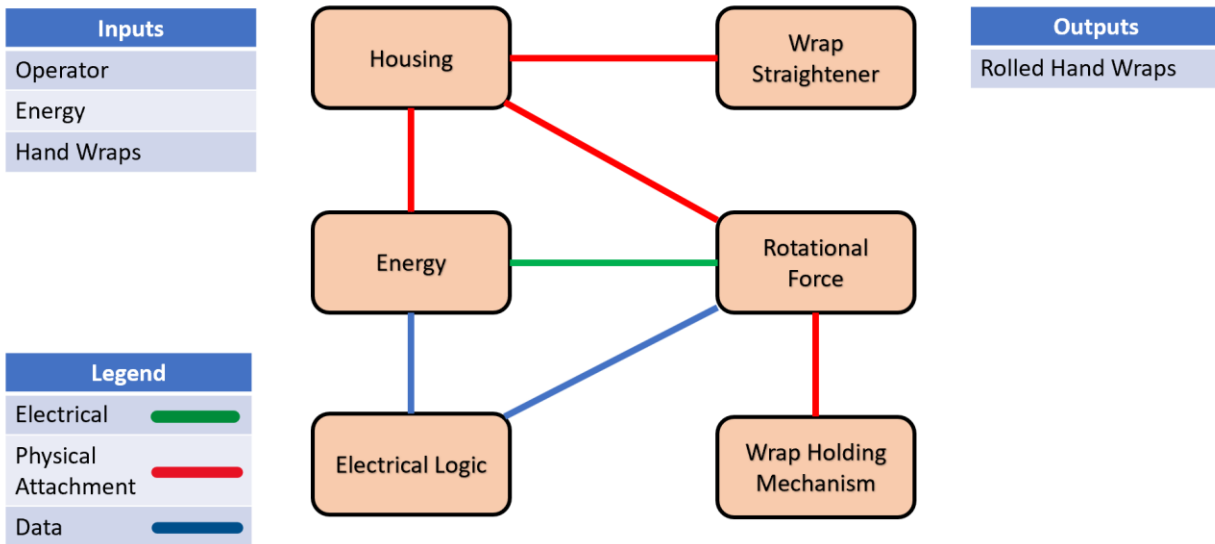


Figure 3-36

3.6.3 Data and Energy Transfer

The functionality of the system relies on the transfer of electrical energy and mechanical output between subsystems. The primary flow of energy begins at the electrical energy subsystem, where the 9.6 V NiMH battery pack stores and supplies electrical power. This energy is transferred through the electrical logic subsystem, which regulates when power is delivered to the motor through the slide switch and momentary pushbutton.

Once activated, electrical energy is supplied to the rotational force subsystem in the form of current delivered to the DC motor. The motor converts this electrical energy into rotational motion, which is transferred directly to the wrap holding mechanism. This mechanical output enables the spindle to rotate and wind the hand wrap.

In addition to energy transfer, user input acts as a form of control data within the system. The position of the slide switch determines whether the system is powered, while the pushbutton provides real-time control over motor activation. This ensures that energy is only transferred when intended, improving both safety and usability.

Subsystem integration required matching the energy supply with motor demand while maintaining compactness and portability. Initial testing with a 3.7 V LiPo battery revealed insufficient power for effective operation. This led to the selection of a 9.6 V NiMH battery pack, which provides improved voltage, resulting in increased motor speed, torque, and overall system performance.

4. Module 4

4.1 Solution Value Proposition

4.1.1 Full Scale Solution Discussion

The full-scale solution meets the established design criteria for ease of use, portability, affordability, durability, and consistent performance. The device operates using a simple user interface consisting of a slide switch and a momentary push button, allowing intuitive control with minimal instruction. Its compact, handheld design enables one-handed operation and convenient use in the gym, home, or training environments.

The system improves rolling consistency using a motor-driven pronged spindle, which maintains alignment and reduces folding during operation. This directly addresses one of the primary shortcomings of existing solutions, where users must manually guide the wrap to prevent uneven rolling. By integrating the straightening and holding mechanisms into a single enclosed system, the device reduces the need for constant user correction.

Compared to manual crank rollers and drill-based solutions, the proposed design is fully self-contained and does not require external tools or attachments. Existing powered solutions often rely on user-owned drills or lack of control over wrap alignment, resulting in inconsistent performance. In contrast, this device provides controlled rotation, consistent tension, and improved reliability through a dedicated motor and internal structure.

The design is also more compelling due to its balance between performance and simplicity. While more complex systems reduce cost, improve reliability, and simplify manufacturing. This makes the device more accessible to the target stakeholders while still meeting functional requirements.

Overall, the proposed solution provides a compact, reliable, and efficient alternative to existing products by combining motorized operation, improved alignment control, and a fully integrated design.

4.1.2 Materials List and Cost for Prototype

Table 4- 1

Component	Description	Quantity	Cost (USD)
12V DC Worm Gear Motor	Provides rotational force	1	\$28.00
3.7V 2000 mAh LiPo Battery	Power source	1	\$9.00
Boost Converter	Steps voltage to 12V	1	\$0.80
USB-C Charging Module	Battery charging	1	\$0.80
Slide Switch	Main power control	1	\$0.35
Push Button (Normally Open)	Motor activation	1	\$0.70
Electrical Wiring	Electrical connections	1 set	\$1.00
PLA Filament	3D printed housing + spindle	1 kg	\$3.00
Misc. Hardware	Screws, connectors, mounts	—	\$1.00

Estimated prototype cost: \$43.95

4.1.3 Materials List and Cost for Final Product

Table 4- 2

Component	Description	Estimated Unit Cost
3.7 V 2000 mAh LiPo Battery	Rechargeable power source	\$6.00
DC Worm Gear Motor	Bulk sourced motor	\$12.00
Boost Converter Circuit	Integrated voltage step-up circuit	\$0.75
USB-C Charging Circuit	Integrated charging circuit	\$0.60
Slide Switch + Push Button	User controls	\$0.75
Wiring and Connectors	Internal electrical connections	\$1.00
Molded Plastic Housing	Final enclosure	\$4.00
Small Hardware	Screws, mounts, and fasteners	\$1.00

Estimated production cost per unit: \$27.60

The final product cost is lower than the prototype cost because production would use bulk component purchasing, integrated circuits instead of individual modules, and molded housing parts instead of individually 3D printed components.

4.1.4 Real Benefits of Solution

The proposed solution provides both measurable and non-measurable benefits to users. The most direct benefit is reduced rolling time. Manual rolling can take around two minutes per wrap, while motor testing shows that powered rolling can complete the process much faster. This saves time for users who train frequently or own multiple pairs of wraps.

The device also improves consistency. The motorized spindle helps create a tighter and more uniform roll compared to hand rolling, especially when the user is tired after training. This makes the wraps easier to store and faster to use during the next training session.

Another benefit is convenience. Unlike drill-based solutions, this device does not require the user to own or carry a separate drill. It is self-contained, rechargeable, and compact enough for home or gym use. The use of a 3.7 V LiPo battery also helps keep the device lightweight and portable.

The final product also has a reasonable cost compared to existing powered solutions. Since the estimated production cost is under \$30 per unit, the product could be priced competitively while still offering improved functionality over manual rollers.

Overall, the real value of the solution comes from reducing time, reducing effort, improving roll quality, and giving users a compact powered device designed specifically for reusable hand wraps

4.2 Risk and Mitigation discussion and table

4.2.1 Full Scale Solution Risks and Mitigations

Due to the nature of the device being motorized with moving parts, there are many possible risks. To bring this product to market a thorough and exhaustive investigation of risk and probability is necessary. Table 4- 3 contains an extensive list of risks along with the potential negative impact from the risks and mitigations taken to prevent them.

Table 4- 3: Risks and mitigation

Risk	Potential impact	Mitigations
Accidentally turning on in storage	• Damage to device • Damage to property	• Separate on/off switch • Recessed power button
Battery meltdown	• Fire • Damage to machine	• Overcharge prevention • Warning label
Spinning mechanism activating during loading	• Hand Injury • Long Hair	• Device will not activate without cylinder cover in place
Short circuit	• Damage to device • Fire	• Secure connections • Insulated conductors
Overcharging	• Battery damage or swelling	• Battery management system
Broken parts	• Damage to device • Sharp edges	• Warning label to dispose if damaged
Heat sensitivity of plastic parts	• Damage to device • New risks from warped device	• Warning label referencing maximum temperature
Fabric jam	• Damage to device • Possible injury	• Motor not powerful enough to jerk too hard

After research into possible risks was completed, a Table 4- 4 was created to determine the severity of each risk, and the probability of occurrence. The Risk Analysis Matrix puts each risk into a category of low, medium, and high risk indicating the severity of the possible outcome. This can include but is not limited to damage to the device and personal injury. Secondly, each risk goes into a probability based on how likely the bad outcome is to occur.

Risk Analysis Matrix

Table 4- 4: Risk Analysis Matrix

Probability				
	Unlikely			
	Moderate			
	Likely			

4.2.2 Risk Mitigation Plan

Precautions were considered or taken for all risks in the final design. Risks which fall into moderate probability, medium severity and higher became an especially high priority to mitigate. An example of risk mitigation includes the inclusion of a cover to prevent hair or hand injury from the turning mechanism.

Additionally, the motor, battery and electrical components are neatly stored in an enclosed housing. This provides protection from dirt debris and prevents fire risk due to component damage. Future iterations of the device will include a warning label referencing the heat tolerance of the device as well as injury risk to the user. Risk with a battery powered electronic device cannot be eliminated. However, careful consideration of the risks can reduce the likelihood of bad outcomes.

5. Module 5

5.1 Final Solution in Real World

5.1.1 Isometric View of Cad Generated Model

The final design of the solution's enclosure is viewed in Figure 5- 1, and a separate isometric view of the wrap holding mechanism is viewed in Figure 5- 2. Orthographic views of the full enclosure and its schematics can be referenced in appendix B.

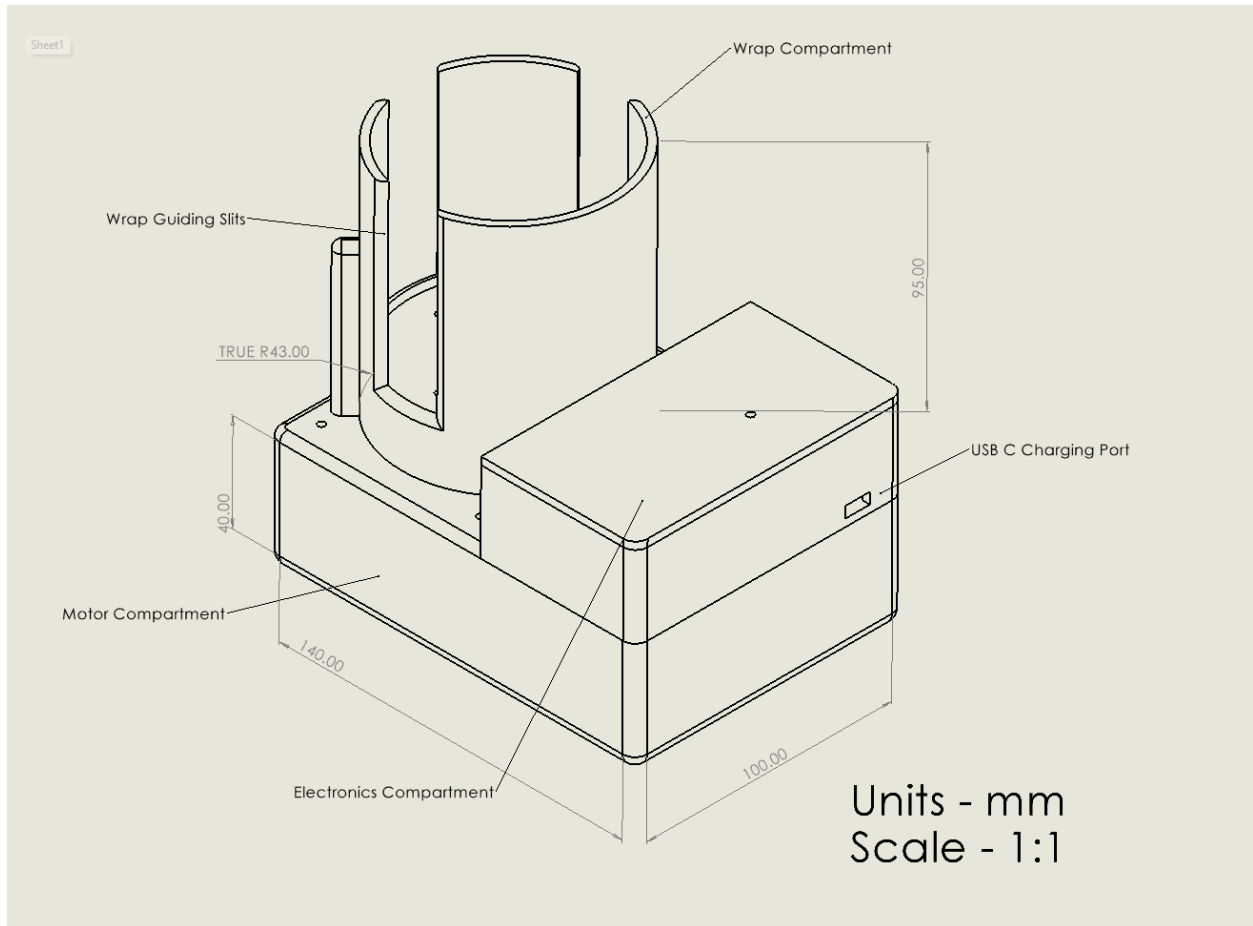


Figure 5- 1: Design

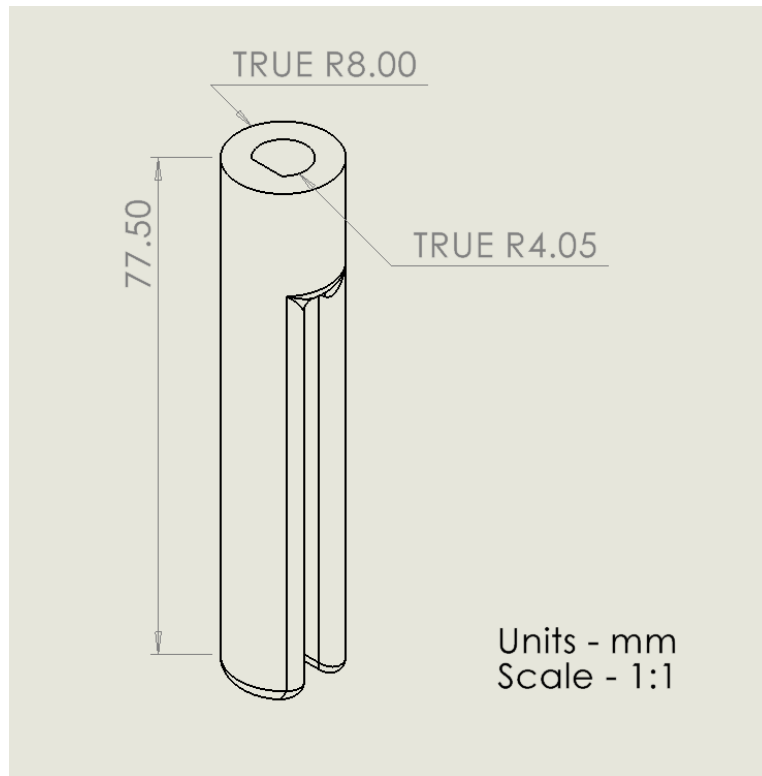


Figure 5- 2: Holding

5.2 Final Solution Design

5.2.1 Materials, Dimensions, and Weight

The final solution will be printed in PETG for its high-impact durability as well as its heat resistance. When printing the enclosure, about 30% infill will be used. Based on the CAD generated model of the enclosure, about 465 grams of filament will be used at 100%. This mass does not include the wrap holding mechanism. At 30% infill, the new total amount of filament used is about 140 grams.

The wrap holding mechanism will utilize PETG filament at 30% infill. This will cause the weight to be about 4 grams. In addition to these subsystems, screws and washers as well as wires will be utilized in the final solution, however their weights are negligible and will not be evaluated further.

Regarding the specifications of the final solution, Figure 5- 3 and Figure 5- 4, both shown below, visualize the broad dimensions of the enclosure.

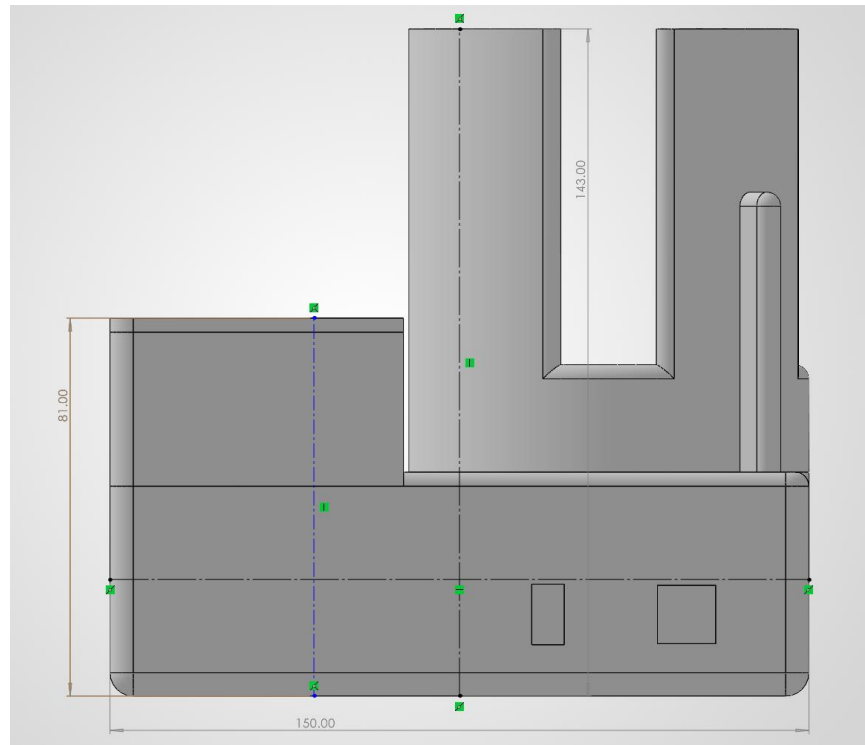


Figure 5- 3: Side

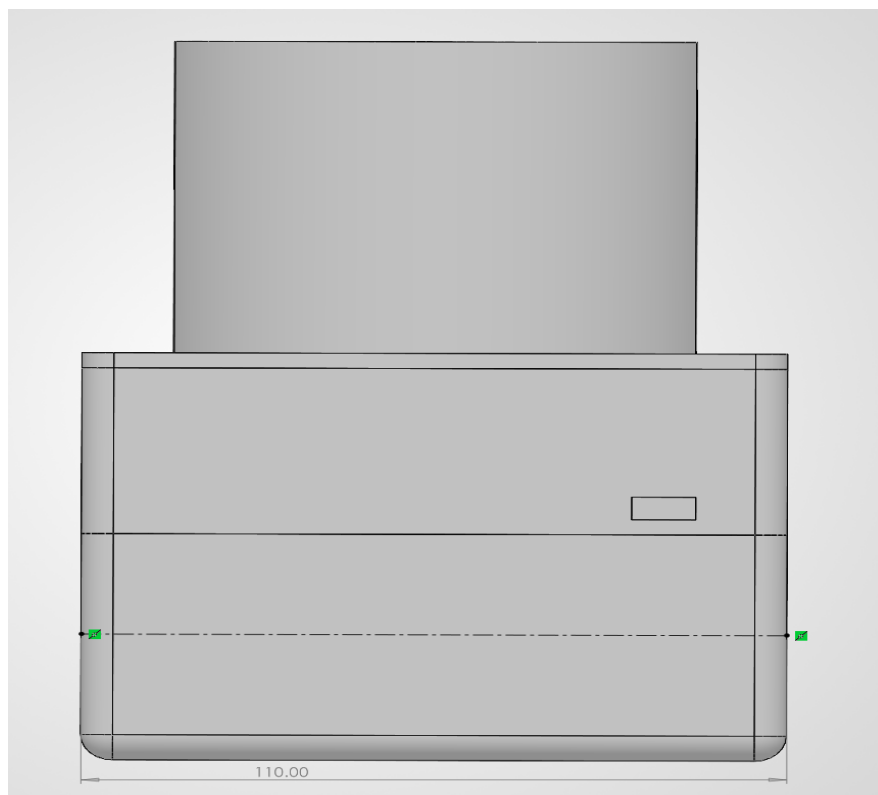


Figure 5- 4: Back

The above figures do not address the radius of the wrap holding compartment, which is 40mm without the walls and 43mm with them. Next, the size of the spindle is observed in Figure 5- 5 below.

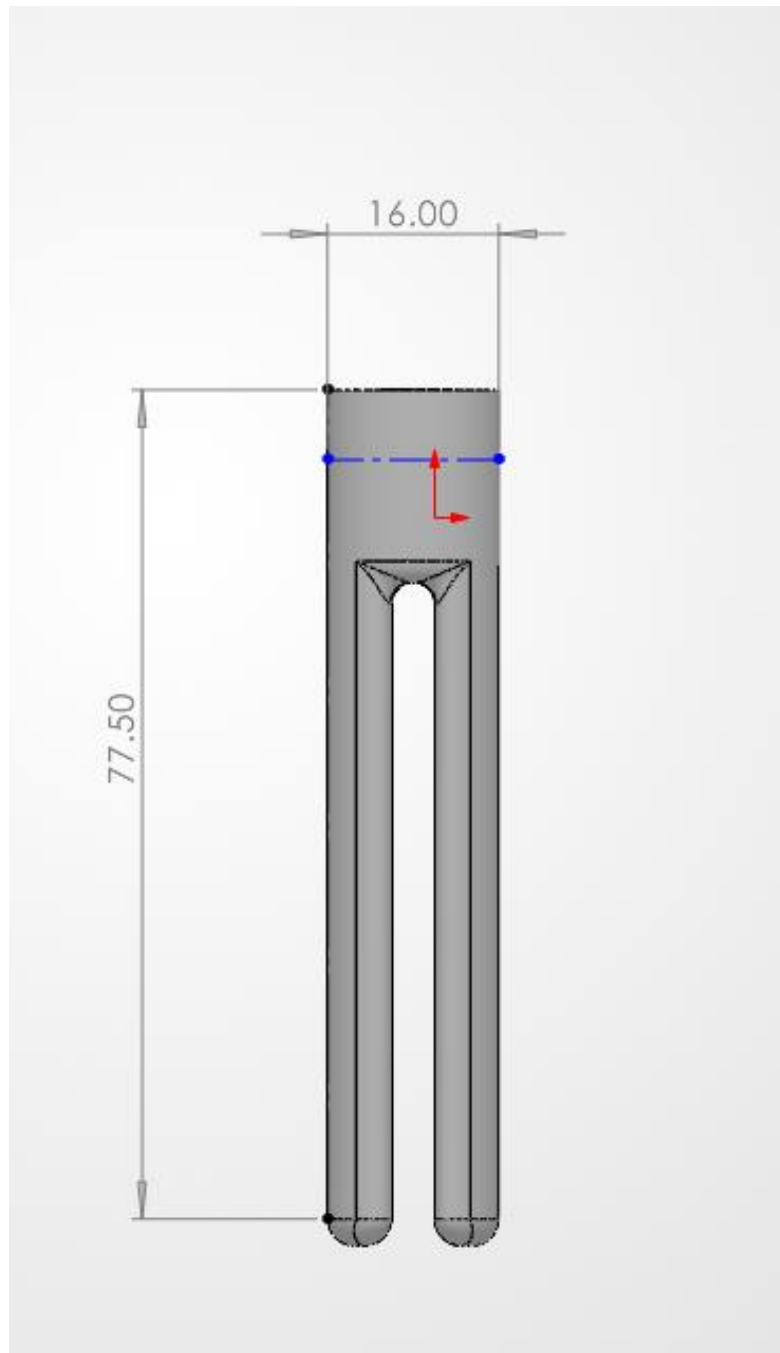


Figure 5- 5: Spindle

5.2.2 Full-Solution CAD Model

Below are two figures representing the CAD model of the final solution: Figure 5- 6 and Figure 5- 7. These models include the enclosure of the solution as well as the wrap holding mechanism. None of the models will include any dimensions or labels.

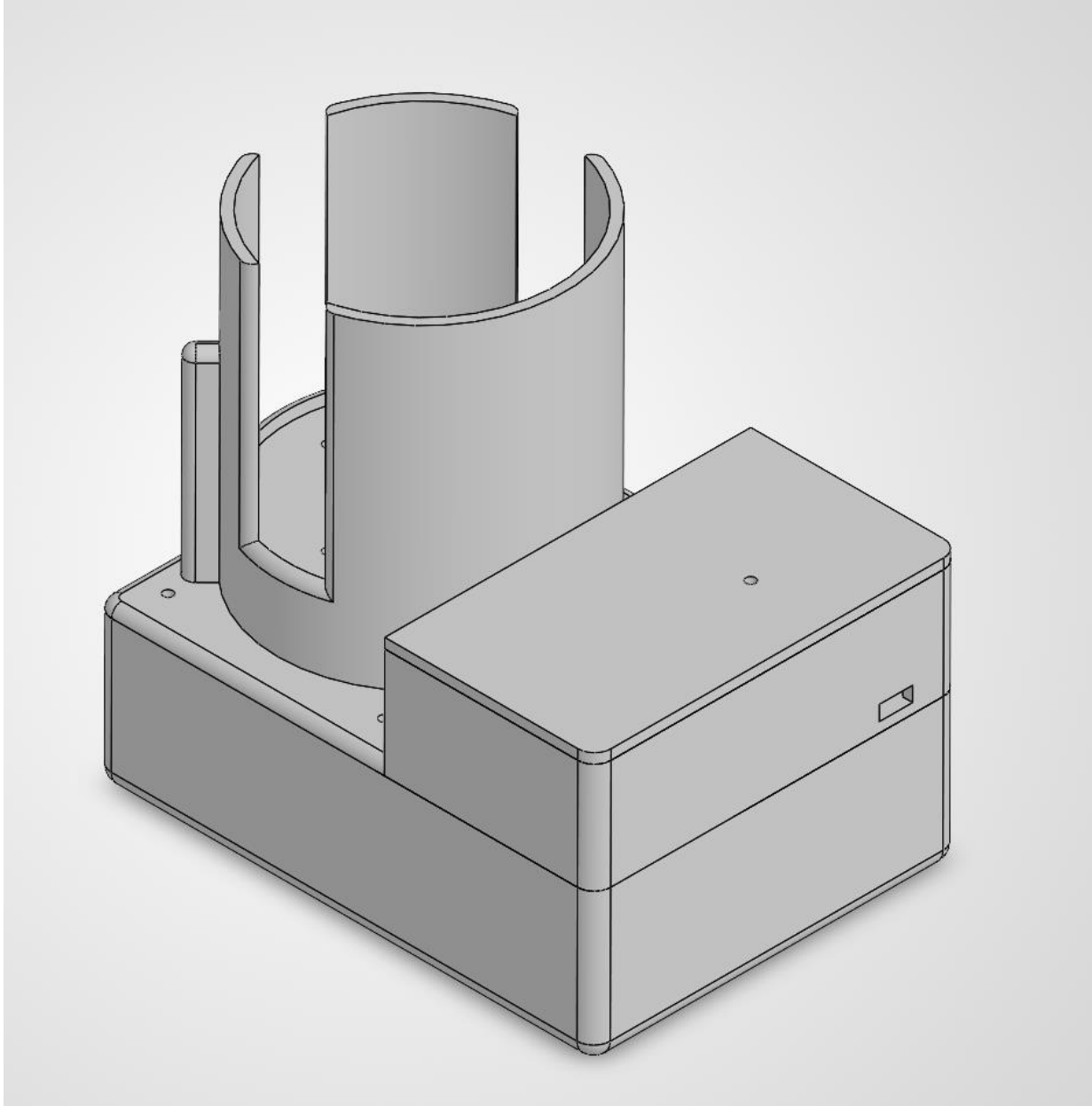


Figure 5- 6: Isometric

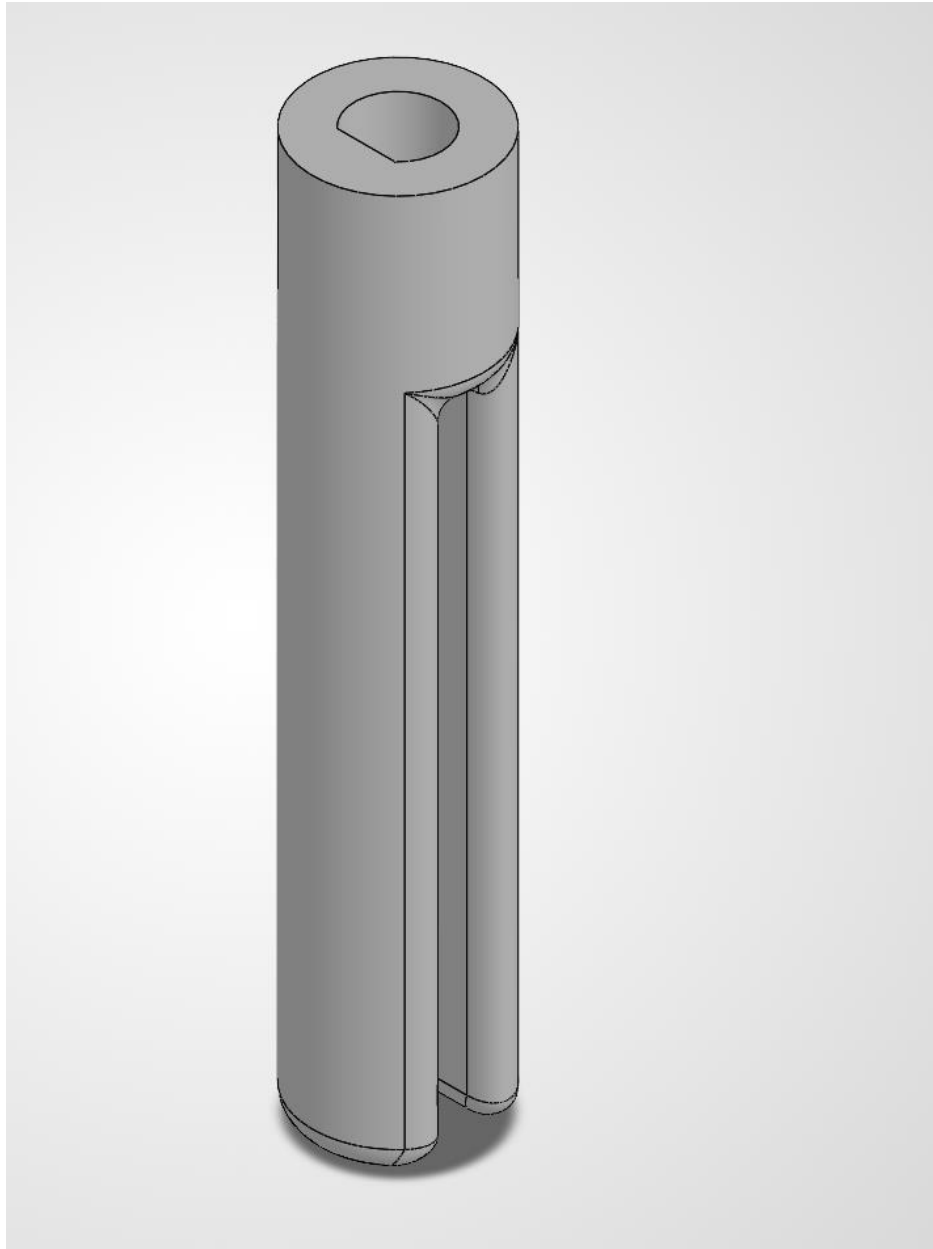


Figure 5- 7: Spindle Isometric

5.3 Concept Validation

5.3.1 Version Control of Final Solution

The final solution changed a lot throughout the design process as each subsystem was tested and reviewed. Early ideas included a wall-mounted roller, a drill-shaped roller, a gun-shaped roller, a circular handheld roller, and multiple versions of a straightening or tensioning mechanism. The final

design kept the strongest parts of these ideas, including the circular handheld shape, powered rolling system, enclosed housing, slide switch, push button, and pronged spindle.

One major change was removing the integrated tensioning mechanism from the final design. This subsystem worked during individual testing and helped keep the wrap straight, but it could not be integrated into the final enclosure in the way originally planned. It added extra moving parts, made the wrap harder to load, and made the device more complicated than needed. Because of this, the final solution uses the pronged spindle and user-guided feeding instead.

Other concepts were also not included in the final version. The wall-mounted design was not used because the final solution needed to be portable. The drill-based design helped show that powered rolling was useful, but requiring the user to own a drill made the design less practical. The detachable claw from the gun-shaped design was also left out because it expanded the scope beyond martial arts hand wraps. The final solution focuses on the main stakeholder needs of reducing rolling time, improving consistency, staying portable, and keeping the device simple.

5.3.2 Scholarly and Authoritative Research

Scholarly and authoritative research helped support both the problem and the design choices. Research on hand and wrist injuries in combat sports helped validate the importance of hand wraps as protective equipment. While there is still limited research on exactly how much protection hand wraps provide, existing studies show that hand wraps can absorb impact and are commonly used by combat athletes [1], [2]. This supports the need for a device that helps keep wraps ready for reuse.

Research on compression bandages and reusable wraps also showed that wrap performance depends on handling, stretch, and tension. If a wrap is stored loosely or unevenly, it may be harder to apply later in a consistent way [4], [5]. This supported the team's goal of creating a tighter and more compact roll.

Research on hygiene also affected the final design. Interviews and textile research showed that moisture and sweat can promote bacterial growth on fabric [14]–[19]. Because of this, the device is intended to be used after wraps have been washed and dried. It helps with rolling and storage, but it does not replace proper cleaning.

Component research also guided the final motor and wiring choices. The original N20 motor was too weak for the system, so the team moved to a stronger 12 V worm gear motor [42], [43]. Research on wire gauge also supported the use of 22 AWG wire because it could handle the expected current while still fitting inside the compact housing [40].

5.3.3 Testing

Testing was used to validate the main functions of the final solution. The first motor tested was a small N20 motor. This motor was useful for early prototyping because it was small and lightweight, but it did not have enough torque to roll the wrap effectively. It struggled to keep the wrap moving when tension was applied, so the team decided that a stronger motor was needed.

The updated worm gear motor was tested at different voltages. At 12 V, the motor produced an average torque of 3.465 Nm and rolled a wrap in an average time of 13.58 seconds. This met the team's goal of rolling a wrap in less than 30 seconds. At 7.4 V, the motor still rolled the wrap in 19.32 seconds,

but 3.7 V took 47.90 seconds and did not meet the target. These results showed that the 12 V setup gave the best overall performance.

The wrap holding spindle was also tested. A hollow printed spindle failed at 37.2 N and 38.7 N, while the solid spindle reached 80 N without breaking. This showed that the spindle should be stronger rather than lighter, because it needs to handle repeated force during rolling.

The straightening mechanism was tested separately through multiple versions. Later versions were able to guide the wrap and reduce folding, but the system was difficult to integrate into the final enclosure. Because of that, the tensioning mechanism was not included in the final design, even though the testing showed it could be useful in a future version.

5.3.4 Analysis and Conclusions

The main analysis focused on whether the device had enough torque and speed to roll a hand wrap. Using the torque equation, the team estimated that the motor needed at least 0.588 Nm of torque, with a target of about 0.980 Nm to allow room for error. The original N20 motor did not meet this need, but the updated worm gear motor produced 3.465 Nm at 12 V. This showed that the new motor had more than enough torque for the design.

The time testing also supported the final motor choice. Since the motor rolled a wrap in 13.58 seconds at 12 V, it met the goal of being faster than manual rolling. This supports the claim that the device can reduce the time needed to prepare wraps for storage or future use.

The spindle testing showed that the final wrap holding mechanism should prioritize strength. The hollow spindle saved material but broke below 40 N. The solid spindle did not break at 80 N, so the stronger printed design was the better option.

Overall, the testing and analysis show that the final design is feasible. The motor has enough torque, the spindle is strong enough, and the simplified enclosure makes the device easier to build. The tensioning system may be useful in the future, but removing it from the current design made the final solution more realistic and easier to use.

5.3.5 Stakeholder Interviews & Feedback

Stakeholder feedback guided the final solution throughout the project. Early interviews showed that hand rolling wraps can be slow and repetitive. One fighter said that rolling by hand takes around one minute per wrap, while using a tool takes around 30 seconds per wrap. This helped make time reduction one of the main design goals.

Other feedback showed that roll quality was also important. Lyam Hassell explained that rolling a 16-foot wrap can be slow because the wrap often folds on itself [24]. This supported the use of a pronged spindle and helped guide the early straightening mechanism designs.

Stakeholders also helped guide the overall form of the product. The wall-mounted design helped with drying and storage, but the final product needed to be more portable. Cost was another concern, so the final design was simplified by removing extra mechanisms and keeping the number of parts lower [27]. Overall, the final design reflects the feedback that the product should be fast, portable, simple, and affordable.

5.4 Conclusion

The final proposed solution is a compact motorized hand wrap roller designed to make re-rolling martial arts hand wraps faster and easier. The device uses a PETG enclosure, a DC worm gear motor, a pronged spindle, and simple electrical controls to roll wraps into a compact shape. It is intended to be used after wraps have been washed and dried, so they can be stored and prepared for the next training session.

The problem was validated through stakeholder interviews, research, and testing. Users explained that hand rolling wraps can be slow, repetitive, and inconsistent, especially when they own multiple sets of wraps. Existing products can help, but many still require careful feeding or outside tools.

The final design was improved through several rounds of testing. The original N20 motor was too weak, so the team selected a stronger 12 V worm gear motor. This motor produced 3.465 Nm of torque and rolled a wrap in 13.58 seconds at 12 V. Spindle testing also showed that a stronger printed spindle was needed, since the hollow design failed below 40 N while the solid version reached 80 N without breaking. Due to time constraints and difficulty in integration the straightening system was not included in the final design.

Overall, the final solution addresses the main project goals of reducing time, reducing effort, improving roll consistency, and staying portable. The enclosed housing also improves safety by covering the moving parts, while the simple switch and button controls make the device easy to use. More full-system testing would be needed before manufacturing, but the current design is a realistic and useful response to the team's problem statement.

6. References

- [1.] M. Lucius Pomerantz, "Hand and Wrist Injuries in Mixed Martial Arts: A Guide to Diagnosis and Treatment," *Hand and Wrist Injuries In Combat Sports*, pp. (pp.63-76), Aug. 2018, doi: https://doi.org/10.1007/978-3-319-52902-8_7.
- [2.] E. M. Gusnowski, Determination of the protective capacity of hand wraps in combat athletes using force analysis, Master's thesis, The University of Western Ontario, London, ON, Canada, 2023. [Online]. Available: <https://ir.lib.uwo.ca/etd/8894/>
- [3.] A. Martin, "Interview regarding hand injury experience and use of protective wraps," in-person interview, Feb. 2026.
- [4] Y. Haririan and A. Asayesh, "Characterization of the structural configuration and tensile properties of medical bandages," *The Journal of The Textile Institute*, vol. 113, no. 8, pp. 1595–1602, 2022.
- [5] C. Kwon, M. Hegarty, W. Oxenham, K. Thoney-Barletta, E. Grant, and L. Reid, "An indirect testing approach for characterizing pressure profiles of compression bandages and hosiery," *The Journal of The Textile Institute*, 2017.
- [6] S. A. Naimer and F. Chemla, "Elastic adhesive dressing treatment of bleeding wounds in trauma victims," *American Journal of Emergency Medicine*, vol. 18, no. 7, pp. 816–819, 2000.
- [7] Mye Choy, "Interview regarding elastic compression bandage storage and deployment practices," personal communication, Feb. 2026.
- [8] Name Withheld, "Interview regarding reusable elastic compression bandage use and storage," personal communication, Feb. 2026.
- [9] Lake, Jason P.; Carden, Patrick J.C.; Shorter, Kath A.. Wearing Knee Wraps Affects Mechanical Output and Performance Characteristics of Back Squat Exercise. *Journal of Strength and Conditioning Research* 26(10):p 2844-2849, October 2012. | DOI: 10.1519/JSC.0b013e3182429840
- [10] The What, Why, and How of POLO WRAPS. *Practical Horseman*. 2006;34(9):82-84. Accessed February 15, 2026. <https://research-ebsco-com.ccco.idm.oclc.org/linkprocessor/plink?id=b9660fe8-bd3c-3291-bac6-5b83dd264acc>
- [11] Altaweel, Ali1; Alabbad, Majed A.2,. The influence of athletic taping and bracing on ankle sprain: A literature review. *Saudi Journal of Sports Medicine* 20(2):p 36-39, May–Aug 2020. | DOI: 10.4103/sjsm.sjsm_15_20
- [12] "Knee Wrap Roller- 01 NEW" by Clay Butler on Youtube.com https://youtu.be/aq_uLHyR34
- [13] "How to Wrap a Horse's Leg" by Michigan State University on YouTube.com <https://www.youtube.com/watch?v=GicfRy-70V4>
- [14] J. Portillo, "Interview regarding boxing hand wrap storage and post-training handling practices," in-person interview, Feb. 2026.

- [15] L. Albert, "Interview regarding microorganisms and hygiene concerns related to athletic hand wraps," in-person interview, Feb. 2026.
- [16] A. Martínez Nova, F. Marcos-Tejedor, B. Gómez Martín, R. Sánchez-Rodríguez, and E. Escamilla-Martínez, "Bioceramic-fiber socks have more benefits than cotton-made socks in controlling bacterial load and the increase of sweat in runners," *Textile Research Journal*, vol. 88, no. 6, pp. 696–703, 2018, doi: 10.1177/0040517516688631.
- [17] L. Teufel, A. Pipal, K. C. Schuster, T. Staudinger, and B. Redl, "Material-dependent growth of human skin bacteria on textiles investigated using challenge tests and DNA genotyping," *Journal of Applied Microbiology*, vol. 107, no. 1, pp. 1–12, 2009.
- [18] H. Yan, Y. Ren, B. Zhou, F. Ye, and Z. Wu, "Microbial profile of T-shirts after a fitness session of Chinese students," *Heliyon*, vol. 8, no. 12, p. e12379, 2022, doi: 10.1016/j.heliyon.2022.e12379.
- [19] F. R. Omi, M. N. S. Shizan, H. B. Habib, and M. S. Uddin, "Microbial contamination and antibiotic resistance in athletic undergarments: A cross-sectional study among university athletes," *Journal of Advances in Microbiology Research*, vol. 6, no. 1, pp. 308–315, 2025, doi: 10.22271/micro.2025.v6.i1d.227.
- [20] - YouTube. https://www.youtube.com/shorts/MthmZb_GZN8?feature=share. Accessed 17 Feb. 2026.
- [21] Anon, "About rolling Muay Thai handwraps...," 2025. https://www.reddit.com/r/MuayThai/comments/1ljeben/about_rolling_muay_thai_handwraps
- [22] Anon, "whats the most easiest and fast way to roll up hand wraps after use," 2024. https://www.reddit.com/r/MuayThai/comments/1c8rz7n/whats_the_most_easiest_and_fast_way_to_roll_up/
- [23] SpeedFast Combat, "How to Use the SpeedFast Recoil Hand Wrap Roller – Quick & Easy!," YouTube, Feb. 04, 2025. <https://www.youtube.com/watch?v=j49g8w8qPxs> (accessed Mar. 13, 2026).
- [24] L. Hassell "Interview regarding hand wraps for martial arts and maintenance between training sessions." In-person interview. Feb. 2026.
- [25] "Boxing: number of participants U.S. 2017," Statista, Feb. 2025. <https://www.statista.com/statistics/191905/participants-in-boxing-in-the-us-since-2006/>
- [26] *List Of Muay Thai boxing gyms in United States*. (2025, October 15). SmartScraper. <https://rentechdigital.com/smartscraper/business-report-details/list-of-muay-thai-boxing-gyms-in-united-states>
- [27] J. Celis-Portillo, "Opinion of Proposed Solution," Mar.2026
- [28] "Prachwork Portable Hand Wrap Roller v2.0 & Laundry Mesh Bag for Boxing, MMA, Muay Thai Wraps," Amazon.com, 2026. https://www.amazon.com/product-reviews/BoBGRCBCJ3/ref=cm_cr_arp_d_viewopt_kywd?ie=UTF8&pageNumber=1&filterByKeyword=portable
- [29] Abby, "Nice Portable Hand Roller," Feb. 02, 2023. https://www.amazon.com/product-reviews/B0BGRCBCJ3/ref=cm_cr_arp_d_viewopt_kywd?ie=UTF8&pageNumber=1&filterByKeyword=portable (accessed Mar. 06, 2026).

[30] T. Micciche-Hall, "Recreated Conditions of Problem," [www.youtube.com](https://www.youtube.com/shorts/vt8S8ldHMJM). 2026. Accessed: Mar. 13, 2026. [Short]. Available: <https://www.youtube.com/shorts/vt8S8ldHMJM>

[31] in Los, "Lobos Boxing Club: The best boxing gym in Los Angeles," *Lobos Boxing Club: The best boxing gym in Los Angeles*, Mar. 17, 2023. <https://www.lobosboxingclub.com/blog/how-to-clean-your-boxing-gloves>

[32] "Meister Omega Portable/Mounted Hand Wrap Roller - Fits All Boxing & MMA Wraps ," *Amazon*. https://www.amazon.com/Meister-Omega-Portable-Mounted-Roller/dp/B0D1DCVQHZ/ref=sr_1_34?crid=1S8YHCQJ38VSS&dib=eyJ2IjoiMSJ9.I07qWssmtSWucFUmp_-ICwM-xl1T6ZBWZBOFOA_8MAuAA7H6ZL0z3wwxqVTI-yBiYjdAzORFoxOBWI8C2bCWx9Kx3YE7aXHkMs7NaY023lak888_oyqTWn_DijptbEUrkvij5a-1RYEkwHPct84RpFapIHXg-B2ySA-aMjXXISKnRH9YRPuX9iPvX5eK9Hk9p3OJ6tDp9diFt5RTQk6wdDh5P0UOLAXEdF0GxZm_lg3XlrPdsKHPeLn_pmdqguvRJ1gsNsS1stJ1oYbBfO3pTEdYj8KLilg6t9VuoLE0aQ.99-3hWyT08GW2_9dkp0Nh-0QBv1Ei2Hps1c5YkF96w8&dib_tag=se&keywords=wrap+roller+meister&qid=1773348359&srefix=wrap+roller+meister%2Caps%2C212&sr=8-34 (accessed Mar. 2026).

[33] J. Celis-Portillo, "Opinion of Wall Mounted Roller," Feb.2026

[34] Juvinall, R. C., & Marshek, K. M. (2020). *Fundamentals of Machine Component Design* (7th ed.). Wiley.

[35] J. Celis-Portillo, "Review of Holding System Final Prototype" Apr.2026.

[36] Ghaznavi Youvalari, A., Alizadeh Kaklar, J. & Mohamadi, M. Investigation of mechanical properties in PLA, ABS and epoxy resin parts fabricated by 3D printing technology. *Sci Rep* **15**, 27777 (2025). <https://doi.org/10.1038/s41598-025-13866-8>.

[37] "Mr. Pen- Wire Hangers, 10 Pack, Metal Hangers, Wire Hangers for Clothes, Wire Coat Hangers, Metal Clothes Hanger, Silver Hangers, Shirt Hangers, Slim Hangers, Thin Hangers, Standard Hangers," *Amazon*. https://www.amazon.com/Mr-Hangers-Clothes-Hanger-Standard/dp/B08KGZY2Z1/ref=sr_1_6?dib=eyJ2IjoiMSJ9.J7_jlsQwEJWel6-ROYURUr2W6Rvl6weG2YdDj-JsCntkDZ5Nlx5g9i2ewHsd_YNa1wiQtrcAcyqIWk00Q-CuzZjJGrgKi3wvTXAvPbSumpEoJqAPEXc6HOjiWmPGLaJX0Bs8F3IZQulHLC3nMnJmwoYF23Umy9jkM62z-WcWx2KSZTzQ49y2UEY9lqO-P6waZaF3qDfxQzOJMYzbJcSE8dLX1yx3KelclR5JZTh1pzRqN8JtbqfyvCsSfW_r6m-lcJUSloUVnXFjqghzHX0HXDLKP7N11Y9J6Asnf_XqAg.NL8fNpXMIj2d9a1yMwKW-GD8qb63JkX7HTs9UnXrzvA&dib_tag=se&keywords=wire%2Bcoat%2Bhangers&qid=1775659972&sr=8-6&th=1

[38] "Scotch Super 33+ Vinyl Electrical Tape, Black, 3/4 in. x 52 ft.," *Walmart*. https://www.walmart.com/ip/Scotch-Super-33-Vinyl-Electrical-Tape-Black-3-4-in-x-52-ft/39138902?wmlspartner=wlp&selectedSellerId=0&wmlspartner=wlp&cn=FY25-ENTP-PMAX_cnv_dps_dsn_dis_ad_entp_e_n&gclsrc=aw.ds&adid=222222229739138902_0000000000_21407473164&wl0=&wl1=x&wl2=c&wl3=&wl4=&wl5=9028749&wl6=&wl7=&wl8=&wl9=pla&wl10=8175035&wl11=online&wl12=39138902&veh=sem&gad_source=1&gad_campaignid=21690411341&gbraid=0AAADmfBIqB7eDxZrN0HAJnD6pJo9SDa&gclid=CjwKCAjw-dfOBhAjEiwAq0RwI5OIBta1d91tG-cBsWLSphW7IVxtkRIntErSbVQQ1WURp8wkbLhNWBocNiUQAvD_BwE

- [39] “Everlast Core Handwraps 120in – Boxing Hand Wraps for Gloves, Wrist & Knuckle Support ,” Amazon. https://www.amazon.com/Everlast-P00003251-Core-3-Pack-Handwraps/dp/B0C7HJLDZY/ref=sr_1_8?c=ts&dib=eyJ2IjojMSJ9.LC5MMMax-meEqrb3N-aylr_0AL4Pac_tegBzzy1AJShQsQvz30LBMmcmZ8cARYD3x-1A4fcS0efTgAh9rkMSF9qpLOFhO8LI8GB_V4OcnppXandiwH00o67AMjGJQe-znRRjsbCQa6_IBZjRiYK7ld_ap8m6qlfxXJSwCStK5HXwfbMHYZtz5V-B4-KdOsv4oOzRfGyBaw6ya-PkWkdyygzoau5iheGufgFq8mcc1fmr40BYCW3RKL_dK1by9NflrGSWa2weV1UK1Z32XzwkRcwQw00JuNrPZ0ZS_UN5AoDw.FOEXrtBSNEsEqrLSrbLgupWRlbn10f6JwgHD5oiKM2M&dib_tag=se&keywords=Martial%2BArts%2BHand%2BWraps&qid=1775661356&s=sports-and-fitness&sr=1-8&ts_id=3415251&th=1
- [40] “American Wire Gauge Chart and AWG Electrical Current Load Limits Table with Ampacities, Wire Sizes, Skin Depth Frequencies and Wire Breaking Strength.” *Powerstream.com*, 2026, www.powerstream.com/Wire_Size.htm?utm_source=chatgpt.com. Accessed 8 Apr. 2026.
- [41] Vaughn, Greg; Golterman, Sam. Demonstration of straightening subsystem. Conducted by Max Golterman, Friday, March 2027.
- [42] “Amazon.com: CYTGOUPER DC Brush Mini Gear Motor,N20 Gear Motor 1218GE-N20,Reduction Mini DC Motor for DIY Engine Toy(DC 6V 70RPM) : Toys & Games,” *Amazon.com*, 2026. https://www.amazon.com/dp/B0DR2XW19N?ref=ppx_yo2ov_dt_b_fed_asin_title.
- [43] “Amazon.com: BRINGSMART 12V 160rpm DC Worm Gear Motor 10kg.cm Self-Locking Engine Reversed Mini Turbine Geared Motor for DIY Robot Rotating Table Door Lock Curtain Machine (12V 160rpm) : Electronics,” *Amazon.com*, 2026. https://www.amazon.com/gp/product/B07F8RYKB8/ref=ox_sc_saved_title_6?smid=A1SVSKKUI8ZCJ7&th=1 (accessed Apr. 11, 2026).
- [44] Evangelos Papadimas, “Boxing- Types of Hand Wraps Are Best for Additional Glove Protection,” *Paragon Elite Fight Group | Premium Boxing & BJJ Equipment*, 2025. <https://paragonelitefight.com/blogs/martial-arts-educational-bjj-boxing/boxing-types-of-hand-wraps-for-glove-protection> (accessed Apr. 04, 2026).
- [45] “TITLE Boxing Wrap Wrench,” *TITLE Boxing*, 2026. <https://www.titleboxing.com/products/title-boxing-wrap-wrench?variant=43566448181463&country=US&srsId=AfmBOosx8AgusKzTA4p6QeIRtqXadaac7PVIW3S9gIV4dD38a50Bt-AInk> (accessed May 10, 2026).

7. Appendices

7.1 Appendix A: CAD Drawings

Within this appendix, an exploded view of the final solution is presented. In addition to this, CAD drawings of each individual component is included. The exploded view is shown in figure 7-1; the Component Cover Top is shown in figure 7-2; the wrap enclosure bottom is shown in figure 7-3; the wrap and component enclosure is shown in figure 7-4; and the motor enclosure is shown in figure 7-5. The spindle used in the final solution is designed to be interchangeable. Because of this, the spindle will be given its own CAD drawing that is separate from the enclosure. This drawing is shown in figure 7-6

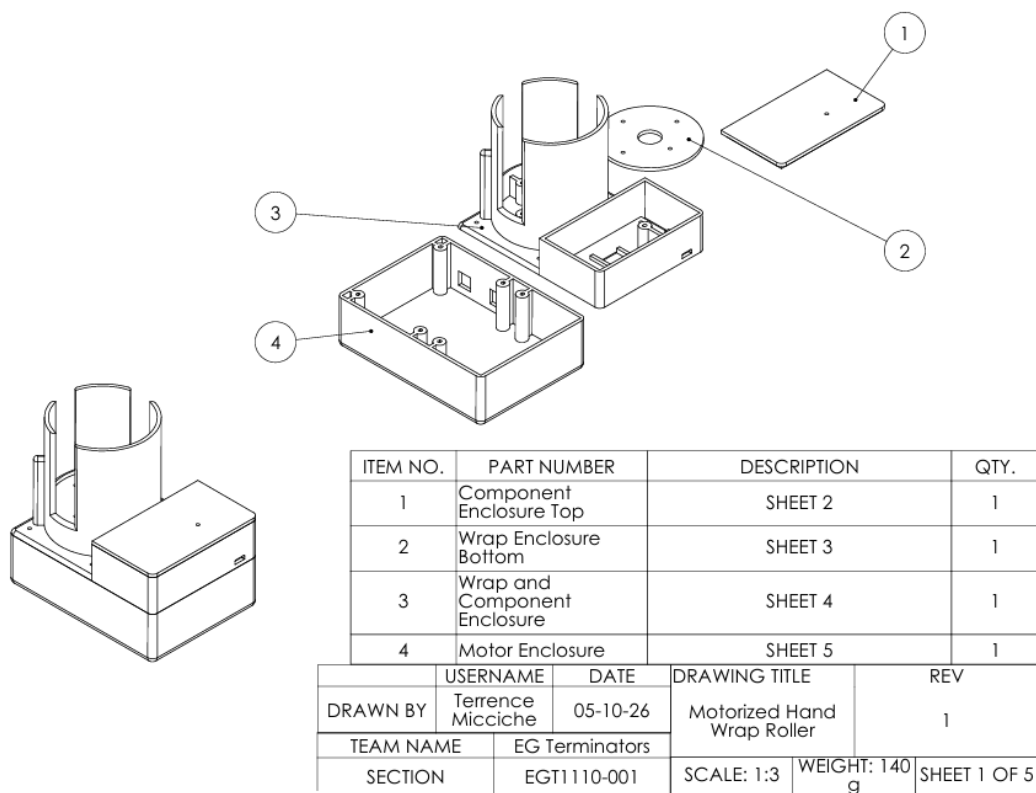
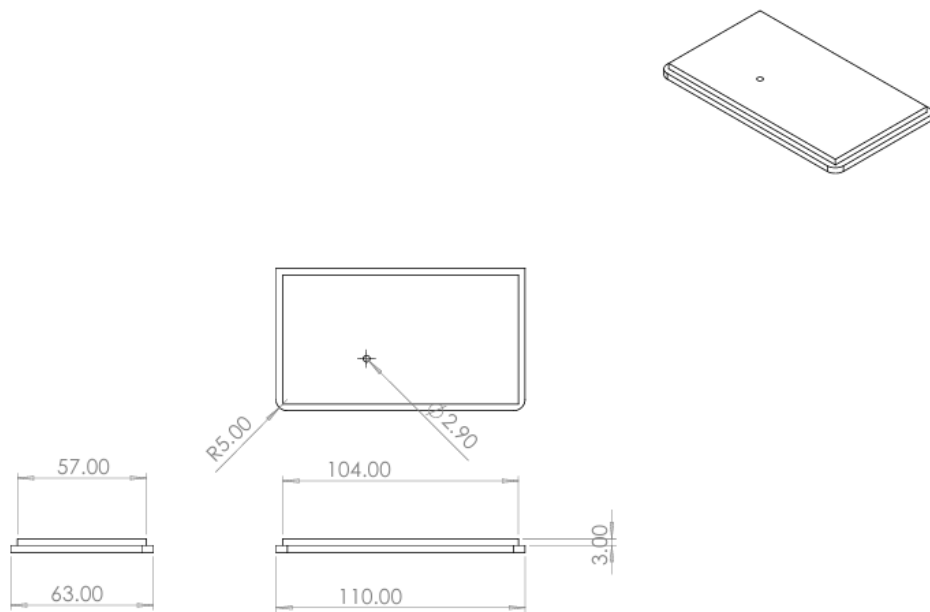


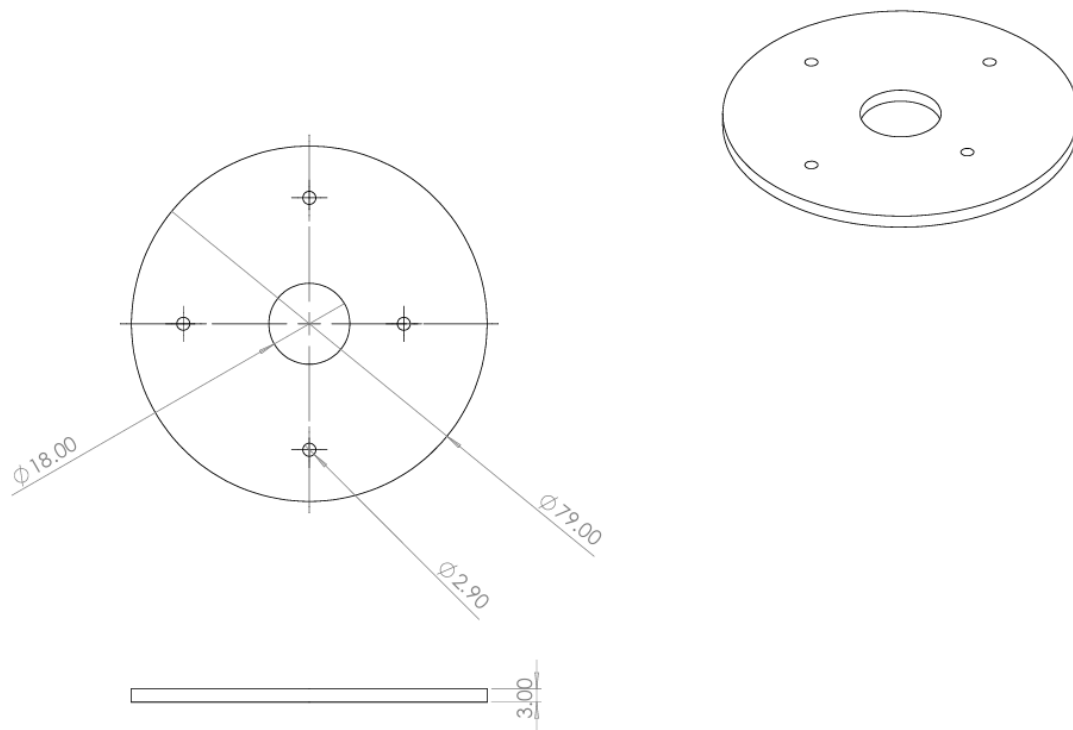
Figure 7-1 – Exploded System Diagram of the Motorized Hand Wrap Roller



	USERNAME	DATE	DRAWING TITLE	REV
DRAWN BY	Terrence Micciche	05-10-26	Component Enclosure Top	1
TEAM NAME	EG Terminators			
SECTION	EGT1110-001	SCALE: 1:2	WEIGHT: 15 g	SHEET 2 OF 5

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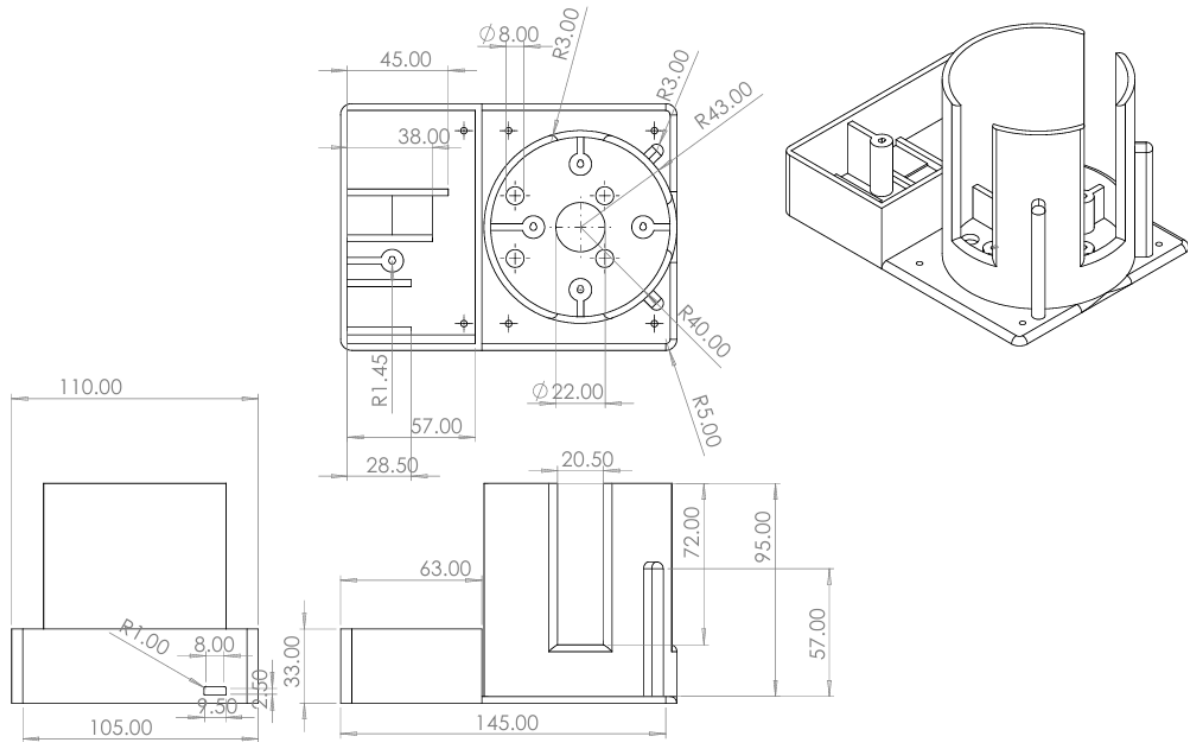
Figure 7-2 – Component Enclosure Top



	USERNAME	DATE	DRAWING TITLE	REV
DRAWN BY	Terrence Micciche	05-10-26	Wrap Enclosure Bottom	1
TEAM NAME		EG Terminators		
SECTION		EGT1110-001	SCALE: 1:1	WEIGHT: 6 g
			SHEET 3 OF 5	

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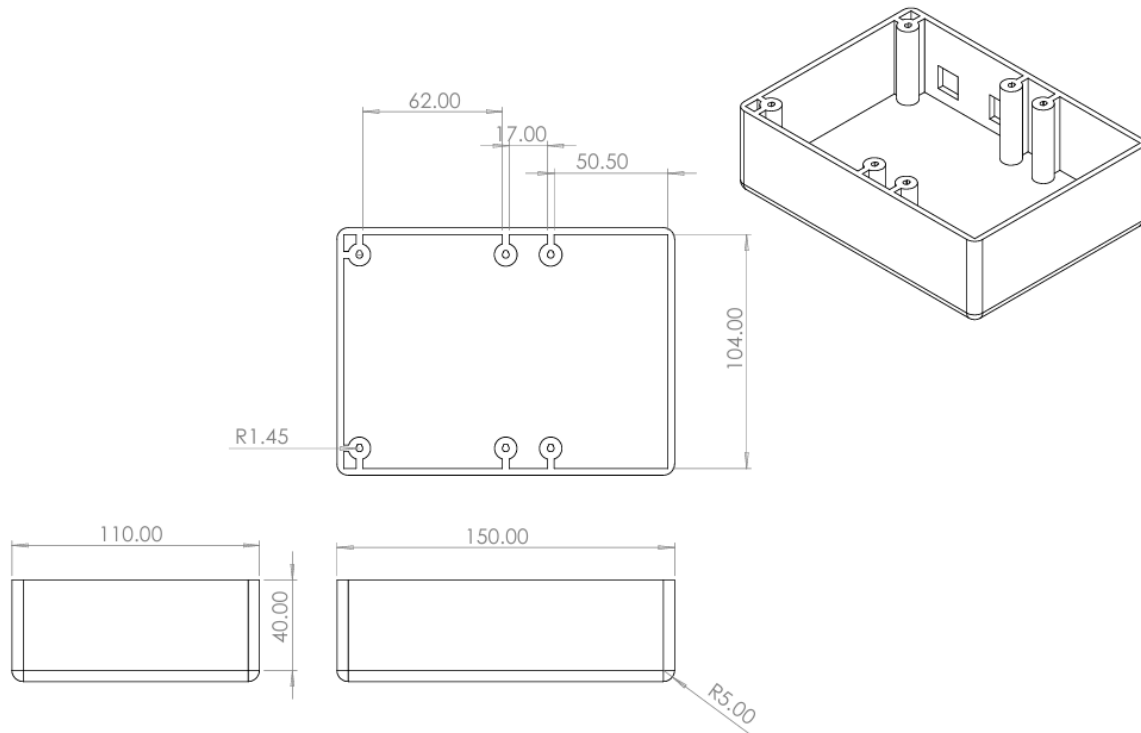
Figure 7-3 – Wrap Enclosure Bottom



	USERNAME	DATE	DRAWING TITLE Wrap and Component Enclosure		REV 1	
DRAWN BY	Terrence Micciche	05-10-26				
TEAM NAME		EG Terminators				
SECTION		EGT1110-001		SCALE: 1:2	WEIGHT: 58 g	SHEET 4 OF 5

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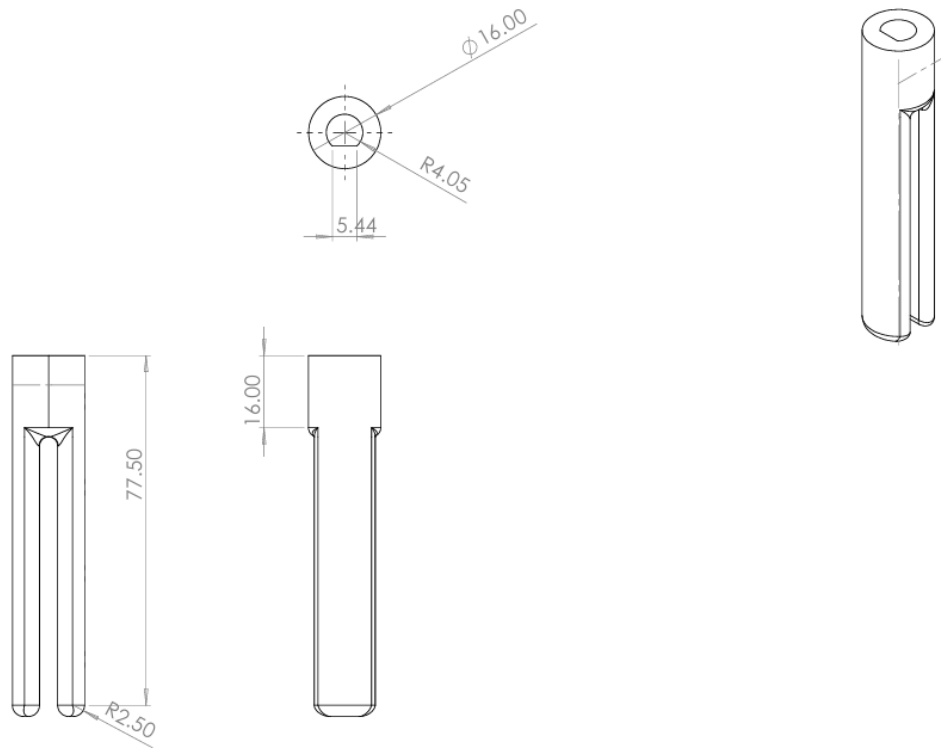
Figure 7-4 – Wrap and Component Enclosure



	USERNAME	DATE	DRAWING TITLE	REV
DRAWN BY	Terrence Micciche	05-10-26	Motor Enclosure	1
TEAM NAME	EG Terminators			
SECTION	EGT1110-001	SCALE: 1:2	WEIGHT: 61 g	SHEET 5 OF 5

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Figure 7-5 – Motor Enclosure



	USERNAME	DATE	DRAWING TITLE	REV
DRAWN BY	Terrence Micciche	05-10-26	Muy Thai Hand Wrap Spindle	1
TEAM NAME	EG Terminators			
SECTION	EGT1110-001		SCALE: 1:1	WEIGHT: 4 g SHEET 1 OF 1

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Figure 7-6 – Muy Thai Hand Wrap Spindle

7.2 Appendix B: Electrical Schematics

Below is figure 7-7, which visualizes the circuit diagram of the Motorized Hand Wrap Roller

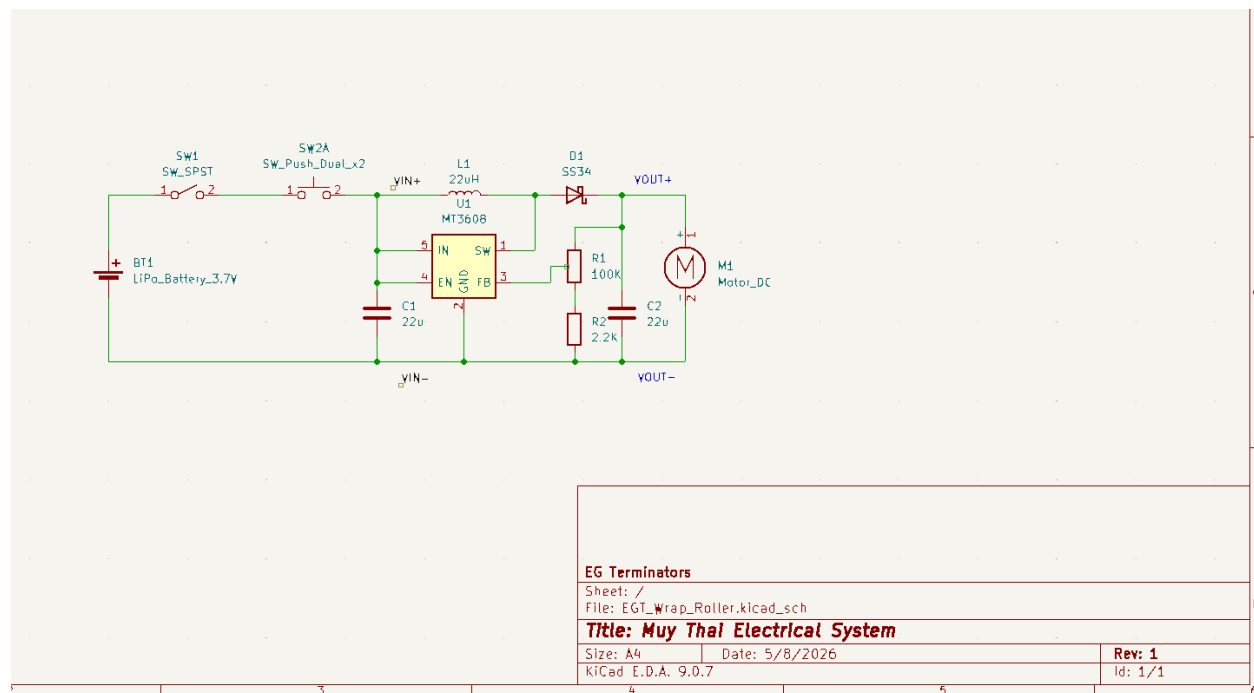


Figure 7-7 Motorized Hand Wrap Roller Electrical Diagram